

Psychological Disorders

History of Mental Illness

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This module is divided into three parts. The first is a brief introduction to various criteria we use to define or distinguish between normality and abnormality. The second, largest part is a history of mental illness from the Stone Age to the 20th century, with a special emphasis on the recurrence of three causal explanations for mental illness; supernatural, somatogenic, and psychogenic factors. This part briefly touches upon trephination, the Greek theory of hysteria within the context of the four bodily humors, witch hunts, asylums, moral treatment, mesmerism, catharsis, the mental hygiene movement, deinstitutionalization, community mental health services, and managed care. The third part concludes with a brief description of the issue of diagnosis.

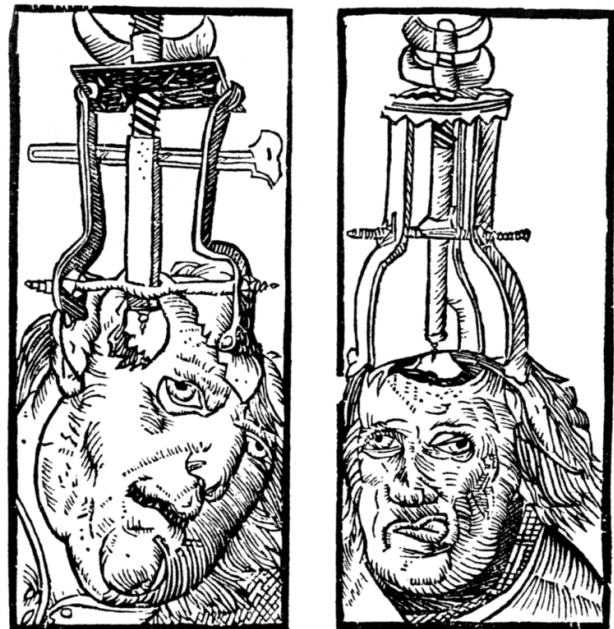
Learning Objectives

- Identify what the criteria used to distinguish normality from abnormality are.
- Understand the difference among the three main etiological theories of mental illness.
- Describe specific beliefs or events in history that exemplify each of these etiological theories (e.g., hysteria, humorism, witch hunts, asylums, moral treatments).
- Explain the differences in treatment facilities for the mentally ill (e.g., mental hospitals, asylums, community mental health centers).
- Describe the features of the “moral treatment” approach used by Chiarughi, Pinel, and Tuke.
- Describe the reform efforts of Dix and Beers and the outcomes of their work.
- Describe Kräpelin’s classification of mental illness and the current DSM system.

History of Mental Illness

References to mental illness can be found throughout history. The evolution of mental illness, however, has not been linear or progressive but rather cyclical. Whether a behavior is considered normal or abnormal depends on the context surrounding the behavior and thus changes as a function of a particular time and culture. In the past, uncommon behavior or behavior that deviated from the sociocultural norms and expectations of a specific culture and period has been used as a way to silence or control certain individuals or groups. As a result, a less cultural relativist view of abnormal behavior has focused instead on whether behavior poses a threat to oneself or others or causes so much pain and suffering that it interferes with one's work responsibilities or with one's relationships with family and friends.

Throughout history there have been three general theories of the etiology of mental illness: supernatural, somatogenic, and psychogenic. Supernatural theories attribute mental illness to possession by evil or demonic spirits, displeasure of gods, eclipses, planetary gravitation, curses, and sin. Somatogenic theories identify disturbances in physical functioning resulting from either illness, genetic inheritance, or brain damage or imbalance. Psychogenic theories focus on traumatic or stressful experiences, maladaptive learned associations and cognitions, or distorted perceptions. Etiological theories of mental illness determine the care and treatment mentally ill individuals receive. As we will see below, an individual believed to be possessed by the devil will be viewed and treated differently from an individual believed to be suffering from an excess of yellow bile. Their treatments will also differ, from exorcism to blood-letting. The theories, however, remain the same. They coexist as well as recycle over time.



Engravings from 1525 showing trephination. It was believed that drilling holes in the skull could cure mental disorders. [Image: Peter Treveris, CC0 Public Domain, <https://goo.gl/m25gce>]

Trephination is an example of the earliest supernatural explanation for mental illness. Examination of prehistoric skulls and cave art from as early as 6500 BC has identified surgical drilling of holes in skulls to treat head injuries and epilepsy as well as to allow evil spirits

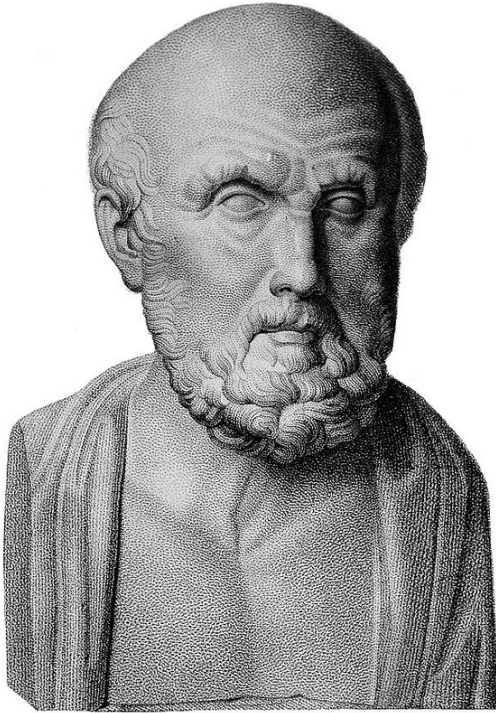
trapped within the skull to be released (Restak, 2000). Around 2700 BC, Chinese medicine's concept of complementary positive and negative bodily forces ("yin and yang") attributed mental (and physical) illness to an imbalance between these forces. As such, a harmonious life that allowed for the proper balance of yin and yang and movement of vital air was essential (Tseng, 1973).

Mesopotamian and Egyptian papyri from 1900 BC describe women suffering from mental illness resulting from a wandering uterus (later named hysteria by the Greeks): The uterus could become dislodged and attached to parts of the body like the liver or chest cavity, preventing their proper functioning or producing varied and sometimes painful symptoms. As a result, the Egyptians, and later the Greeks, also employed a somatogenic treatment of strong smelling substances to guide the uterus back to its proper location (pleasant odors to lure and unpleasant ones to dispel).

Throughout classical antiquity we see a return to supernatural theories of demonic possession or godly displeasure to account for abnormal behavior that was beyond the person's control. Temple attendance with religious healing ceremonies and incantations to the gods were employed to assist in the healing process. Hebrews saw madness as punishment from God, so treatment consisted of confessing sins and repenting. Physicians were also believed to be able to comfort and cure madness, however.

Greek physicians rejected supernatural explanations of mental disorders. It was around 400 BC that Hippocrates (460–370 BC) attempted to separate superstition and religion from medicine by systematizing the belief that a deficiency in or especially an excess of one of the four essential bodily fluids (i.e., humors)—blood, yellow bile, black bile, and phlegm—was responsible for physical and mental illness. For example, someone who was too temperamental suffered from too much blood and thus blood-letting would be the necessary treatment. Hippocrates classified mental illness into one of four categories—epilepsy, mania, melancholia, and brain fever—and like other prominent physicians and philosophers of his time, he did not believe mental illness was shameful or that mentally ill individuals should be held accountable for their behavior. Mentally ill individuals were cared for at home by family members and the state shared no responsibility for their care. Humorism remained a recurrent somatogenic theory up until the 19th century.

While Greek physician Galen (AD 130–201) rejected the notion of a uterus having an animistic soul, he agreed with the notion that an imbalance of the four bodily fluids could cause mental illness. He also opened the door for psychogenic explanations for mental illness, however, by allowing for the experience of psychological stress as a potential cause of abnormality. Galen's psychogenic theories were ignored for centuries, however, as physicians attributed mental



Many of Hippocrates' medical theories are no longer practiced today. However, he pioneered medicine as an empirical practice and came up with the "Hippocratic oath," which all doctors must swear to before joining the profession (i.e., the promise to never intentionally harm a patient). [Image: Wellcome Images, <https://goo.gl/dX21yj>, CC BY 4.0, <https://goo.gl/FJluOM>]

illness to physical causes throughout most of the millennium.

By the late Middle Ages, economic and political turmoil threatened the power of the Roman Catholic church. Between the 11th and 15th centuries, supernatural theories of mental disorders again dominated Europe, fueled by natural disasters like plagues and famines that lay people interpreted as brought about by the devil. Superstition, astrology, and alchemy took hold, and common treatments included prayer rites, relic touching, confessions, and atonement. Beginning in the 13th century the mentally ill, especially women, began to be persecuted as witches who were possessed. At the height of the witch hunts during the 15th through 17th centuries, with the Protestant Reformation having plunged Europe into religious strife, two Dominican monks wrote the *Malleus Maleficarum* (1486) as the ultimate manual

to guide witch hunts. Johann Weyer and Reginald Scot tried to convince people in the mid- to late-16th century that accused witches were actually women with mental illnesses and that mental illness was not due to demonic possession but to faulty metabolism and disease, but the Church's Inquisition banned both of their writings. Witch-hunting did not decline until the 17th and 18th centuries, after more than 100,000 presumed witches had been burned at the stake (Schoeneman, 1977; Zilboorg & Henry, 1941).

Modern treatments of mental illness are most associated with the establishment of hospitals and **asylums** beginning in the 16th century. Such institutions' mission was to house and confine the mentally ill, the poor, the homeless, the unemployed, and the criminal. War and economic depression produced vast numbers of undesirables and these were separated from society and sent to these institutions. Two of the most well-known institutions, St. Mary of Bethlehem in London, known as Bedlam, and the Hôpital Général of Paris—which included La Salpêtrière, La Pitié, and La Bicêtre—began housing mentally ill patients in the mid-16th and 17th centuries. As confinement laws focused on protecting the public *from* the mentally ill, governments became responsible for housing and feeding undesirables in exchange for

their personal liberty. Most inmates were institutionalized against their will, lived in filth and chained to walls, and were commonly exhibited to the public for a fee. Mental illness was nonetheless viewed somatogenically, so treatments were similar to those for physical illnesses: purges, bleedings, and emetics.

While inhumane by today's standards, the view of insanity at the time likened the mentally ill to animals (i.e., animalism) who did not have the capacity to reason, could not control themselves, were capable of violence without provocation, did not have the same physical sensitivity to pain or temperature, and could live in miserable conditions without complaint. As such, instilling fear was believed to be the best way to restore a disordered mind to reason.

By the 18th century, protests rose over the conditions under which the mentally ill lived, and the 18th and 19th centuries saw the growth of a more humanitarian view of mental illness. In 1785 Italian physician Vincenzo Chiarughi (1759–1820) removed the chains of patients at his St. Boniface hospital in Florence, Italy, and encouraged good hygiene and recreational and occupational training. More well known, French physician Philippe Pinel (1745–1826) and former patient Jean-Baptiste Pussin created a “traitement moral” at La Bicêtre and the Salpêtrière in 1793 and 1795 that also included unshackling patients, moving them to well-aired, well-lit rooms, and encouraging purposeful activity and freedom to move about the grounds (Micale, 1985).

In England, humanitarian reforms rose from religious concerns. William Tuke (1732–1822) urged the Yorkshire Society of (Quaker) Friends to establish the York Retreat in 1796, where patients were guests, not prisoners, and where the standard of care depended on dignity and courtesy as well as the therapeutic and moral value of physical work (Bell, 1980).

While America had asylums for the mentally ill—such as the Pennsylvania Hospital in Philadelphia and the Williamsburg Hospital, established in 1756 and 1773—the somatogenic theory of mental illness of the time—promoted especially by the father of America psychiatry, Benjamin Rush (1745–1813)—had led to treatments



Dorothea Dix worked to change the negative perceptions of people with mental illness and helped create institutions where they could receive compassionate care. [Image: State Archives of North Carolina, <https://goo.gl/wRgGsi>, no known copyright restrictions]

such as blood-letting, gyrators, and tranquilizer chairs. When Tuke's York Retreat became the model for half of the new private asylums established in the United States, however, psychogenic treatments such as compassionate care and physical labor became the hallmarks of the new American asylums, such as the Friends Asylum in Frankford, Pennsylvania, and the Bloomingdale Asylum in New York City, established in 1817 and 1821 (Grob, 1994).

Moral treatment had to be abandoned in America in the second half of the 19th century, however, when these asylums became overcrowded and custodial in nature and could no longer provide the space nor attention necessary. When retired school teacher Dorothea Dix discovered the negligence that resulted from such conditions, she advocated for the establishment of state hospitals. Between 1840 and 1880, she helped establish over 30 mental institutions in the United States and Canada (Viney & Zorich, 1982). By the late 19th century, moral treatment had given way to the mental hygiene movement, founded by former patient Clifford Beers with the publication of his 1908 memoir *A Mind That Found Itself*. Riding on Pasteur's breakthrough germ theory of the 1860s and 1870s and especially on the early 20th century discoveries of vaccines for cholera, syphilis, and typhus, the mental hygiene movement reverted to a somatogenic theory of mental illness.

European psychiatry in the late 18th century and throughout the 19th century, however, struggled between somatogenic and psychogenic explanations of mental illness, particularly hysteria, which caused physical symptoms such as blindness or paralysis with no apparent physiological explanation. Franz Anton Mesmer (1734–1815), influenced by contemporary discoveries in electricity, attributed hysterical symptoms to imbalances in a universal magnetic fluid found in individuals, rather than to a wandering uterus (Forrest, 1999). James Braid (1795–1860) shifted this belief in **mesmerism** to one in hypnosis, thereby proposing a psychogenic treatment for the removal of symptoms. At the time, famed Salpêtrière Hospital neurologist Jean-Martin Charcot (1825–1893), and Ambroise Auguste Liébault (1823–1904) and Hyppolyte Bernheim (1840–1919) of the Nancy School in France, were engaged in a bitter etiological battle over hysteria, with Charcot maintaining that the hypnotic suggestibility underlying hysteria was a neurological condition while Liébault and Bernheim believed it to be a general trait that varied in the population. Josef Breuer (1842–1925) and Sigmund Freud (1856–1939) would resolve this dispute in favor of a psychogenic explanation for mental illness by treating hysteria through hypnosis, which eventually led to the **cathartic method** that became the precursor for psychoanalysis during the first half of the 20th century.

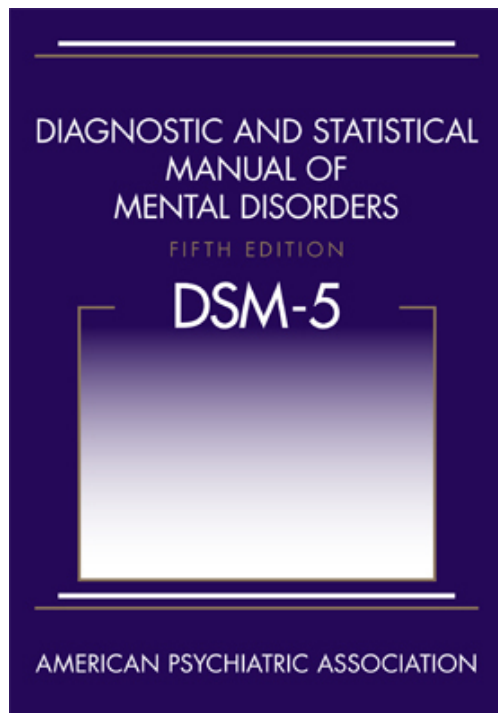
Psychoanalysis was the dominant psychogenic treatment for mental illness during the first half of the 20th century, providing the launching pad for the more than 400 different schools of psychotherapy found today (Magnavita, 2006). Most of these schools cluster around broader behavioral, cognitive, cognitive-behavioral, psychodynamic, and client-centered

approaches to psychotherapy applied in individual, marital, family, or group formats. Negligible differences have been found among all these approaches, however; their efficacy in treating mental illness is due to factors shared among all of the approaches (not particular elements specific to each approach): the therapist-patient alliance, the therapist's allegiance to the therapy, therapist competence, and placebo effects (Luborsky et al., 2002; Messer & Wampold, 2002).

In contrast, the leading somatogenic treatment for mental illness can be found in the establishment of the first psychotropic medications in the mid-20th century. Restraints, electro-convulsive shock therapy, and lobotomies continued to be employed in American state institutions until the 1970s, but they quickly made way for a burgeoning pharmaceutical industry that has viewed and treated mental illness as a chemical imbalance in the brain.

Both etiological theories coexist today in what the psychological discipline holds as the **biopsychosocial model** of explaining human behavior. While individuals may be born with a genetic predisposition for a certain psychological disorder, certain psychological stressors

need to be present for them to develop the disorder. Sociocultural factors such as sociopolitical or economic unrest, poor living conditions, or problematic interpersonal relationships are also viewed as contributing factors. However much we want to believe that we are above the treatments described above, or that the present is always the most enlightened time, let us not forget that our thinking today continues to reflect the same underlying somatogenic and psychogenic theories of mental illness discussed throughout this cursory 9,000-year history.



Up until the 1970's, homosexuality was included in the DSM as a psychological disorder. Thankfully, society and clinical understanding changed to recognize it didn't belong. [Image: Rene Walter, <https://goo.gl/CcJAA1>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Diagnosis of Mental Illness

Progress in the treatment of mental illness necessarily implies improvements in the diagnosis of mental illness. A standardized diagnostic classification system with agreed-upon definitions of psychological disorders creates a shared language among mental-health providers and aids in clinical research. While diagnoses were recognized as far

back as the Greeks, it was not until 1883 that German psychiatrist Emil Kräpelin (1856–1926) published a comprehensive system of psychological disorders that centered around a pattern of symptoms (i.e., **syndrome**) suggestive of an underlying physiological cause. Other clinicians also suggested popular classification systems but the need for a single, shared system paved the way for the American Psychiatric Association's 1952 publication of the first *Diagnostic and Statistical Manual* (DSM).

The DSM has undergone various revisions (in 1968, 1980, 1987, 1994, 2000, 2013), and it is the 1980 DSM-III version that began a multiaxial classification system that took into account the entire individual rather than just the specific problem behavior. Axes I and II contain the clinical diagnoses, including intellectual disability and personality disorders. Axes III and IV list any relevant medical conditions or psychosocial or environmental stressors, respectively. Axis V provides a global assessment of the individual's level of functioning. The most recent version -- the DSM-5-- has combined the first three axes and removed the last two. These revisions reflect an attempt to help clinicians streamline diagnosis and work better with other diagnostic systems such as health diagnoses outlined by the World Health Organization.

While the DSM has provided a necessary shared language for clinicians, aided in clinical research, and allowed clinicians to be reimbursed by insurance companies for their services, it is not without criticism. The DSM is based on clinical and research findings from Western culture, primarily the United States. It is also a medicalized categorical classification system that assumes disordered behavior does not differ in degree but in kind, as opposed to a dimensional classification system that would plot disordered behavior along a continuum. Finally, the number of diagnosable disorders has tripled since it was first published in 1952, so that almost half of Americans will have a diagnosable disorder in their lifetime, contributing to the continued concern of labeling and stigmatizing mentally ill individuals. These concerns appear to be relevant even in the DSM-5 version that came out in May of 2013.

Outside Resources

Video: An introduction to and overview of psychology, from its origins in the nineteenth century to current study of the brain's biochemistry.

<http://www.learner.org/series/discoveringpsychology/01/e01expand.html>

Video: The BBC provides an overview of ancient Greek approaches to health and medicine.

<https://www.tes.com/teaching-resource/ancient-greek-approaches-to-health-and-medicine-6176019>

Web: Images from the History of Medicine. Search "\"mental illness\""

<http://ihm.nlm.nih.gov/luna/servlet/view/all>

Web: Science Museum Brought to Life

<http://www.sciencemuseum.org.uk/broughttolife/themes/mentalhealthandillness.aspx>

Web: The Social Psychology Network provides a number of links and resources.

<https://www.socialpsychology.org/history.htm>

Web: The Wellcome Library. Search "\"mental illness\"".

<http://wellcomelibrary.org/>

Web: UCL Department of Science and Technology Studies

<https://www.ucl.ac.uk/sts/>

Web: US National Library of Medicine

<http://vsearch.nlm.nih.gov/vivisimo/cgi-bin/query-meta?query=mental+illness&v:project.nlm-main-website>

Discussion Questions

1. What does it mean to say that someone is mentally ill? What criteria are usually considered to determine whether someone is mentally ill?
2. Describe the difference between supernatural, somatogenic, and psychogenic theories of mental illness and how subscribing to a particular etiological theory determines the type of treatment used.
3. How did the Greeks describe hysteria and what treatment did they prescribe?
4. Describe humorism and how it explained mental illness.

5. Describe how the witch hunts came about and their relationship to mental illness.
6. Describe the development of treatment facilities for the mentally insane, from asylums to community mental health centers.
7. Describe the humane treatment of the mentally ill brought about by Chiarughi, Pinel, and Tuke in the late 18th and early 19th centuries and how it differed from the care provided in the centuries preceding it.
8. Describe William Tuke's treatment of the mentally ill at the York Retreat within the context of the Quaker Society of Friends. What influence did Tuke's treatment have in other parts of the world?
9. What are the 20th-century treatments resulting from the psychogenic and somatogenic theories of mental illness?
10. Describe why a classification system is important and how the leading classification system used in the United States works. Describe some concerns with regard to this system.

Vocabulary

Animism

The belief that everyone and everything had a “soul” and that mental illness was due to animistic causes, for example, evil spirits controlling an individual and his/her behavior.

Asylum

A place of refuge or safety established to confine and care for the mentally ill; forerunners of the mental hospital or psychiatric facility.

Biopsychosocial model

A model in which the interaction of biological, psychological, and sociocultural factors is seen as influencing the development of the individual.

Cathartic method

A therapeutic procedure introduced by Breuer and developed further by Freud in the late 19th century whereby a patient gains insight and emotional relief from recalling and reliving traumatic events.

Cultural relativism

The idea that cultural norms and values of a society can only be understood on their own terms or in their own context.

Etiology

The causal description of all of the factors that contribute to the development of a disorder or illness.

Humorism (or humoralism)

A belief held by ancient Greek and Roman physicians (and until the 19th century) that an excess or deficiency in any of the four bodily fluids, or humors—blood, black bile, yellow bile, and phlegm—directly affected their health and temperament.

Hysteria

Term used by the ancient Greeks and Egyptians to describe a disorder believed to be caused by a woman’s uterus wandering throughout the body and interfering with other organs (today referred to as conversion disorder, in which psychological problems are expressed in physical form).

Maladaptive

Term referring to behaviors that cause people who have them physical or emotional harm, prevent them from functioning in daily life, and/or indicate that they have lost touch with reality and/or cannot control their thoughts and behavior (also called dysfunctional).

Mesmerism

Derived from Franz Anton Mesmer in the late 18th century, an early version of hypnotism in which Mesmer claimed that hysterical symptoms could be treated through animal magnetism emanating from Mesmer's body and permeating the universe (and later through magnets); later explained in terms of high suggestibility in individuals.

Psychogenesis

Developing from psychological origins.

Somatogenesis

Developing from physical/bodily origins.

Supernatural

Developing from origins beyond the visible observable universe.

Syndrome

Involving a particular group of signs and symptoms.

"Traitement moral" (moral treatment)

A therapeutic regimen of improved nutrition, living conditions, and rewards for productive behavior that has been attributed to Philippe Pinel during the French Revolution, when he released mentally ill patients from their restraints and treated them with compassion and dignity rather than with contempt and denigration.

Trephination

The drilling of a hole in the skull, presumably as a way of treating psychological disorders.

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Therapeutic Orientations

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In the past century, a number of psychotherapeutic orientations have gained popularity for treating mental illnesses. This module outlines some of the best-known therapeutic approaches and explains the history, techniques, advantages, and disadvantages associated with each. The most effective modern approach is cognitive behavioral therapy (CBT). We also discuss psychoanalytic therapy, person-centered therapy, and mindfulness-based approaches. Drug therapy and emerging new treatment strategies will also be briefly explored.

Learning Objectives

- Become familiar with the most widely practiced approaches to psychotherapy.
- For each therapeutic approach, consider: history, goals, key techniques, and empirical support.
- Consider the impact of emerging treatment strategies in mental health.

Introduction

The history of mental illness can be traced as far back as 1500 BCE, when the ancient Egyptians noted cases of “distorted concentration” and “emotional distress in the heart or mind” (Nasser, 1987). Today, nearly half of all Americans will experience mental illness at some point in their lives, and mental health problems affect more than one-quarter of the population in any given year (Kessler et al., 2005). Fortunately, a range of psychotherapies exist to treat mental illnesses. This module provides an overview of some of the best-known schools of thought in

psychotherapy. Currently, the most effective approach is called Cognitive Behavioral Therapy (CBT); however, other approaches, such as psychoanalytic therapy, person-centered therapy, and mindfulness-based therapies are also used—though the effectiveness of these treatments aren't as clear as they are for CBT. Throughout this module, note the advantages and disadvantages of each approach, paying special attention to their support by empirical research.



CBT is an approach to treating mental illness that involves work with a therapist as well as homework assignments between sessions. It has proven to be very effective for virtually all psychiatric illnesses. [Image: DFAT, <https://goo.gl/bWmzaa>, CC BY 2.0, <https://goo.gl/BRvSA7>]

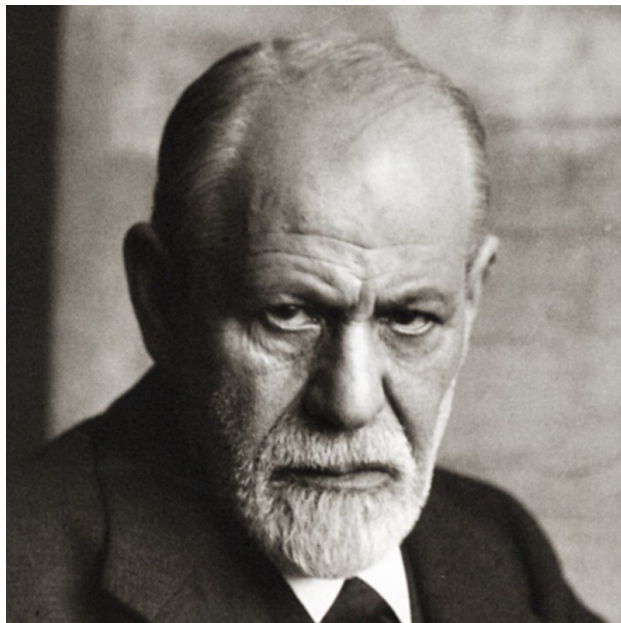
Psychoanalysis and Psychodynamic Therapy

The earliest organized therapy for mental disorders was psychoanalysis. Made famous in the early 20th century by one of the best-known clinicians of all time, Sigmund Freud, this approach stresses that mental health problems are rooted in unconscious conflicts and desires. In order to resolve the mental illness, then, these unconscious struggles must be identified and addressed. Psychoanalysis often does this through exploring one's early childhood experiences that may have continuing repercussions on one's mental health in the present and later in life. Psychoanalysis is an intensive, long-term approach in which patients and therapists may meet multiple times per week, often for many years.

History of Psychoanalytic Therapy

Freud initially suggested that mental health problems arise from efforts to push inappropriate sexual urges out of conscious awareness (Freud, 1895/1955). Later, Freud suggested more generally that psychiatric problems are the result of tension between different parts of the mind: the id, the superego, and the ego. In Freud's *structural model*, the id represents pleasure-driven unconscious urges (e.g., our animalistic desires for sex and aggression), while the superego is the semi-conscious part of the mind where morals and societal judgment are internalized (e.g., the part of you that automatically knows how society expects you to behave). The ego—also partly conscious—mediates between the id and superego. Freud believed that bringing unconscious struggles like these (where the id demands one thing and the superego another) into conscious awareness would relieve the stress of the conflict (Freud, 1920/1955)—which became the goal of psychoanalytic therapy.

Although psychoanalysis is still practiced today, it has largely been replaced by the more broadly defined psychodynamic therapy. This latter approach has the same basic tenets as psychoanalysis, but is briefer, makes more of an effort to put clients in their social and interpersonal context, and focuses more on relieving psychological distress than on changing the person.



Building on the work of Josef Breuer and others, Sigmund Freud developed psychotherapeutic theories and techniques that became widely known as psychoanalysis or psychoanalytic therapy. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Techniques in Psychoanalysis

Psychoanalysts and psychodynamic therapists employ several techniques to explore patients' unconscious mind. One common technique is called free association. Here, the patient shares any and all thoughts that come to mind, without attempting to organize or censor them in any way. For example, if you took a pen and paper and just wrote down whatever came into your head, letting one thought lead to the next without allowing conscious criticism to shape what you were writing, you would be doing free association. The analyst then uses his or her expertise to discern patterns or underlying meaning in the patient's

thoughts.

Sometimes, free association exercises are applied specifically to childhood recollections. That is, psychoanalysts believe a person's childhood relationships with caregivers often determine the way that person relates to others, and predicts later psychiatric difficulties. Thus, exploring these childhood memories, through free association or otherwise, can provide therapists with insights into a patient's psychological makeup.

Because we don't always have the ability to consciously recall these deep memories, psychoanalysts also discuss their patients' dreams. In Freudian theory, dreams contain not only *manifest* (or literal) content, but also *latent* (or symbolic) content (Freud, 1900; 1955). For example, someone may have a dream that his/her teeth are falling out—the manifest or actual content of the dream. However, dreaming that one's teeth are falling out could be a reflection of the person's unconscious concern about losing his or her physical attractiveness—the latent or metaphorical content of the dream. It is the therapist's job to help discover the latent content underlying one's manifest content through dream analysis.

In psychoanalytic and psychodynamic therapy, the therapist plays a receptive role—interpreting the patient's thoughts and behavior based on clinical experience and psychoanalytic theory. For example, if during therapy a patient begins to express unjustified anger toward the therapist, the therapist may recognize this as an act of *transference*. That is, the patient may be displacing feelings for people in his or her life (e.g., anger toward a parent) onto the therapist. At the same time, though, the therapist has to be aware of his or her own thoughts and emotions, for, in a related process, called *countertransference*, the therapist may displace his/her own emotions onto the patient.

The key to psychoanalytic theory is to have patients uncover the buried, conflicting content of their mind, and therapists use various tactics—such as seating patients to face away from them—to promote a freer self-disclosure. And, as a therapist spends more time with a patient, the therapist can come to view his or her relationship with the patient as another reflection of the patient's mind.

Advantages and Disadvantages of Psychoanalytic Therapy

Psychoanalysis was once the only type of psychotherapy available, but presently the number of therapists practicing this approach is decreasing around the world. Psychoanalysis is not appropriate for some types of patients, including those with severe psychopathology or intellectual disability. Further, psychoanalysis is often expensive because treatment usually

lasts many years. Still, some patients and therapists find the prolonged and detailed analysis very rewarding.

Perhaps the greatest disadvantage of psychoanalysis and related approaches is the lack of empirical support for their effectiveness. The limited research that has been conducted on these treatments suggests that they do not reliably lead to better mental health outcomes (e.g., Driessen et al., 2010). And, although there are some reviews that seem to indicate that long-term psychodynamic therapies might be beneficial (e.g., Leichsenring & Rabung, 2008), other researchers have questioned the validity of these reviews. Nevertheless, psychoanalytic theory was history's first attempt at formal treatment of mental illness, setting the stage for the more modern approaches used today.

Humanistic and Person-Centered Therapy

One of the next developments in therapy for mental illness, which arrived in the mid-20th century, is called humanistic or **person-centered therapy** (PCT). Here, the belief is that mental health problems result from an inconsistency between patients' behavior and their true personal identity. Thus, the goal of PCT is to create conditions under which patients can discover their self-worth, feel comfortable exploring their own identity, and alter their behavior to better reflect this identity.

History of Person-Centered Therapy

PCT was developed by a psychologist named Carl Rogers, during a time of significant growth in the movements of humanistic theory and human potential. These perspectives were based on the idea that humans have an inherent drive to realize and express their own capabilities and creativity. Rogers, in particular, believed that all people have the potential to change and improve, and that the role of therapists is to foster self-understanding in an environment where adaptive change is most likely to occur (Rogers, 1951). Rogers suggested that the therapist and patient must engage in a



The quality of the relationship between therapist and patient is of great importance in person-centered therapy. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

genuine, egalitarian relationship in which the therapist is nonjudgmental and empathetic. In PCT, the patient should experience both a vulnerability to anxiety, which motivates the desire to change, and an appreciation for the therapist's support.

Techniques in Person-Centered Therapy

Humanistic and person-centered therapy, like psychoanalysis, involves a largely unstructured conversation between the therapist and the patient. Unlike psychoanalysis, though, a therapist using PCT takes a passive role, guiding the patient toward his or her own self-discovery. Rogers's original name for PCT was *non-directive therapy*, and this notion is reflected in the flexibility found in PCT. Therapists do not try to change patients' thoughts or behaviors directly. Rather, their role is to provide the therapeutic relationship as a platform for personal growth. In these kinds of sessions, the therapist tends only to ask questions and doesn't provide any judgment or interpretation of what the patient says. Instead, the therapist is present to provide a safe and encouraging environment for the person to explore these issues for him- or herself.

An important aspect of the PCT relationship is the therapist's unconditional positive regard for the patient's feelings and behaviors. That is, the therapist is never to condemn or criticize the patient for what s/he has done or thought; the therapist is only to express warmth and empathy. This creates an environment free of approval or disapproval, where patients come to appreciate their value and to behave in ways that are congruent with their own identity.

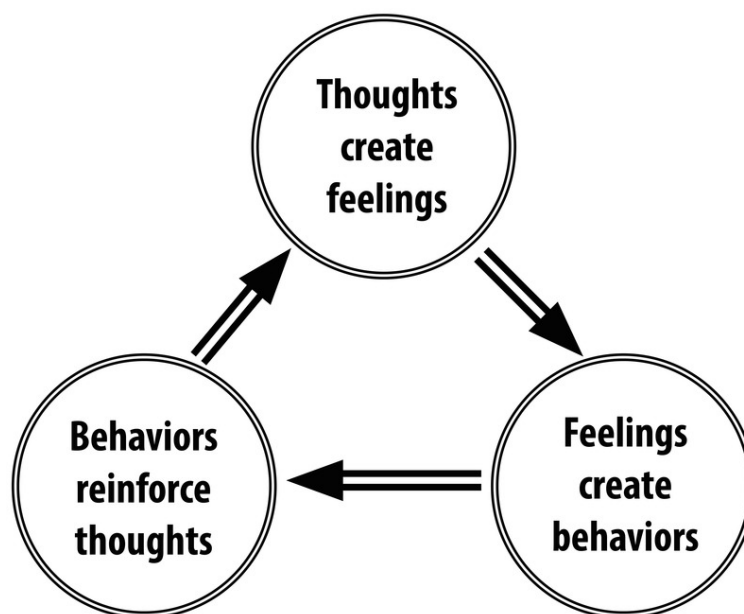
Advantages and Disadvantages of Person-Centered Therapy

One key advantage of person-centered therapy is that it is highly acceptable to patients. In other words, people tend to find the supportive, flexible environment of this approach very rewarding. Furthermore, some of the themes of PCT translate well to other therapeutic approaches. For example, most therapists of any orientation find that clients respond well to being treated with nonjudgmental empathy. The main disadvantage to PCT, however, is that findings about its effectiveness are mixed. One possibility for this could be that the treatment is primarily based on *unspecific treatment factors*. That is, rather than using therapeutic techniques that are specific to the patient and the mental problem (i.e., *specific treatment factors*), the therapy focuses on techniques that can be applied to anyone (e.g., establishing a good relationship with the patient) (Cuijpers et al., 2012; Friedli, King, Lloyd, & Horder, 1997). Similar to how "one-size-fits-all" doesn't really fit every person, PCT uses the same practices for everyone, which may work for some people but not others. Further research is necessary to evaluate its utility as a therapeutic approach.

Cognitive Behavioral Therapy

Although both psychoanalysis and PCT are still used today, another therapy, cognitive-behavioral therapy (CBT), has gained more widespread support and practice. CBT refers to a family of therapeutic approaches whose goal is to alleviate psychological symptoms by changing their underlying cognitions and behaviors. The premise of CBT is that thoughts, behaviors, and emotions interact and contribute to various mental disorders. For example, let's consider how a CBT therapist would view a patient who compulsively washes her hands for hours every day. First, the therapist would identify the patient's maladaptive thought: "If I don't wash my hands like this, I will get a disease and die." The therapist then identifies how this maladaptive *thought* leads to a maladaptive *emotion*: the feeling of anxiety when her hands aren't being washed. And finally, this maladaptive emotion leads to the maladaptive behavior: the patient washing her hands for hours every day.

CBT is a present-focused therapy (i.e., focused on the "now" rather than causes from the past, such as childhood relationships) that uses behavioral goals to improve one's mental illness. Often, these behavioral goals involve between-session homework assignments. For example, the therapist may give the hand-washing patient a worksheet to take home; on this worksheet, the woman is to write down every time she feels the urge to wash her hands, how she deals with the urge, and what behavior she replaces that urge with. When the patient has her next therapy session, she and the therapist review her "homework" together. CBT is a relatively



Pattern of thoughts, feelings, and behaviors addressed through cognitive-behavioral therapy.

brief intervention of 12 to 16 weekly sessions, closely tailored to the nature of the psychopathology and treatment of the specific mental disorder. And, as the empirical data shows, CBT has proven to be highly efficacious for virtually all psychiatric illnesses (Hofmann, Asnaani, Vonk, Sawyer, & Fang, 2012).

History of Cognitive Behavioral Therapy

CBT developed from clinical work conducted in the mid-20th century by Dr. Aaron T. Beck, a psychiatrist, and Albert Ellis, a psychologist. Beck used the term **automatic thoughts** to refer to the thoughts depressed patients report experiencing spontaneously. He observed that these thoughts arise from three belief systems, or **schemas**: beliefs about the self, beliefs about the world, and beliefs about the future. In treatment, therapy initially focuses on identifying automatic thoughts (e.g., “If I don’t wash my hands constantly, I’ll get a disease”), testing their validity, and replacing maladaptive thoughts with more adaptive thoughts (e.g., “Washing my hands three times a day is sufficient to prevent a disease”). In later stages of treatment, the patient’s maladaptive schemas are examined and modified. Ellis (1957) took a comparable approach, in what he called rational-emotive-behavioral therapy (REBT), which also encourages patients to evaluate their own thoughts about situations.

Techniques in CBT

Beck and Ellis strove to help patients identify maladaptive appraisals, or the untrue judgments and evaluations of certain thoughts. For example, if it’s your first time meeting new people, you may have the automatic thought, “These people won’t like me because I have nothing interesting to share.” That thought itself is not what’s troublesome; the appraisal (or evaluation) that it might have merit is what’s troublesome. The goal of CBT is to help people make adaptive, instead of maladaptive, appraisals (e.g., “I do know interesting things!”). This technique of **reappraisal, or cognitive restructuring**, is a fundamental aspect of CBT. With cognitive restructuring, it is the therapist’s job to help point out when a person has an inaccurate or maladaptive thought, so that the patient can either eliminate it or modify it to be more adaptive.

In addition to *thoughts*, though, another important treatment target of CBT is maladaptive *behavior*. Every time a person engages in maladaptive behavior (e.g., never speaking to someone in new situations), he or she reinforces the validity of the maladaptive thought, thus maintaining or perpetuating the psychological illness. In treatment, the therapist and patient work together to develop healthy behavioral habits (often tracked with worksheet-like homework), so that the patient can break this cycle of maladaptive thoughts and behaviors.

For many mental health problems, especially anxiety disorders, CBT incorporates what is known as **exposure therapy**. During exposure therapy, a patient confronts a problematic situation and fully engages in the experience instead of avoiding it. For example, imagine a man who is terrified of spiders. Whenever he encounters one, he immediately screams and panics. In exposure therapy, the man would be forced to confront and interact with spiders, rather than simply avoiding them as he usually does. The goal is to reduce the fear associated with the situation through *extinction learning*, a neurobiological and cognitive process by which the patient “unlearns” the irrational fear. For example, exposure therapy for someone terrified of spiders might begin with him looking at a cartoon of a spider, followed by him looking at pictures of real spiders, and later, him handling a plastic spider. After weeks of this incremental exposure, the patient may even be able to hold a live spider. After repeated exposure (starting small and building one’s way up), the patient experiences less physiological fear and maladaptive thoughts about spiders, breaking his tendency for anxiety and subsequent avoidance.

Advantages and Disadvantages of CBT

CBT interventions tend to be relatively brief, making them cost-effective for the average consumer. In addition, CBT is an intuitive treatment that makes logical sense to patients. It can also be adapted to suit the needs of many different populations. One disadvantage, however, is that CBT does involve significant effort on the patient’s part, because the patient is an active participant in treatment. Therapists often assign “homework” (e.g., worksheets for recording one’s thoughts and behaviors) between sessions to maintain the cognitive and behavioral habits the patient is working on. The greatest strength of CBT is the abundance of empirical support for its effectiveness. Studies have consistently found CBT to be equally or more effective than other forms of treatment, including medication and other therapies (Butler, Chapman, Forman, & Beck, 2006; Hofmann et al., 2012). For this reason, CBT is considered a first-line treatment for many mental disorders.

Focus Topic: Pioneers of CBT

The central notion of CBT is the idea that a person’s behavioral and emotional responses are causally influenced by one’s thinking. The stoic Greek philosopher Epictetus is quoted as saying, “men are not moved by things, but by the view they take of them.” Meaning, it is not the event per se, but rather one’s assumptions (including interpretations and perceptions) of the event that are responsible for one’s emotional response to it. Beck calls these assumptions

about events and situations automatic thoughts (Beck, 1979), whereas Ellis (1962) refers to these assumptions as self-statements. The cognitive model assumes that these cognitive processes cause the emotional and behavioral responses to events or stimuli. This causal chain is illustrated in Ellis's ABC model, in which A stands for the antecedent event, B stands for belief, and C stands for consequence. During CBT, the person is encouraged to carefully observe the sequence of events and the response to them, and then explore the validity of the underlying beliefs through behavioral experiments and reasoning, much like a detective or scientist.

Acceptance and Mindfulness-Based Approaches

Unlike the preceding therapies, which were developed in the 20th century, this next one was born out of age-old Buddhist and yoga practices. **Mindfulness**, or a process that tries to cultivate a nonjudgmental, yet attentive, mental state, is a therapy that focuses on one's awareness of bodily sensations, thoughts, and the outside environment. Whereas other therapies work to modify or eliminate these sensations and thoughts, mindfulness focuses on nonjudgmentally accepting them (Kabat-Zinn, 2003; Baer, 2003). For example, whereas CBT may actively confront and work to change a maladaptive thought, mindfulness therapy works to acknowledge and accept the thought, understanding that the thought is spontaneous and not what the person truly believes. There are two important components of mindfulness: (1) self-regulation of attention, and (2) orientation toward the present moment (Bishop et al., 2004). Mindfulness is thought to improve mental health because it draws attention away from past and future stressors, encourages acceptance of troubling thoughts and feelings, and promotes physical relaxation.

Techniques in Mindfulness-Based Therapy

Psychologists have adapted the practice of mindfulness as a form of psychotherapy, generally called **mindfulness-based therapy** (MBT). Several types of MBT have become popular in recent years, including *mindfulness-based stress reduction* (MBSR) (e.g., Kabat-Zinn, 1982) and *mindfulness-based cognitive therapy* (MBCT) (e.g., Segal, Williams, & Teasdale, 2002).

MBSR uses meditation, yoga, and attention to physical experiences to reduce stress. The hope is that reducing a person's overall stress will allow that person to more objectively evaluate his or her thoughts. In MBCT, rather than reducing one's general stress to address a specific problem, attention is focused on one's thoughts and their associated emotions. For example,



One of the most important advantages of mindfulness based therapy is its level of accessibility to patients. [Image: Wayne MacPhail, <https://goo.gl/aSZanf>, CC BY-NC SA 2.0, <https://goo.gl/Toc0ZF>]

MBCT helps prevent relapses in depression by encouraging patients to evaluate their own thoughts objectively and without value judgment (Baer, 2003). Although cognitive behavioral therapy (CBT) may seem similar to this, it focuses on “pushing out” the maladaptive thought, whereas mindfulness-based cognitive therapy focuses on “not getting caught up” in it. The treatments used in MBCT have been used to address a wide range of illnesses, including depression, anxiety, chronic pain, coronary artery disease, and fibromyalgia (Hofmann, Sawyer, Witt & Oh, 2010).

Mindfulness and acceptance—in addition to being therapies in their own right—have also been used as “tools” in other cognitive-

behavioral therapies, particularly in **dialectical behavior therapy (DBT)** (e.g., Linehan, Armstrong, Suarez, Allmon, & Heard, 1991). DBT, often used in the treatment of borderline personality disorder, focuses on skills training. That is, it often employs mindfulness and cognitive behavioral therapy practices, but it also works to teach its patients “skills” they can use to correct maladaptive tendencies. For example, one skill DBT teaches patients is called *distress tolerance*—or, ways to cope with maladaptive thoughts and emotions in the moment. For example, people who feel an urge to cut themselves may be taught to snap their arm with a rubber band instead. The primary difference between DBT and CBT is that DBT employs techniques that address the symptoms of the problem (e.g., cutting oneself) rather than the problem itself (e.g., understanding the psychological motivation to cut oneself). CBT does not teach such skills training because of the concern that the skills—even though they may help in the short-term—may be harmful in the long-term, by maintaining maladaptive thoughts and behaviors.

DBT is founded on the perspective of a **dialectical worldview**. That is, rather than thinking of the world as “black and white,” or “only good and only bad,” it focuses on accepting that some things can have characteristics of both “good” and “bad.” So, in a case involving maladaptive thoughts, instead of teaching that a thought is entirely bad, DBT tries to help patients be less judgmental of their thoughts (as with mindfulness-based therapy) and encourages change through therapeutic progress, using cognitive-behavioral techniques as well as mindfulness

exercises.

Another form of treatment that also uses mindfulness techniques is acceptance and commitment therapy (ACT) (Hayes, Strosahl, & Wilson, 1999). In this treatment, patients are taught to observe their thoughts from a detached perspective (Hayes et al., 1999). ACT encourages patients *not* to attempt to change or avoid thoughts and emotions they observe in themselves, but to recognize which are beneficial and which are harmful. However, the differences among ACT, CBT, and other mindfulness-based treatments are a topic of controversy in the current literature.

Advantages and Disadvantages of Mindfulness-Based Therapy

Two key advantages of mindfulness-based therapies are their acceptability and accessibility to patients. Because yoga and meditation are already widely known in popular culture, consumers of mental healthcare are often interested in trying related psychological therapies. Currently, psychologists have not come to a consensus on the efficacy of MBT, though growing evidence supports its effectiveness for treating mood and anxiety disorders. For example, one review of MBT studies for anxiety and depression found that mindfulness-based interventions generally led to moderate symptom improvement (Hofmann et al., 2010).

Emerging Treatment Strategies



Recent improvements in video chat technology along with the proliferation of mobile devices like smartphones and tablets has made online delivery of therapy more commonplace. [Image: Noba, CC BY 2.0, <https://goo.gl/BRvSA7>]

With growth in research and technology, psychologists have been able to develop new treatment strategies in recent years. Often, these approaches focus on enhancing existing treatments, such as cognitive-behavioral therapies, through the use of technological advances. For example, *internet-* and *mobile-delivered therapies* make psychological treatments more available, through smartphones and online access. Clinician-supervised online CBT modules allow patients to access treatment from home on their own schedule—an opportunity particularly important for patients with less geographic or socioeconomic access to traditional

treatments. Furthermore, smartphones help extend therapy to patients' daily lives, allowing for symptom tracking, homework reminders, and more frequent therapist contact.

Another benefit of technology is cognitive bias modification. Here, patients are given exercises, often through the use of video games, aimed at changing their problematic thought processes. For example, researchers might use a mobile app to train alcohol abusers to avoid stimuli related to alcohol. One version of this game flashes four pictures on the screen—three alcohol cues (e.g., a can of beer, the front of a bar) and one health-related image (e.g., someone drinking water). The goal is for the patient to tap the healthy picture as fast as s/he can. Games like these aim to target patients' automatic, subconscious thoughts that may be difficult to direct through conscious effort. That is, by repeatedly tapping the healthy image, the patient learns to “ignore” the alcohol cues, so when those cues are encountered in the environment, they will be less likely to trigger the urge to drink. Approaches like these are promising because of their accessibility, however they require further research to establish their effectiveness.

Yet another emerging treatment employs *CBT-enhancing pharmaceutical agents*. These are drugs used to improve the effects of therapeutic interventions. Based on research from animal experiments, researchers have found that certain drugs influence the biological processes known to be involved in learning. Thus, if people take these drugs while going through psychotherapy, they are better able to “learn” the techniques for improvement. For example, the antibiotic d-cycloserine improves treatment for anxiety disorders by facilitating the learning processes that occur during exposure therapy. Ongoing research in this exciting area may prove to be quite fruitful.

Pharmacological Treatments

Up until this point, all the therapies we have discussed have been talk-based or meditative practices. However, psychiatric medications are also frequently used to treat mental disorders, including schizophrenia, bipolar disorder, depression, and anxiety disorders. Psychiatric drugs are commonly used, in part, because they can be prescribed by general medical practitioners, whereas only trained psychologists are qualified to deliver effective psychotherapy. While drugs and CBT therapies tend to be almost equally effective, choosing the best intervention depends on the disorder and individual being treated, as well as other factors—such as treatment availability and comorbidity (i.e., having multiple mental or physical disorders at once). Although many new drugs have been introduced in recent decades, there is still much we do not understand about their mechanism in the brain. Further research is needed to refine our understanding of both pharmacological and behavioral treatments before we can make firm claims about their effectiveness.

Integrative and Eclectic Psychotherapy

In discussing therapeutic orientations, it is important to note that some clinicians incorporate techniques from multiple approaches, a practice known as integrative or eclectic psychotherapy. For example, a therapist may employ distress tolerance skills from DBT (to resolve short-term problems), cognitive reappraisal from CBT (to address long-standing issues), and mindfulness-based meditation from MBCT (to reduce overall stress). And, in fact, between 13% and 42% of therapists have identified their own approaches as integrative or eclectic (Norcross & Goldfried, 2005).

Conclusion

Throughout human history we have had to deal with mental illness in one form or another. Over time, several schools of thought have emerged for treating these problems. Although various therapies have been shown to work for specific individuals, cognitive behavioral therapy is currently the treatment most widely supported by empirical research. Still, practices like psychodynamic therapies, person-centered therapy, mindfulness-based treatments, and acceptance and commitment therapy have also shown success. And, with recent advances in research and technology, clinicians are able to enhance these and other therapies to treat more patients more effectively than ever before. However, what is important in the end is that people actually seek out mental health specialists to help them with their problems. One of the biggest deterrents to doing so is that people don't understand what psychotherapy really entails. Through understanding how current practices work, not only can we better educate people about how to get the help they need, but we can continue to advance our treatments to be more effective in the future.

Outside Resources

Article: A personal account of the benefits of mindfulness-based therapy

<https://www.theguardian.com/lifeandstyle/2014/jan/11/julie-myerson-mindfulness-based-cognitive-therapy>

Article: The Effect of Mindfulness-Based Therapy on Anxiety and Depression: A Meta-Analytic Review

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2848393/>

Video: An example of a person-centered therapy session.

<https://www.youtube.com/watch?v=4wTVbzbvBH0k>

Video: Carl Rogers, the founder of the humanistic, person-centered approach to psychology, discusses the position of the therapist in PCT.

<https://www.youtube.com/watch?v=o0neRQzudzw>

Video: CBT (cognitive behavioral therapy) is one of the most common treatments for a range of mental health problems, from anxiety, depression, bipolar, OCD or schizophrenia. This animation explains the basics and how you can decide whether it's best for you or not.

https://www.youtube.com/watch?v=9c_Bv_FBE-c

Web: An overview of the purpose and practice of cognitive behavioral therapy (CBT)

<http://psychcentral.com/lib/in-depth-cognitive-behavioral-therapy/>

Web: The history and development of psychoanalysis

<http://www.freudfile.org/psychoanalysis/history.html>

Discussion Questions

1. Psychoanalytic theory is no longer the dominant therapeutic approach. Yet many consumers continue to seek psychoanalytic or psychodynamic treatments. Do you think psychoanalysis still has a place in mental health treatment? If so, why?
2. What might be some advantages and disadvantages of technological advances in psychological treatment? What will psychotherapy look like 100 years from now?
3. Some people have argued that all therapies are about equally effective, and that they all affect change through common factors such as the involvement of a supportive therapist.

Does this claim sound reasonable to you? Why or why not?

4. When choosing a psychological treatment for a specific patient, what factors besides the treatment's demonstrated efficacy should be taken into account?

Vocabulary

Acceptance and commitment therapy

A therapeutic approach designed to foster nonjudgmental observation of one's own mental processes.

Automatic thoughts

Thoughts that occur spontaneously; often used to describe problematic thoughts that maintain mental disorders.

Cognitive bias modification

Using exercises (e.g., computer games) to change problematic thinking habits.

Cognitive-behavioral therapy (CBT)

A family of approaches with the goal of changing the thoughts and behaviors that influence psychopathology.

Comorbidity

Describes a state of having more than one psychological or physical disorder at a given time.

Dialectical behavior therapy (DBT)

A treatment often used for borderline personality disorder that incorporates both cognitive-behavioral and mindfulness elements.

Dialectical worldview

A perspective in DBT that emphasizes the joint importance of change and acceptance.

Exposure therapy

A form of intervention in which the patient engages with a problematic (usually feared) situation without avoidance or escape.

Free association

In psychodynamic therapy, a process in which the patient reports all thoughts that come to mind without censorship, and these thoughts are interpreted by the therapist.

Integrative or eclectic psychotherapy

Also called integrative psychotherapy, this term refers to approaches combining multiple orientations (e.g., CBT with psychoanalytic elements).

Integrative or eclectic psychotherapy

Also called integrative psychotherapy, this term refers to approaches combining multiple orientations (e.g., CBT with psychoanalytic elements).

Mindfulness

A process that reflects a nonjudgmental, yet attentive, mental state.

Mindfulness-based therapy

A form of psychotherapy grounded in mindfulness theory and practice, often involving meditation, yoga, body scan, and other features of mindfulness exercises.

Person-centered therapy

A therapeutic approach focused on creating a supportive environment for self-discovery.

Psychoanalytic therapy

Sigmund Freud's therapeutic approach focusing on resolving unconscious conflicts.

Psychodynamic therapy

Treatment applying psychoanalytic principles in a briefer, more individualized format.

Reappraisal, or Cognitive restructuring

The process of identifying, evaluating, and changing maladaptive thoughts in psychotherapy.

Schema

A mental representation or set of beliefs about something.

Unconditional positive regard

In person-centered therapy, an attitude of warmth, empathy and acceptance adopted by the therapist in order to foster feelings of inherent worth in the patient.

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Schizophrenia Spectrum Disorders

Deanna M. Barch

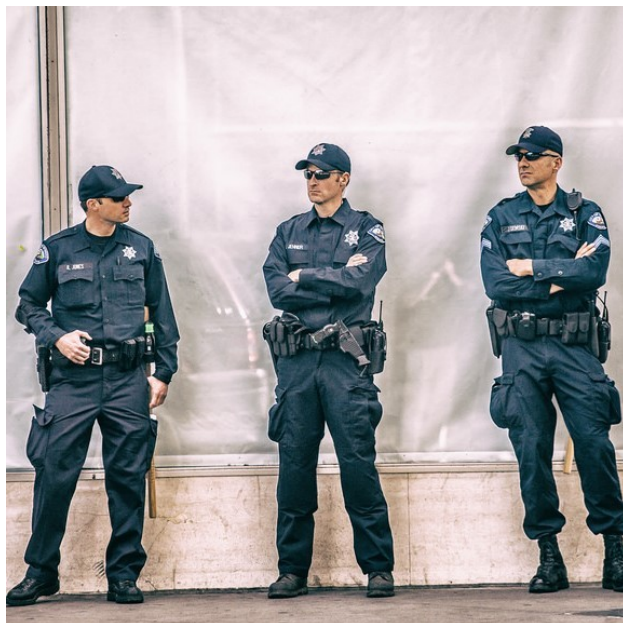
Schizophrenia and the other psychotic disorders are some of the most impairing forms of psychopathology, frequently associated with a profound negative effect on the individual's educational, occupational, and social function. Sadly, these disorders often manifest right at time of the transition from adolescence to adulthood, just as young people should be evolving into independent young adults. The spectrum of psychotic disorders includes schizophrenia, schizoaffective disorder, delusional disorder, schizotypal personality disorder, schizophreniform disorder, brief psychotic disorder, as well as psychosis associated with substance use or medical conditions. In this module, we summarize the primary clinical features of these disorders, describe the known cognitive and neurobiological changes associated with schizophrenia, describe potential risk factors and/or causes for the development of schizophrenia, and describe currently available treatments for schizophrenia.

Learning Objectives

- Describe the signs and symptoms of schizophrenia and related psychotic disorders.
- Describe the most well-replicated cognitive and neurobiological changes associated with schizophrenia.
- Describe the potential risk factors for the development of schizophrenia.
- Describe the controversies associated with “clinical high risk” approaches to identifying individuals at risk for the development of schizophrenia.
- Describe the treatments that work for some of the symptoms of schizophrenia.

The phenomenology of schizophrenia and related psychotic disorders

Most of you have probably had the experience of walking down the street in a city and seeing a person you thought was acting oddly. They may have been dressed in an unusual way, perhaps disheveled or wearing an unusual collection of clothes, makeup, or jewelry that did not seem to fit any particular group or subculture. They may have been talking to themselves or yelling at someone you could not see. If you tried to speak to them, they may have been difficult to follow or understand, or they may have acted paranoid or started telling a bizarre story about the people who were plotting against them. If so, chances are that you have encountered an individual with schizophrenia or another type of psychotic disorder. If you have watched the movie *A Beautiful Mind* or *The Fisher King*, you have also seen a portrayal of someone thought to have schizophrenia. Sadly, a few of the individuals who have committed some of the recently highly publicized mass murders may have had schizophrenia, though most people who commit such crimes do not have schizophrenia. It is also likely that you have met people with schizophrenia without ever knowing it, as they may suffer in silence or stay isolated to protect themselves from the horrors they see, hear, or believe are operating in the outside world. As these examples begin to illustrate, psychotic disorders involve many different types of symptoms, including delusions, hallucinations, disorganized speech and behavior, abnormal motor behavior (including catatonia), and negative symptoms such as anhedonia/amotivation and blunted affect/reduced speech.



Under Surveillance: Abstract groups like the police or the government are commonly the focus of a schizophrenic's persecutory delusions. [Image: Thomas Hawk, <https://goo.gl/qsrqiR>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

Delusions are false beliefs that are often fixed, hard to change even when the person is presented with conflicting information, and are often culturally influenced in their content (e.g., delusions involving Jesus in Judeo-Christian cultures, delusions involving Allah in Muslim cultures). They can be terrifying for the person, who may remain convinced that they are true even when loved ones and friends present them with clear information that they cannot be true. There are many different types or themes to delusions.

The most common delusions are persecutory and involve the belief that individuals or

groups are trying to hurt, harm, or plot against the person in some way. These can be people that the person knows (people at work, the neighbors, family members), or more abstract groups (the FBI, the CIA, aliens, etc.). Other types of delusions include grandiose delusions, where the person believes that they have some special power or ability (e.g., I am the new Buddha, I am a rock star); referential delusions, where the person believes that events or objects in the environment have special meaning for them (e.g., that song on the radio is being played *specifically* for me); or other types of delusions where the person may believe that others are controlling their thoughts and actions, their thoughts are being broadcast aloud, or that others can read their mind (or they can read other people's minds).

When you see a person on the street talking to themselves or shouting at other people, they are experiencing **hallucinations**. These are perceptual experiences that occur even when there is no stimulus in the outside world generating the experiences. They can be auditory, visual, olfactory (smell), gustatory (taste), or somatic (touch). The most common hallucinations in psychosis (at least in adults) are auditory, and can involve one or more voices talking about the person, commenting on the person's behavior, or giving them orders. The content of the hallucinations is frequently negative ("you are a loser," "that drawing is stupid," "you should go kill yourself") and can be the voice of someone the person knows or a complete stranger. Sometimes the voices sound as if they are coming from outside the person's head. Other times the voices seem to be coming from inside the person's head, but are not experienced the same as the person's inner thoughts or inner speech.

Talking to someone with schizophrenia is sometimes difficult, as their speech may be difficult to follow, either because their answers do not clearly flow from your questions, or because one sentence does not logically follow from another. This is referred to as **disorganized speech**, and it can be present even when the person is writing. **Disorganized behavior** can include odd dress, odd makeup (e.g., lipstick outlining a mouth for 1 inch), or unusual rituals (e.g., repetitive hand gestures). Abnormal motor behavior can include catatonia, which refers to a variety of behaviors that seem to reflect a reduction in responsiveness to the external environment. This can include



People who suffer from schizophrenia may see the world differently. This can include hallucinations, delusions, and disorganized thinking. [Image: Noba Project CCBYNC-SA 4.0 <https://tinyurl.com/y3k6qoz4>]

holding unusual postures for long periods of time, failing to respond to verbal or motor prompts from another person, or excessive and seemingly purposeless motor activity.

Some of the most debilitating symptoms of schizophrenia are difficult for others to see. These include what people refer to as “negative symptoms” or the absence of certain things we typically expect most people to have. For example, anhedonia or amotivation reflect a lack of apparent interest in or drive to engage in social or recreational activities. These symptoms can manifest as a great amount of time spent in physical immobility. Importantly, anhedonia and amotivation do not seem to reflect a lack of enjoyment in pleasurable activities or events (Cohen & Minor, 2010; Kring & Moran, 2008; Llerena, Strauss, & Cohen, 2012) but rather a reduced drive or ability to take the steps necessary to obtain the potentially positive outcomes (Barch & Dowd, 2010). **Flat affect** and reduced speech (**alogia**) reflect a lack of showing emotions through facial expressions, gestures, and speech intonation, as well as a reduced amount of speech and increased pause frequency and duration.

In many ways, the types of symptoms associated with psychosis are the most difficult for us to understand, as they may seem far outside the range of our normal experiences. Unlike depression or anxiety, many of us may not have had experiences that we think of as on the same continuum as psychosis. However, just like many of the other forms of **psychopathology** described in this book, the types of psychotic symptoms that characterize disorders like schizophrenia are on a continuum with “normal” mental experiences. For example, work by Jim van Os in the Netherlands has shown that a surprisingly large percentage of the general population (10%+) experience psychotic-like symptoms, though many fewer have multiple experiences and most will not continue to experience these symptoms in the long run (Verdoux & van Os, 2002). Similarly, work in a general population of adolescents and young adults in Kenya has also shown that a relatively high percentage of individuals experience one or more psychotic-like experiences (~19%) at some point in their lives (Mamah et al., 2012; Ndeti et al., 2012), though again most will not go on to develop a full-blown psychotic disorder.

Schizophrenia is the primary disorder that comes to mind when we discuss “psychotic” disorders (see Table 1 for **diagnostic criteria**), though there are a number of other disorders that share one or more features with schizophrenia. In the remainder of this module, we will use the terms “psychosis” and “schizophrenia” somewhat interchangeably, given that most of the research has focused on schizophrenia. In addition to schizophrenia (see Table 1), other psychotic disorders include schizophreniform disorder (a briefer version of schizophrenia), schizoaffective disorder (a mixture of psychosis and depression/mania symptoms), delusional disorder (the experience of only delusions), and brief psychotic disorder (psychotic symptoms that last only a few days or weeks).

Schizophrenia (Lifetime prevalence about 0.3% to 0.7% [APA, 2013])
• Two or more of the following for at least 1 month: hallucinations, delusions, disorganized speech, grossly disorganized or catatonic behavior, negative symptoms. • Impairment in one or more areas of function (social, occupational, educational self-care) for a significant period of time since the onset of the illness. • Continuous signs of the illness for at least 6 months (this can include prodromal or residual symptoms, which are attenuated forms of the symptoms described above).
Schizophreniform Disorder (Lifetime prevalence similar to Schizophrenia [APA, 2013])
• The same symptoms of schizophrenia described above that are present for at least 1 month but less than 6 months.
Schizoaffective Disorder (Lifetime prevalence about 0.3% [APA, 2013])
• A period of illness where the person has both the psychotic symptoms necessary to meet criteria for schizophrenia and either a major depression or manic episode. • The person experiences either delusions or hallucinations for at least 2 weeks when they are not having a depressive or manic episode. • The symptoms that meet criteria for depressive or manic episodes are present for over half of the illness duration.
Delusional Disorder (Lifetime prevalence about 0.2% [APA, 2013])
• The presence of at least one delusion for at least a month. • The person has never met criteria for schizophrenia. • The person's function is not impaired outside the specific impact of the delusion. • The duration of any depressive or manic episodes have been brief relative to the duration of the delusion(s).
Brief Psychotic Disorder (Lifetime prevalence unclear [APA, 2013])
• One or more of the following symptoms present for at least 1 day but less than 1 month: delusions, hallucinations, disorganized speech, grossly disorganized or catatonic behavior.
Attenuated Psychotic Disorder (In Section III of the [APA, 2013]-V, Lifetime presence unclear [APA, 2013])
• One or more of the following symptoms in an "attenuated" form: delusions, hallucinations, or disorganized speech. • The symptoms must have occurred at least once a week for the past month and must have started or gotten worse in the past year. • The symptoms must be severe enough to distress or disable the individual or to suggest to others that the person needs clinical help. • The person has never met the diagnostic criteria for a psychotic disorder, and the symptoms are not better attributed to another disorder, to substance use, or to a medical condition.

Table 1: Types of Psychotic Disorders (Simplified from the Diagnostic and Statistical Manual - 5th Edition (DSM-5) (APA, 2013)

The Cognitive Neuroscience of Schizophrenia

As described above, when we think of the core symptoms of psychotic disorders such as schizophrenia, we think of people who hear voices, see visions, and have false beliefs about reality (i.e., delusions). However, problems in cognitive function are also a critical aspect of psychotic disorders and of schizophrenia in particular. This emphasis on cognition in

schizophrenia is in part due to the growing body of research suggesting that cognitive problems in schizophrenia are a major source of disability and loss of **functional capacity** (Green, 2006; Nuechterlein et al., 2011). The cognitive deficits that are present in schizophrenia are widespread and can include problems with **episodic memory** (the ability to learn and retrieve new information or episodes in one's life), **working memory** (the ability to maintain information over a short period of time, such as 30 seconds), and other tasks that require one to "control" or regulate one's behavior (Barch & Ceaser, 2012; Bora, Yucel, & Pantelis, 2009a; Fioravanti, Carlone, Vitale, Cinti, & Clare, 2005; Forbes, Carrick, McIntosh, & Lawrie, 2009; Mesholam-Gately, Giuliano, Goff, Faraone, & Seidman, 2009). Individuals with schizophrenia also have difficulty with what is referred to as "**processing speed**" and are frequently slower than healthy individuals on almost all tasks. Importantly, these cognitive deficits are present prior to the onset of the illness (Fusar-Poli et al., 2007) and are also present, albeit in a milder form, in the first-degree relatives of people with schizophrenia (Snitz, Macdonald, & Carter, 2006). This suggests that cognitive impairments in schizophrenia reflect part of the risk for the development of psychosis, rather than being an outcome of developing psychosis. Further, people with schizophrenia who have more severe cognitive problems also tend to have more severe negative symptoms and more disorganized speech and behavior (Barch, Carter, & Cohen, 2003; Barch et al., 1999; Dominguez Mde, Viechtbauer, Simons, van Os, & Krabbendam, 2009; Ventura, Helleman, Thames, Koellner, & Nuechterlein, 2009; Ventura, Thames, Wood, Guzik, & Helleman, 2010). In addition, people with more cognitive problems have worse function in everyday life (Bowie et al., 2008; Bowie, Reichenberg, Patterson, Heaton, & Harvey, 2006; Fett et al., 2011).



Some with schizophrenia suffer from difficulty with social cognition. They may not be able to detect the meaning of facial expressions or other subtle cues that most other people rely on to navigate the social world. [Image: Ralph Buckley, <https://goo.gl/KuBzsD>, CC BY-SA 2.0, <https://goo.gl/i4GXf5>]

Some people with schizophrenia also show deficits in what is referred to as social cognition, though it is not clear whether such problems are separate from the cognitive problems described above or the result of them (Hoe, Nakagami, Green, & Brekke, 2012; Kerr & Neale, 1993; van Hooren et al., 2008). This includes problems with the recognition of emotional expressions on the faces of other individuals (Kohler, Walker, Martin, Healey, & Moberg, 2010) and problems inferring the intentions of other people (theory of mind) (Bora, Yucel, & Pantelis, 2009b).

Individuals with schizophrenia who have more problems with social cognition also tend to have more negative and disorganized symptoms (Ventura, Wood, & Helleman, 2011), as well as worse community function (Fett et al., 2011).

The advent of neuroimaging techniques such as structural and functional magnetic resonance imaging and positron emission tomography opened up the ability to try to understand the brain mechanisms of the symptoms of schizophrenia as well as the cognitive impairments found in psychosis. For example, a number of studies have suggested that delusions in psychosis may be associated with problems in “salience” detection mechanisms supported by the ventral striatum (Jensen & Kapur, 2009; Jensen et al., 2008; Kapur, 2003; Kapur, Mizrahi, & Li, 2005; Murray et al., 2008) and the anterior prefrontal cortex (Corlett et al., 2006; Corlett, Honey, & Fletcher, 2007; Corlett, Murray, et al., 2007a, 2007b). These are regions of the brain that normally increase their activity when something important (aka “salient”) happens in the environment. If these brain regions misfire, it may lead individuals with psychosis to mistakenly attribute importance to irrelevant or unconnected events. Further, there is good evidence that problems in working memory and cognitive control in schizophrenia are related to problems in the function of a region of the brain called the dorsolateral prefrontal cortex (DLPFC) (Minzenberg, Laird, Thelen, Carter, & Glahn, 2009; Ragland et al., 2009). These problems include changes in how the DLPFC works when people are doing working-memory or cognitive-control tasks, and problems with how this brain region is connected to other brain regions important for working memory and cognitive control, including the posterior parietal cortex (e.g., Karlsgodt et al., 2008; J. J. Kim et al., 2003; Schlosser et al., 2003), the anterior cingulate (Repovs & Barch, 2012), and temporal cortex (e.g., Fletcher et al., 1995; Meyer-Lindenberg et al., 2001). In terms of understanding episodic memory problems in schizophrenia, many researchers have focused on medial temporal lobe deficits, with a specific focus on the hippocampus (e.g., Heckers & Konradi, 2010). This is because there is much data from humans and animals showing that the hippocampus is important for the creation of new memories (Squire, 1992). However, it has become increasingly clear that problems with the DLPFC also make important contributions to episodic memory deficits in schizophrenia (Ragland et al., 2009), probably because this part of the brain is important for controlling our use of memory.

In addition to problems with regions such as the DLPFC and medial temporal lobes in schizophrenia described above, magnitude resonance neuroimaging studies have also identified changes in cellular architecture, white matter connectivity, and gray matter volume in a variety of regions that include the prefrontal and temporal cortices (Bora et al., 2011). People with schizophrenia also show reduced overall brain volume, and reductions in brain volume as people get older may be larger in those with schizophrenia than in healthy people (Olabi et al., 2011). Taking antipsychotic medications or taking drugs such as marijuana,

alcohol, and tobacco may cause some of these structural changes. However, these structural changes are not completely explained by medications or substance use alone. Further, both functional and structural brain changes are seen, again to a milder degree, in the first-degree relatives of people with schizophrenia (Boos, Aleman, Cahn, Pol, & Kahn, 2007; Brans et al., 2008; Fusar-Poli et al., 2007; MacDonald, Thermenos, Barch, & Seidman, 2009). This again suggests that that neural changes associated with schizophrenia are related to a genetic risk for this illness.

Risk Factors for Developing Schizophrenia

It is clear that there are important genetic contributions to the likelihood that someone will develop schizophrenia, with consistent evidence from family, twin, and adoption studies. (Sullivan, Kendler, & Neale, 2003). However, there is no “schizophrenia gene” and it is likely that the genetic risk for schizophrenia reflects the summation of many different genes that each contribute something to the likelihood of developing psychosis (Gottesman & Shields, 1967; Owen, Craddock, & O'Donovan, 2010). Further, schizophrenia is a very heterogeneous disorder, which means that two different people with “schizophrenia” may each have very different symptoms (e.g., one has hallucinations and delusions, the other has disorganized

speech and negative symptoms). This makes it even more challenging to identify specific genes associated with risk for psychosis. Importantly, many studies also now suggest that at least some of the genes potentially associated with schizophrenia are also associated with other mental health conditions, including bipolar disorder, depression, and autism (Gejman, Sanders, & Kendler, 2011; Y. Kim, Zerwas, Trace, & Sullivan, 2011; Owen et al., 2010; Rutter, Kim-Cohen, & Maughan, 2006).



There are a number of genetic and environmental risk factors associated with higher likelihood of developing schizophrenia including older fathers, complications during pregnancy/delivery, family history of schizophrenia, and growing up in an urban environment. [Image: CC0 Public Domain]

There are also a number of environmental factors that are associated with an increased risk of developing schizophrenia. For example, problems during pregnancy such as increased stress, infection, malnutrition, and/or diabetes have been associated with increased risk of schizophrenia.

In addition, complications that occur at the time of birth and which cause hypoxia (lack of oxygen) are also associated with an increased risk for developing schizophrenia (M. Cannon, Jones, & Murray, 2002; Miller et al., 2011). Children born to older fathers are also at a somewhat increased risk of developing schizophrenia. Further, using cannabis increases risk for developing psychosis, especially if you have other risk factors (Casadio, Fernandes, Murray, & Di Forti, 2011; Luzi, Morrison, Powell, di Forti, & Murray, 2008). The likelihood of developing schizophrenia is also higher for kids who grow up in urban settings (March et al., 2008) and for some minority ethnic groups (Bourque, van der Ven, & Malla, 2011). Both of these factors may reflect higher social and environmental stress in these settings. Unfortunately, none of these risk factors is specific enough to be particularly useful in a clinical setting, and most people with these “risk” factors do not develop schizophrenia. However, together they are beginning to give us clues as the neurodevelopmental factors that may lead someone to be at an increased risk for developing this disease.

An important research area on risk for psychosis has been work with individuals who may be at “clinical high risk.” These are individuals who are showing attenuated (milder) symptoms of psychosis that have developed recently and who are experiencing some distress or disability associated with these symptoms. When people with these types of symptoms are followed over time, about 35% of them develop a psychotic disorder (T. D. Cannon et al., 2008), most frequently schizophrenia (Fusar-Poli, McGuire, & Borgwardt, 2012). In order to identify these individuals, a new category of diagnosis, called “Attenuated Psychotic Syndrome,” was added to Section III (the section for disorders in need of further study) of the DSM-5 (see Table 1 for symptoms) (APA, 2013). However, adding this diagnostic category to the DSM-5 created a good deal of controversy (Batstra & Frances, 2012; Fusar-Poli & Yung, 2012). Many scientists and clinicians have been worried that including “risk” states in the DSM-5 would create mental disorders where none exist, that these individuals are often already seeking treatment for other problems, and that it is not clear that we have good treatments to stop these individuals from developing to psychosis. However, the counterarguments have been that there is evidence that individuals with high-risk symptoms develop psychosis at a much higher rate than individuals with other types of psychiatric symptoms, and that the inclusion of Attenuated Psychotic Syndrome in Section III will spur important research that might have clinical benefits. Further, there is some evidence that non-invasive treatments such as omega-3 fatty acids and intensive family intervention may help reduce the development of full-blown psychosis (Preti & Cella, 2010) in people who have high-risk symptoms.

Treatment of Schizophrenia

The currently available treatments for schizophrenia leave much to be desired, and the search

for more effective treatments for both the psychotic symptoms of schizophrenia (e.g., hallucinations and delusions) as well as cognitive deficits and negative symptoms is a highly active area of research. The first line of treatment for schizophrenia and other psychotic disorders is the use of antipsychotic medications. There are two primary types of antipsychotic medications, referred to as “typical” and “atypical.” The fact that “typical” antipsychotics helped some symptoms of schizophrenia was discovered serendipitously more than 60 years ago (Carpenter & Davis, 2012; Lopez-Munoz et al., 2005). These are drugs that all share a common feature of being a strong block of the D2 type dopamine receptor. Although these drugs can help reduce hallucinations, delusions, and disorganized speech, they do little to improve cognitive deficits or negative symptoms and can be associated with distressing motor side effects. The newer generation of antipsychotics is referred to as “atypical” antipsychotics. These drugs have more mixed mechanisms of action in terms of the receptor types that they influence, though most of them also influence D2 receptors. These newer antipsychotics are not necessarily more helpful for schizophrenia but have fewer motor side effects. However, many of the atypical antipsychotics are associated with side effects referred to as the “metabolic syndrome,” which includes weight gain and increased risk for cardiovascular illness, Type-2 diabetes, and mortality (Lieberman et al., 2005).

The evidence that cognitive deficits also contribute to functional impairment in schizophrenia has led to an increased search for treatments that might enhance cognitive function in schizophrenia. Unfortunately, as of yet, there are no pharmacological treatments that work consistently to improve cognition in schizophrenia, though many new types of drugs are currently under exploration. However, there is a type of psychological intervention, referred to as cognitive remediation, which has shown some evidence of helping cognition and function in schizophrenia. In particular, a version of this treatment called Cognitive Enhancement Therapy (CET) has been shown to improve cognition, functional outcome, social cognition, and to protect against gray matter loss (Eack et al., 2009; Eack, Greenwald, Hogarty, & Keshavan, 2010; Eack et al., 2010; Eack, Pogue-Geile, Greenwald, Hogarty, & Keshavan, 2010; Hogarty, Greenwald, & Eack, 2006) in young individuals with schizophrenia. The development of new treatments such as Cognitive Enhancement Therapy provides some hope that we will be able to develop new and better approaches to improving the lives of individuals with this serious mental health condition and potentially even prevent it some day.

Outside Resources

Book: *Ben Behind His Voices: One family's journal from the chaos of schizophrenia to hope* (2011). Randy Kaye. Rowman and Littlefield.

Book: *Conquering Schizophrenia: A father, his son, and a medical breakthrough* (1997). Peter Wyden. Knopf.

Book: *Henry's Demons: Living with schizophrenia, a father and son's story* (2011). Henry and Patrick Cockburn. Scribner Macmillan.

Book: *My Mother's Keeper: A daughter's memoir of growing up in the shadow of schizophrenia* (1997). Tara Elgin Holley. William Morrow Co.

Book: *Recovered, Not Cured: A journey through schizophrenia* (2005). Richard McLean. Allen and Unwin.

Book: *The Center Cannot Hold: My journey through madness* (2008). Elyn R. Saks. Hyperion.

Book: *The Quiet Room: A journal out of the torment of madness* (1996). Lori Schiller. Grand Central Publishing.

Book: *Welcome Silence: My triumph over schizophrenia* (2003). Carol North. CSS Publishing.

Web: National Alliance for the Mentally Ill. This is an excellent site for learning more about advocacy for individuals with major mental illnesses such as schizophrenia.

<http://www.nami.org/>

Web: National Institute of Mental Health. This website has information on NIMH-funded schizophrenia research.

<http://www.nimh.nih.gov/health/topics/schizophrenia/index.shtml>

Web: Schizophrenia Research Forum. This is an excellent website that contains a broad array of information about current research on schizophrenia.

<http://www.schizophreniaforum.org/>

Discussion Questions

1. Describe the major differences between the major psychotic disorders.
2. How would one be able to tell when an individual is “delusional” versus having non-delusional beliefs that differ from the societal normal? How should cultural and sub-cultural variation be taken into account when assessing psychotic symptoms?
3. Why are cognitive impairments important to understanding schizophrenia?
4. Why has the inclusion of a new diagnosis (Attenuated Psychotic Syndrome) in Section III of the DSM-5 created controversy?
5. What are some of the factors associated with increased risk for developing schizophrenia? If we know whether or not someone has these risk factors, how well can we tell whether they will develop schizophrenia?
6. What brain changes are most consistent in schizophrenia?
7. Do antipsychotic medications work well for all symptoms of schizophrenia? If not, which symptoms respond better to antipsychotic medications?
8. Are there any treatments besides antipsychotic medications that help any of the symptoms of schizophrenia? If so, what are they?

Vocabulary

Alogia

A reduction in the amount of speech and/or increased pausing before the initiation of speech.

Anhedonia/amotivation

A reduction in the drive or ability to take the steps or engage in actions necessary to obtain the potentially positive outcome.

Catatonia

Behaviors that seem to reflect a reduction in responsiveness to the external environment. This can include holding unusual postures for long periods of time, failing to respond to verbal or motor prompts from another person, or excessive and seemingly purposeless motor activity.

Delusions

False beliefs that are often fixed, hard to change even in the presence of conflicting information, and often culturally influenced in their content.

Diagnostic criteria

The specific criteria used to determine whether an individual has a specific type of psychiatric disorder. Commonly used diagnostic criteria are included in the Diagnostic and Statistical Manual of Mental Disorder, 5th Edition (DSM-5) and the International Classification of Disorders, Version 9 (ICD-9).

Disorganized behavior

Behavior or dress that is outside the norm for almost all subcultures. This would include odd dress, odd makeup (e.g., lipstick outlining a mouth for 1 inch), or unusual rituals (e.g., repetitive hand gestures).

Disorganized speech

Speech that is difficult to follow, either because answers do not clearly follow questions or because one sentence does not logically follow from another.

Dopamine

A neurotransmitter in the brain that is thought to play an important role in regulating the function of other neurotransmitters.

Episodic memory

The ability to learn and retrieve new information or episodes in one's life.

Flat affect

A reduction in the display of emotions through facial expressions, gestures, and speech intonation.

Functional capacity

The ability to engage in self-care (cook, clean, bathe), work, attend school, and/or engage in social relationships.

Hallucinations

Perceptual experiences that occur even when there is no stimulus in the outside world generating the experiences. They can be auditory, visual, olfactory (smell), gustatory (taste), or somatic (touch).

Magnetic resonance imaging

A set of techniques that uses strong magnets to measure either the structure of the brain (e.g., gray matter and white matter) or how the brain functions when a person performs cognitive tasks (e.g., working memory or episodic memory) or other types of tasks.

Neurodevelopmental

Processes that influence how the brain develops either in utero or as the child is growing up.

Positron emission tomography

A technique that uses radio-labelled ligands to measure the distribution of different neurotransmitter receptors in the brain or to measure how much of a certain type of neurotransmitter is released when a person is given a specific type of drug or does a particularly cognitive task.

Processing speed

The speed with which an individual can perceive auditory or visual information and respond to it.

Psychopathology

Illnesses or disorders that involve psychological or psychiatric symptoms.

Working memory

The ability to maintain information over a short period of time, such as 30 seconds or less.

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Anxiety and Related Disorders

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Anxiety is a natural part of life and, at normal levels, helps us to function at our best. However, for people with anxiety disorders, anxiety is overwhelming and hard to control. Anxiety disorders develop out of a blend of biological (genetic) and psychological factors that, when combined with stress, may lead to the development of ailments. Primary anxiety-related diagnoses include generalized anxiety disorder, panic disorder, specific phobia, social anxiety disorder (social phobia), post traumatic stress disorder, and obsessive-compulsive disorder. In this module, we summarize the main clinical features of each of these disorders and discuss their similarities and differences with everyday experiences of anxiety.

Learning Objectives

- Understand the relationship between anxiety and anxiety disorders.
- Identify key vulnerabilities for developing anxiety and related disorders.
- Identify main diagnostic features of specific anxiety-related disorders.
- Differentiate between disordered and non-disordered functioning.

Introduction

What is anxiety? Most of us feel some anxiety almost every day of our lives. Maybe you have an important test coming up for school. Or maybe there's that big game next Saturday, or that first date with someone new you are hoping to impress. Anxiety can be defined as a negative mood state that is accompanied by bodily symptoms such as increased heart rate,

muscle tension, a sense of unease, and apprehension about the future (APA, 2013; Barlow, 2002).

Anxiety is what motivates us to plan for the future, and in this sense, anxiety is actually a good thing. It's that nagging feeling that motivates us to study for that test, practice harder for that game, or be at our very best on that date. But some people experience anxiety so intensely that it is no longer helpful or useful. They may become so overwhelmed and distracted by anxiety that they actually fail their test, fumble the ball, or spend the whole date fidgeting and avoiding eye contact. If anxiety begins to interfere in the person's life in a significant way, it is considered a disorder.



Anxiety and closely related disorders emerge from “triple vulnerabilities,” a combination of biological, psychological, and specific factors that increase our risk for developing a disorder (Barlow, 2002; Suárez, Bennett, Goldstein, & Barlow,

2009). **Biological vulnerabilities** refer to specific genetic and neurobiological factors that might predispose someone to develop anxiety disorders. No single gene directly causes anxiety or panic, but our genes may make us more susceptible to anxiety and influence how our brains react to stress (Drabant et al., 2012; Gelernter & Stein, 2009; Smoller, Block, & Young, 2009). **Psychological vulnerabilities** refer to the influences that our early experiences have on how we view the world. If we were confronted with unpredictable stressors or traumatic experiences at younger ages, we may come to view the world as unpredictable and uncontrollable, even dangerous (Chorpita & Barlow, 1998; Gunnar & Fisher, 2006). **Specific vulnerabilities** refer to how our experiences lead us to focus and channel our anxiety (Suárez et al., 2009). If we learned that physical illness is dangerous, maybe through witnessing our family's reaction whenever anyone got sick, we may focus our anxiety on physical sensations. If we learned that disapproval from others has negative, even dangerous consequences, such as being yelled at or severely punished for even the slightest offense, we might focus our anxiety on social evaluation. If we learn that the “other shoe might drop” at any moment, we may focus our anxiety on worries about the future. None of these vulnerabilities directly

While everyone may experience some level of anxiety at one time or another, those with anxiety disorders experience it consistently and so intensely that it has a significantly negative impact on their quality of life. [Image: Bada Bing, <https://goo.gl/aawyLi>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

causes anxiety disorders on its own—instead, when all of these vulnerabilities are present, and we experience some triggering life stress, an anxiety disorder may be the result (Barlow, 2002; Suárez et al., 2009). In the next sections, we will briefly explore each of the major anxiety based disorders, found in the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) (APA, 2013).

Generalized Anxiety Disorder

Most of us worry some of the time, and this worry can actually be useful in helping us to plan for the future or make sure we remember to do something important. Most of us can set aside our worries when we need to focus on other things or stop worrying altogether whenever a problem has passed. However, for someone with **generalized anxiety disorder (GAD)**, these worries become difficult, or even impossible, to turn off. They may find themselves worrying excessively about a number of different things, both minor and catastrophic. Their worries also come with a host of other symptoms such as muscle tension, fatigue, agitation or restlessness, irritability, difficulties with sleep (either falling asleep, staying asleep, or both), or difficulty concentrating. The *DSM-5* criteria specify that at least six months of excessive anxiety and worry of this type must be ongoing, happening more days than not for a good proportion of the day, to receive a diagnosis of GAD. About 5.7% of the population has met criteria for GAD at some point during their lifetime (Kessler, Berglund, et al., 2005), making it one of the most common anxiety disorders (see Table 1).

Disorder	1-Year Prevalence Rates ¹	Lifetime Prevalence Rates ²	Prevalence by Gender	Median Age of Onset
Generalized Anxiety Disorder	3.1%	5.7%	67% female	31 yrs.
OCD	1%	1.6%	55% female	19 yrs.
Panic Disorder	2.7%	4.7%	67% female	24 yrs.
PTSD	3.5%	6.8%	52% female ³	23 yrs.
Social Anxiety	6.8%	12.1%	50% female	13 yrs.
Specific Phobia	8.7%	12.5%	60% - 90% female ⁴	7-9 yrs.

Table 1: Prevalence rates for major anxiety disorders. [1] Kessler et al. (2005), [2]Kessler, Chiu, Demler, Merikangas, & Walters (2005), [3]Kessler, Sonnega, Bromet, Hughes, & Nelson (1995), [4]Craske et al. (1996).

What makes a person with GAD worry more than the average person? Research shows that individuals with GAD are more sensitive and vigilant toward possible threats than people who are not anxious (Aikins & Craske, 2001; Barlow, 2002; Bradley, Mogg, White, Groom, & de Bono, 1999). This may be related to early stressful experiences, which can lead to a view of the world as an unpredictable, uncontrollable, and even dangerous place. Some have suggested that people with GAD worry as a way to gain some control over these otherwise uncontrollable or unpredictable experiences and against uncertain outcomes (Dugas, Gagnon, Ladouceur, & Freeston, 1998). By repeatedly going through all of the possible “What if?” scenarios in their mind, the person might feel like they are less vulnerable to an unexpected outcome, giving them the sense that they have *some* control over the situation (Wells, 2002). Others have suggested people with GAD worry as a way to avoid feeling distressed (Borkovec, Alcaine, & Behar, 2004). For example, Borkovec and Hu (1990) found that those who worried when confronted with a stressful situation had less physiological arousal than those who didn’t worry, maybe because the worry “distracted” them in some way.

The problem is, all of this “what if?”-ing doesn’t get the person any closer to a solution or an answer and, in fact, might take them away from important things they should be paying attention to in the moment, such as finishing an important project. Many of the catastrophic outcomes people with GAD worry about are very unlikely to happen, so when the catastrophic event doesn’t materialize, the act of worrying gets **reinforced** (Borkovec, Hazlett-Stevens, & Diaz, 1999). For example, if a mother spends all night worrying about whether her teenage daughter will get home safe from a night out and the daughter returns home without incident, the mother could easily attribute her daughter’s safe return to her successful “vigil.” What the mother hasn’t learned is that her daughter would have returned home just as safe if she had been focusing on the movie she was watching with her husband, rather than being preoccupied with worries. In this way, the cycle of worry is perpetuated, and, subsequently, people with GAD often miss out on many otherwise enjoyable events in their lives.

Panic Disorder and Agoraphobia

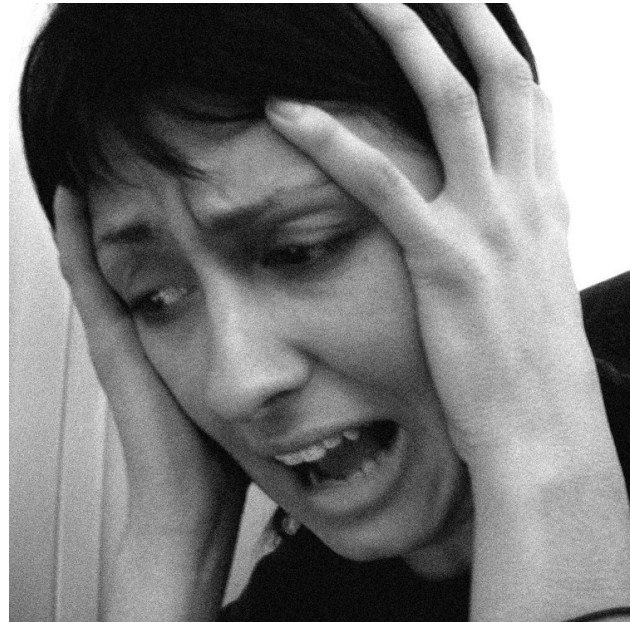
Have you ever gotten into a near-accident or been taken by surprise in some way? You may have felt a flood of physical sensations, such as a racing heart, shortness of breath, or tingling sensations. This alarm reaction is called the **“fight or flight”** response (Cannon, 1929) and is your body’s natural reaction to fear, preparing you to either fight or escape in response to threat or danger. It’s likely you weren’t too concerned with these sensations, because you knew what was causing them. But imagine if this alarm reaction came “out of the blue,” for no apparent reason, or in a situation in which you didn’t expect to be anxious or fearful. This is called an “unexpected” panic attack or a false alarm. Because there is no apparent reason

or cue for the alarm reaction, you might react to the sensations with intense fear, maybe thinking you are having a heart attack, or going crazy, or even dying. You might begin to associate the physical sensations you felt during this attack with this fear and may start to go out of your way to avoid having those sensations again.

Unexpected panic attacks such as these are at the heart of **panic disorder (PD)**. However, to receive a diagnosis of PD, the person must not only have unexpected panic attacks but also must experience continued intense anxiety and avoidance related to the attack for at least one month, causing significant distress or interference in their lives. People with panic disorder tend to interpret even normal physical sensations in a catastrophic way, which triggers more anxiety and, ironically, more physical sensations, creating a vicious cycle of panic (Clark, 1986, 1996). The person may begin to avoid a number of situations or activities that produce the same physiological arousal that was present during the beginnings of a panic attack. For example, someone who experienced a racing heart during a panic attack might avoid exercise or caffeine. Someone who

experienced choking sensations might avoid wearing high-necked sweaters or necklaces. Avoidance of these **internal bodily or somatic cues** for panic has been termed **interoceptive avoidance** (Barlow & Craske, 2007; Brown, White, & Barlow, 2005; Craske & Barlow, 2008; Shear et al., 1997).

The individual may also have experienced an overwhelming urge to escape during the unexpected panic attack. This can lead to a sense that certain places or situations—particularly situations where escape might not be possible—are not “safe.” These situations become **external cues** for panic. If the person begins to avoid several places or situations, or still endures these situations but does so with a significant amount of apprehension and anxiety, then the person also has **agoraphobia** (Barlow, 2002; Craske & Barlow, 1988; Craske & Barlow, 2008). Agoraphobia can cause significant disruption to a person’s life, causing them to go out of their way to avoid situations, such as adding hours to a commute to avoid taking the train



Panic disorder is a debilitating condition that leaves sufferers with acute anxiety that persists long after a specific panic attack has subsided. When this anxiety leads to deliberate avoidance of particular places and situations a person may be given a diagnosis of agoraphobia. [Image: Nate Steiner, <https://goo.gl/dUYWdf>, Public Domain]

or only ordering take-out to avoid having to enter a grocery store. In one tragic case seen by our clinic, a woman suffering from agoraphobia had not left her apartment for 20 years and had spent the past 10 years confined to one small area of her apartment, away from the view of the outside. In some cases, agoraphobia develops in the absence of panic attacks and therefor is a separate disorder in DSM-5. But agoraphobia often accompanies panic disorder.

About 4.7% of the population has met criteria for PD or agoraphobia over their lifetime (Kessler, Chiu, Demler, Merikangas, & Walters, 2005; Kessler et al., 2006) (see Table 1). In all of these cases of panic disorder, what was once an adaptive natural alarm reaction now becomes a learned, and much feared, false alarm.

Specific Phobia

The majority of us might have certain things we fear, such as bees, or needles, or heights (Myers et al., 1984). But what if this fear is so consuming that you can't go out on a summer's day, or get vaccines needed to go on a special trip, or visit your doctor in her new office on the 26th floor? To meet criteria for a diagnosis of specific phobia, there must be an irrational fear of a specific object or situation that substantially interferes with the person's ability to function. For example, a patient at our clinic turned down a prestigious and coveted artist residency because it required spending time near a wooded area, bound to have insects. Another patient purposely left her house two hours early each morning so she could walk past her neighbor's fenced yard before they let their dog out in the morning.



Elevators can be a trigger for sufferers of claustrophobia or agoraphobia. [Image: srgpicker, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

The list of possible phobias is staggering, but four major subtypes of specific phobia are recognized: blood-injury-injection (BII) type, situational type (such as planes, elevators, or enclosed places), natural environment type for events one may encounter in nature (for example, heights, storms, and water), and animal type.

A fifth category "other" includes phobias that do not fit any of the four major subtypes (for example, fears of choking, vomiting, or contracting an illness). Most phobic reactions cause a surge of activity in the sympathetic nervous system and

increased heart rate and blood pressure, maybe even a panic attack. However, people with BII type phobias usually experience a marked *drop* in heart rate and blood pressure and may even faint. In this way, those with BII phobias almost always differ in their physiological reaction from people with other types of phobia (Barlow & Liebowitz, 1995; Craske, Antony, & Barlow, 2006; Hofmann, Alpers, & Pauli, 2009; Ost, 1992). BII phobia also runs in families more strongly than any phobic disorder we know (Antony & Barlow, 2002; Page & Martin, 1998). Specific phobia is one of the most common psychological disorders in the United States, with 12.5% of the population reporting a lifetime history of fears significant enough to be considered a “phobia” (Arrindell et al., 2003; Kessler, Berglund, et al., 2005) (see Table 1). Most people who suffer from specific phobia tend to have multiple phobias of several types (Hofmann, Lehman, & Barlow, 1997).

Social Anxiety Disorder (Social Phobia)

Many people consider themselves shy, and most people find social evaluation uncomfortable at best, or giving a speech somewhat mortifying. Yet, only a small proportion of the population fear these types of situations significantly enough to merit a diagnosis of **social anxiety disorder (SAD)** (APA, 2013). SAD is more than exaggerated shyness (Bogels et al., 2010; Schneier et al., 1996). To receive a diagnosis of SAD, the fear and anxiety associated with social situations must be so strong that the person avoids them entirely, or if avoidance is not possible, the person endures them with a great deal of distress. Further, the fear and avoidance of social situations must get in the way of the person’s daily life, or seriously limit their academic or occupational functioning. For example, a patient at our clinic compromised her perfect 4.0 grade point average because she could not complete a required oral presentation in one of her classes, causing her to fail the course. Fears of negative evaluation might make someone repeatedly turn down invitations to social events or avoid having conversations with people, leading to greater and greater isolation.

The specific social situations that trigger anxiety and fear range from one-on-one interactions, such as starting or maintaining a conversation; to performance-based situations, such as giving a speech or performing on stage; to assertiveness, such as asking someone to change disruptive or undesirable behaviors. Fear of social evaluation might even extend to such things as using public restrooms, eating in a restaurant, filling out forms in a public place, or even reading on a train. Any type of situation that could potentially draw attention to the person can become a feared social situation. For example, one patient of ours went out of her way to avoid any situation in which she might have to use a public restroom for fear that someone would hear her in the bathroom stall and think she was disgusting. If the fear is limited to performance-based situations, such as public speaking, a diagnosis of **SAD performance only**

is assigned.

What causes someone to fear social situations to such a large extent? The person may have learned growing up that social evaluation in particular can be dangerous, creating a specific psychological vulnerability to develop social anxiety (Bruch & Heimberg, 1994; Lieb et al., 2000; Rapee & Melville, 1997). For example, the person's caregivers may have harshly criticized and punished them for even the smallest mistake, maybe even punishing them physically.

Or, someone might have experienced a social trauma that had lasting effects, such as being bullied or humiliated. Interestingly, one group of researchers found that 92% of adults in their study sample with social phobia experienced severe teasing and bullying in childhood, compared with only 35% to 50% among people with other anxiety disorders (McCabe, Antony, Summerfeldt, Liss, & Swinson, 2003). Someone else might react so strongly to the anxiety provoked by a social situation that they have an unexpected panic attack. This panic attack then becomes associated (**conditioned response**) with the social

situation, causing the person to fear they will panic the next time they are in that situation. This is not considered PD, however, because the person's fear is more focused on social evaluation than having unexpected panic attacks, and the fear of having an attack is limited to social situations. As many as 12.1% of the general population suffer from social phobia at some point in their lives (Kessler, Berglund, et al., 2005), making it one of the most common anxiety disorders, second only to specific phobia (see Table 1).



Social trauma in childhood may have long-lasting effects. [Image: ihtatho, <https://goo.gl/dTzrdj>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

Posttraumatic Stress Disorder

With stories of war, natural disasters, and physical and sexual assault dominating the news, it is clear that trauma is a reality for many people. Many individual traumas that occur every day never even make the headlines, such as a car accident, domestic abuse, or the death of a loved one. Yet, while many people face traumatic events, not everyone who faces a trauma

develops a disorder. Some, with the help of family and friends, are able to recover and continue on with their lives (Friedman, 2009). For some, however, the months and years following a trauma are filled with intrusive reminders of the event, a sense of intense fear that another traumatic event might occur, or a sense of isolation and emotional numbing. They may engage in a host of behaviors intended to protect themselves from being vulnerable or unsafe, such as constantly scanning their surroundings to look for signs of potential danger, never sitting with their back to the door, or never allowing themselves to be anywhere alone. This lasting reaction to trauma is what characterizes posttraumatic stress disorder (PTSD).

A diagnosis of PTSD begins with the traumatic event itself. An individual must have been exposed to an event that involves actual or threatened death, serious injury, or sexual violence. To receive a diagnosis of PTSD, exposure to the event must include either directly experiencing the event, witnessing the event happening to someone else, learning that the event occurred to a close relative or friend, or having repeated or extreme exposure to details of the event (such as in the case of first responders). The person subsequently re-experiences the event through both intrusive memories and nightmares. Some memories may come back so vividly that the person feels like they are experiencing the event all over again, what is known as having a flashback. The individual may avoid anything that reminds them of the trauma, including conversations, places, or even specific types of people. They may feel emotionally numb or restricted in their ability to feel, which may interfere in their interpersonal relationships. The person may not be able to remember certain aspects of what happened during the event. They may feel a sense of a foreshortened future, that they will never marry, have a family, or live a long, full life. They may be jumpy or easily startled, hypervigilant to their surroundings, and quick to anger. The prevalence of PTSD among the population as a whole is relatively low, with 6.8% having experienced PTSD at some point in their life (Kessler, Berglund, et al., 2005) (see Table 1). Combat and sexual assault are the most common precipitating traumas (Kessler, Sonnega, Bromet, Hughes, & Nelson, 1995). Whereas PTSD was previously categorized as an Anxiety Disorder, in the most recent version of the DSM (DSM-5; APA, 2013) it has been reclassified under the more specific category of Trauma- and Stressor-Related Disorders.

A person with PTSD is particularly sensitive to both internal and external cues that serve as reminders of their traumatic experience. For example, as we saw in PD, the physical sensations of arousal present during the initial trauma can become threatening in and of themselves, becoming a powerful reminder of the event. Someone might avoid watching intense or emotional movies in order to prevent the experience of emotional arousal. Avoidance of conversations, reminders, or even of the experience of emotion itself may also be an attempt to avoid triggering internal cues. External stimuli that were present during the trauma can also become strong triggers. For example, if a woman is raped by a man wearing a red t-shirt,

she may develop a strong alarm reaction to the sight of red shirts, or perhaps even more indiscriminately to anything with a similar color red. A combat veteran who experienced a strong smell of gasoline during a roadside bomb attack may have an intense alarm reaction when pumping gas back at home. Individuals with a psychological vulnerability toward viewing the world as uncontrollable and unpredictable may particularly struggle with the possibility of additional future, unpredictable traumatic events, fueling their need for hypervigilance and avoidance, and perpetuating the symptoms of PTSD.

Obsessive-Compulsive Disorder

Have you ever had a strange thought pop into your mind, such as picturing the stranger next to you naked? Or maybe you walked past a crooked picture on the wall and couldn't resist straightening it. Most people have occasional strange thoughts and may even engage in some "compulsive" behaviors, especially when they are stressed (Boyer & Liénard, 2008; Fullana et al., 2009). But for most people, these thoughts are nothing more than a passing oddity, and the behaviors are done (or not done) without a second thought. For someone with **obsessive-compulsive disorder (OCD)**, however, these thoughts and compulsive behaviors don't just come and go. Instead, strange or unusual thoughts are taken to mean something much more important and real, maybe even something dangerous or frightening. The urge to engage in some behavior, such as straightening a picture, can become so intense that it is nearly impossible *not* to carry it out, or causes significant anxiety if it can't be carried out. Further, someone with OCD might become preoccupied with the possibility that the behavior wasn't carried out to completion and feel compelled to repeat the behavior again and again, maybe several times before they are "satisfied."

To receive a diagnosis of OCD, a person must experience obsessive thoughts and/or compulsions that seem irrational or nonsensical, but that keep coming into their mind. Some examples of obsessions include doubting thoughts (such as doubting a door is locked or an appliance is turned off), thoughts of contamination (such as thinking that touching almost



People suffering from OCD may have an irrational fear of germs and "becoming contaminated". [Image: benchilada, <https://goo.gl/qemgDm>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

anything might give you cancer), or aggressive thoughts or images that are unprovoked or nonsensical. Compulsions may be carried out in an attempt to neutralize some of these thoughts, providing temporary relief from the anxiety the obsessions cause, or they may be nonsensical in and of themselves. Either way, compulsions are distinct in that they must be repetitive or excessive, the person feels “driven” to carry out the behavior, and the person feels a great deal of distress if they can’t engage in the behavior. Some examples of compulsive behaviors are repetitive washing (often in response to contamination obsessions), repetitive checking (locks, door handles, appliances often in response to doubting obsessions), ordering and arranging things to ensure symmetry, or doing things according to a specific ritual or sequence (such as getting dressed or ready for bed in a specific order). To meet diagnostic criteria for OCD, engaging in obsessions and/or compulsions must take up a significant amount of the person’s time, at least an hour per day, and must cause significant distress or impairment in functioning. About 1.6% of the population has met criteria for OCD over the course of a lifetime (Kessler, Berglund, et al., 2005) (see Table 1). Whereas OCD was previously categorized as an Anxiety Disorder, in the most recent version of the DSM (DSM-5; APA, 2013) it has been reclassified under the more specific category of Obsessive-Compulsive and Related Disorders.

People with OCD often confuse having an intrusive thought with their potential for carrying out the thought. Whereas most people when they have a strange or frightening thought are able to let it go, a person with OCD may become “stuck” on the thought and be intensely afraid that they might somehow lose control and act on it. Or worse, they believe that having the thought is just as bad as doing it. This is called thought-action fusion. For example, one patient of ours was plagued by thoughts that she would cause harm to her young daughter. She experienced intrusive images of throwing hot coffee in her daughter’s face or pushing her face underwater when she was giving her a bath. These images were so terrifying to the patient that she would no longer allow herself any physical contact with her daughter and would leave her daughter in the care of a babysitter if her husband or another family was not available to “supervise” her. In reality, the last thing she wanted to do was harm her daughter, and she had no intention or desire to act on the aggressive thoughts and images, nor does anybody with OCD act on these thoughts, but these thoughts were so horrifying to her that she made every attempt to prevent herself from the potential of carrying them out, even if it meant not being able to hold, cradle, or cuddle her daughter. These are the types of struggles people with OCD face every day.

Treatments for Anxiety and Related Disorders

Many successful treatments for anxiety and related disorders have been developed over the years. Medications (anti-anxiety drugs and antidepressants) have been found to be beneficial

for disorders other than specific phobia, but relapse rates are high once medications are stopped (Heimberg et al., 1998; Hollon et al., 2005), and some classes of medications (minor tranquilizers or benzodiazepines) can be habit forming.



Exposure-based CBT aims to help patients recognize and change problematic thoughts and behaviors in real-life situations. A person with a fear of elevators would be encouraged to practice exposure exercises that might involve approaching or riding elevators to attempt to overcome their anxiety. [Image: Mag3737, <https://goo.gl/j9L5AQ>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Exposure-based cognitive behavioral therapies (CBT) are effective psychosocial treatments for anxiety disorders, and many show greater treatment effects than medication in the long term (Barlow, Allen, & Basden, 2007; Barlow, Gorman, Shear, & Woods, 2000). In CBT, patients are taught skills to help identify and change problematic thought processes, beliefs, and behaviors that tend to worsen symptoms of anxiety, and practice applying these skills to real-life situations through exposure exercises. Patients learn how the automatic “appraisals” or thoughts they have about a situation affect both how they feel and how they behave. Similarly, patients learn how engaging in certain behaviors, such as avoiding situations, tends to strengthen the belief that the situation is something to be feared. A key aspect of CBT is exposure exercises, in which the patient learns to gradually approach situations

they find fearful or distressing, in order to challenge their beliefs and learn new, less fearful associations about these situations.

Typically 50% to 80% of patients receiving drugs or CBT will show a good initial response, with the effect of CBT more durable. Newer developments in the treatment of anxiety disorders are focusing on novel interventions, such as the use of certain medications to enhance learning during CBT (Otto et al., 2010), and transdiagnostic treatments targeting core, underlying vulnerabilities (Barlow et al., 2011). As we advance our understanding of anxiety and related disorders, so too will our treatments advance, with the hopes that for the many people suffering from these disorders, anxiety can once again become something useful and adaptive, rather than something debilitating.

Outside Resources

American Psychological Association (APA)

<http://www.apa.org/topics/anxiety/index.aspx>

National Institutes of Mental Health (NIMH)

<http://www.nimh.nih.gov/health/topics/anxiety-disorders/index.shtml>

Web: Anxiety and Depression Association of America (ADAA)

<http://www.adaa.org/>

Web: Center for Anxiety and Related Disorders (CARD)

<http://www.bu.edu/card/>

Discussion Questions

1. Name and describe the three main vulnerabilities contributing to the development of anxiety and related disorders. Do you think these disorders could develop out of biological factors alone? Could these disorders develop out of learning experiences alone?
2. Many of the symptoms in anxiety and related disorders overlap with experiences most people have. What features differentiate someone with a disorder versus someone without?
3. What is an “alarm reaction?” If someone experiences an alarm reaction when they are about to give a speech in front of a room full of people, would you consider this a “true alarm” or a “false alarm?”
4. Many people are shy. What differentiates someone who is shy from someone with social anxiety disorder? Do you think shyness should be considered an anxiety disorder?
5. Is anxiety ever helpful? What about worry?

Vocabulary

Agoraphobia

A sort of anxiety disorder distinguished by feelings that a place is uncomfortable or may be unsafe because it is significantly open or crowded.

Anxiety

A mood state characterized by negative affect, muscle tension, and physical arousal in which a person apprehensively anticipates future danger or misfortune.

Biological vulnerability

A specific genetic and neurobiological factor that might predispose someone to develop anxiety disorders.

Conditioned response

A learned reaction following classical conditioning, or the process by which an event that automatically elicits a response is repeatedly paired with another neutral stimulus (conditioned stimulus), resulting in the ability of the neutral stimulus to elicit the same response on its own.

External cues

Stimuli in the outside world that serve as triggers for anxiety or as reminders of past traumatic events.

Fight or flight response

A biological reaction to alarming stressors that prepares the body to resist or escape a threat.

Flashback

Sudden, intense re-experiencing of a previous event, usually trauma-related.

Generalized anxiety disorder (GAD)

Excessive worry about everyday things that is at a level that is out of proportion to the specific causes of worry.

Internal bodily or somatic cues

Physical sensations that serve as triggers for anxiety or as reminders of past traumatic events.

Interoceptive avoidance

Avoidance of situations or activities that produce sensations of physical arousal similar to those occurring during a panic attack or intense fear response.

Obsessive-compulsive disorder (OCD)

A disorder characterized by the desire to engage in certain behaviors excessively or compulsively in hopes of reducing anxiety. Behaviors include things such as cleaning, repeatedly opening and closing doors, hoarding, and obsessing over certain thoughts.

Panic disorder (PD)

A condition marked by regular strong panic attacks, and which may include significant levels of worry about future attacks.

Posttraumatic stress disorder (PTSD)

A sense of intense fear, triggered by memories of a past traumatic event, that another traumatic event might occur. PTSD may include feelings of isolation and emotional numbing.

Psychological vulnerabilities

Influences that our early experiences have on how we view the world.

Reinforced response

Following the process of operant conditioning, the strengthening of a response following either the delivery of a desired consequence (positive reinforcement) or escape from an aversive consequence.

SAD performance only

Social anxiety disorder which is limited to certain situations that the sufferer perceives as requiring some type of performance.

Social anxiety disorder (SAD)

A condition marked by acute fear of social situations which lead to worry and diminished day to day functioning.

Specific vulnerabilities

How our experiences lead us to focus and channel our anxiety.

Thought-action fusion

The tendency to overestimate the relationship between a thought and an action, such that one mistakenly believes a “bad” thought is the equivalent of a “bad” action.

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Mood Disorders

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Everyone feels down or euphoric from time to time, but this is different from having a mood disorder such as major depressive disorder or bipolar disorder. Mood disorders are extended periods of depressed, euphoric, or irritable moods that in combination with other symptoms cause the person significant distress and interfere with his or her daily life, often resulting in social and occupational difficulties. In this module, we describe major mood disorders, including their symptom presentations, general prevalence rates, and how and why the rates of these disorders tend to vary by age, gender, and race. In addition, biological and environmental risk factors that have been implicated in the development and course of mood disorders, such as heritability and stressful life events, are reviewed. Finally, we provide an overview of treatments for mood disorders, covering treatments with demonstrated effectiveness, as well as new treatment options showing promise.

Learning Objectives

- Describe the diagnostic criteria for mood disorders.
- Understand age, gender, and ethnic differences in prevalence rates of mood disorders.
- Identify common risk factors for mood disorders.
- Know effective treatments of mood disorders.

The actress Brooke Shields published a memoir titled *Down Came the Rain: My Journey through Postpartum Depression* in which she described her struggles with depression following the birth of her daughter. Despite the fact that about one in 20 women experience



Perinatal depression following child birth afflicts about 5% of all mothers. An unfortunate social stigma regarding this form of depression compounds the problem for the women who suffer its effects. [Image: CC0 Public Domain]

depression after the birth of a baby (American Psychiatric Association [APA], 2013), postpartum depression—recently renamed “perinatal depression”—continues to be veiled by stigma, owing in part to a widely held expectation that motherhood should be a time of great joy. In an opinion piece in the *New York Times*, Shields revealed that entering motherhood was a profoundly overwhelming experience for her. She vividly describes experiencing a sense of “doom” and “dread” in response to her newborn baby. Because motherhood is conventionally thought of as a joyous event and not associated with sadness and hopelessness, responding to a newborn baby in this way can be shocking to the new mother as well as those close to her. It may also involve a

great deal of shame for the mother, making her reluctant to divulge her experience to others, including her doctors and family.

Feelings of shame are not unique to perinatal depression. Stigma applies to other types of depressive and bipolar disorders and contributes to people not always receiving the necessary support and treatment for these disorders. In fact, the World Health Organization ranks both major depressive disorder (MDD) and bipolar disorder (BD) among the top 10 leading causes of disability worldwide. Further, MDD and BD carry a high risk of suicide. It is estimated that 25%–50% of people diagnosed with BD will attempt suicide at least once in their lifetimes (Goodwin & Jamison, 2007).

What Are Mood Disorders?

Mood Episodes

Everyone experiences brief periods of sadness, irritability, or euphoria. This is different than having a mood disorder, such as MDD or BD, which are characterized by a constellation of symptoms that causes people significant distress or impairs their everyday functioning.

Major Depressive Episode

A major depressive episode (MDE) refers to symptoms that co-occur for at least two weeks and cause significant distress or impairment in functioning, such as interfering with work, school, or relationships. Core symptoms include feeling down or depressed or experiencing **anhedonia**—loss of interest or pleasure in things that one typically enjoys. According to the fifth edition of the *Diagnostic and Statistical Manual (DSM-5; APA, 2013)*, the criteria for an MDE require five or more of the following nine symptoms, including one or both of the first two symptoms, for most of the day, nearly every day:

1. depressed mood
2. diminished interest or pleasure in almost all activities
3. significant weight loss or gain or an increase or decrease in appetite
4. insomnia or **hypersomnia**
5. **psychomotor agitation** or **retardation**
6. fatigue or loss of energy
7. feeling worthless or excessive or inappropriate guilt
8. diminished ability to concentrate or indecisiveness
9. recurrent thoughts of death, **suicidal ideation**, or a suicide attempt

These symptoms cannot be caused by physiological effects of a substance or a general medical condition (e.g., hypothyroidism).

Manic or Hypomanic Episode

The core criterion for a manic or hypomanic episode is a distinct period of abnormally and persistently euphoric, expansive, or irritable mood and persistently increased goal-directed activity or energy. The mood disturbance must be present for one week or longer in mania (unless hospitalization is required) or four days or longer in hypomania. Concurrently, at least three of the following symptoms must be present in the context of euphoric mood (or at least four in the context of irritable mood):

1. inflated self-esteem or **grandiosity**
2. increased goal-directed activity or psychomotor agitation

3. reduced need for sleep
4. racing thoughts or flight of ideas
5. distractibility
6. increased talkativeness
7. excessive involvement in risky behaviors

Manic episodes are distinguished from hypomanic episodes by their duration and associated impairment; whereas manic episodes must last one week and are defined by a significant impairment in functioning, hypomanic episodes are shorter and not necessarily accompanied by impairment in functioning.

Mood Disorders

Unipolar Mood Disorders

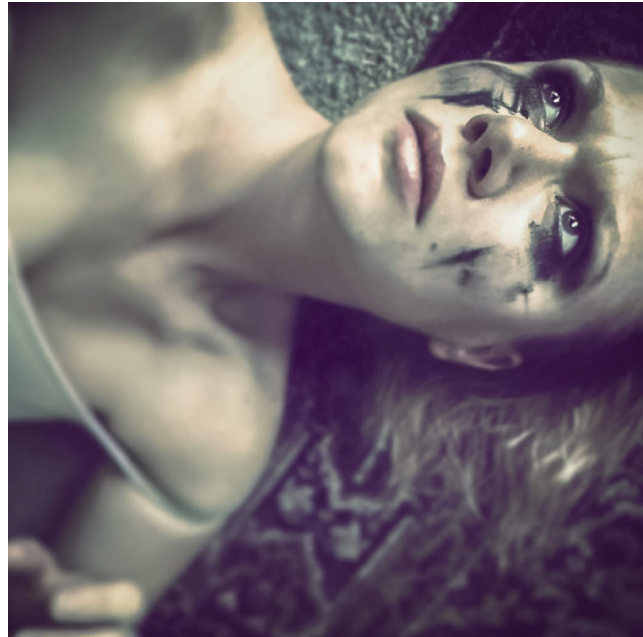
Two major types of unipolar disorders described by the *DSM-5* (APA, 2013) are major depressive disorder and persistent depressive disorder (PDD; dysthymia). MDD is defined by one or more MDEs, but no history of manic or hypomanic episodes. Criteria for PDD are feeling depressed most of the day for more days than not, for at least two years. At least two of the following symptoms are also required to meet criteria for PDD:

1. poor appetite or overeating
2. insomnia or hypersomnia
3. low energy or fatigue
4. low self-esteem
5. poor concentration or difficulty making decisions
6. feelings of hopelessness

Like MDD, these symptoms need to cause significant distress or impairment and cannot be due to the effects of a substance or a general medical condition. To meet criteria for PDD, a person cannot be without symptoms for more than two months at a time. PDD has overlapping symptoms with MDD. If someone meets criteria for an MDE during a PDD episode, the person will receive diagnoses of PDD and MDD.

Bipolar Mood Disorders

Three major types of BDs are described by the *DSM-5* (APA, 2013). Bipolar I Disorder (BD I), which was previously known as manic-depression, is characterized by a single (or recurrent) manic episode. A depressive episode is not necessary but commonly present for the diagnosis of BD I. Bipolar II Disorder is characterized by single (or recurrent) hypomanic episodes and depressive episodes. Another type of BD is cyclothymic disorder, characterized by numerous and alternating periods of hypomania and depression, lasting at least two years. To qualify for cyclothymic disorder, the periods of depression cannot meet full diagnostic criteria for an MDE; the person must experience symptoms at least half the time with no more than two consecutive symptom-free months; and the symptoms must cause significant distress or impairment.



Bipolar disorders are characterized by cycles of high energy and depression. [Image: Brett Whaley, <https://goo.gl/k4HTR7>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

It is important to note that the *DSM-5* was published in 2013, and findings based on the updated manual will be forthcoming. Consequently, the research presented below was largely based on a similar, but not identical, conceptualization of mood disorders drawn from the *DSM-IV* (APA, 2000).

How Common Are Mood Disorders? Who Develops Mood Disorders?

Depressive Disorders

In a nationally representative sample, lifetime prevalence rate for MDD is 16.6% (Kessler, Berglund, Demler, Jin, Merikangas, & Walters, 2005). This means that nearly one in five Americans will meet the criteria for MDD during their lifetime. The 12-month prevalence—the proportion of people who meet criteria for a disorder during a 12-month period—for PDD is approximately 0.5% (APA, 2013).

Although the onset of MDD can occur at any time throughout the lifespan, the average age of onset is mid-20s, with the age of onset decreasing with people born more recently (APA, 2000). Prevalence of MDD among older adults is much lower than it is for younger cohorts (Kessler, Birnbaum, Bromet, Hwang, Sampson, & Shahly, 2010). The duration of MDEs varies widely. Recovery begins within three months for 40% of people with MDD and within 12 months for 80% (APA, 2013). MDD tends to be a recurrent disorder with about 40%–50% of those who experience one MDE experiencing a second MDE (Monroe & Harkness, 2011). An earlier age of onset predicts a worse course. About 5%–10% of people who experience

an MDE will later experience a manic episode (APA, 2000), thus no longer meeting criteria for MDD but instead meeting them for BD I. Diagnoses of other disorders across the lifetime are common for people with MDD: 59% experience an anxiety disorder; 32% experience an impulse control disorder, and 24% experience a substance use disorder (Kessler, Merikangas, & Wang, 2007).

Women experience two to three times higher rates of MDD than do men (Nolen-Hoeksema & Hilt, 2009). This gender difference emerges during puberty (Conley & Rudolph, 2009). Before puberty, boys exhibit similar or higher prevalence rates of MDD than do girls (Twenge & Nolen-Hoeksema, 2002). MDD is inversely correlated with **socioeconomic status** (SES), a person's economic and social position based on income, education, and occupation. Higher prevalence rates of MDD are associated with lower SES (Lorant, Deliege, Eaton, Robert, Philippot, & Anseau, 2003), particularly for adults over 65 years old (Kessler et al., 2010). Independent of SES, results from a nationally representative sample found that European Americans had a higher prevalence rate of MDD than did African Americans and Hispanic Americans, whose rates were similar (Breslau, Aguilar-Gaxiola, Kendler, Su, Williams, & Kessler, 2006). The course of MDD for African Americans is often more severe and less often treated than it is for European Americans, however (Williams et al., 2007). Native Americans have a higher prevalence rate than do European Americans, African Americans, or Hispanic Americans (Hasin, Goodwin, Stinson & Grant, 2005). Depression is not limited to industrialized or western cultures; it is

Box 1. Specifiers

Both MDEs and manic episodes can be further described using standardized tags based on the timing of, or other symptoms that are occurring during, the mood episode, to increase diagnostic specificity and inform treatment. Psychotic features is specified when the episodes are accompanied by delusions (rigidly held beliefs that are false) or hallucinations (perceptual disturbances that are not based in reality). Seasonal pattern is specified when a mood episode occurs at the same time of the year for two consecutive years—most commonly occurring in the fall and winter. Peripartum onset is specified when a mood episode has an onset during pregnancy or within four weeks of the birth of a child. Approximately 3%–6% of women who have a child experience an MDE with peripartum onset (APA, 2013). This is less frequent and different from the baby blues or when women feel transient mood symptoms usually within 10 days of giving birth, which are experienced by most women (Nolen-Hoeksema & Hilt, 2009).

found in all countries that have been examined, although the symptom presentation as well as prevalence rates vary across cultures (Chentsova-Dutton & Tsai, 2009).

Bipolar Disorders



Adolescents experience a higher incidence of bipolar spectrum disorders than do adults. Making matters worse, those who are diagnosed with BD at a younger age seem to suffer symptoms more intensely than those with adult onset. [Image: CC0 Public Domain]

national study sample of more than 60,000 adults from 11 countries, estimated the worldwide prevalence of BD at 2.4%, with BD I constituting 0.6% of this rate (Merikangas et al., 2011). In this study, the prevalence of BD varied somewhat by country. Whereas the United States had the highest lifetime prevalence (4.4%), India had the lowest (0.1%). Variation in prevalence rates was not necessarily related to SES, as in the case of Japan, a high-income country with a very low prevalence rate of BD (0.7%).

With regard to ethnicity, data from studies not confounded by SES or inaccuracies in diagnosis are limited, but available reports suggest rates of BD among European Americans are similar to those found among African Americans (Blazer et al., 1985) and Hispanic Americans (Breslau, Kendler, Su, Gaxiola-Aguilar, & Kessler, 2005). Another large community-based study found that although prevalence rates of mood disorders were similar across ethnic groups, Hispanic Americans and African Americans with a mood disorder were more likely to remain

The lifetime prevalence rate of bipolar spectrum disorders in the general U.S. population is estimated at approximately 4.4%, with BD I constituting about 1% of this rate (Merikangas et al., 2007). Prevalence estimates, however, are highly dependent on the diagnostic procedures used (e.g., interviews vs. self-report) and whether or not sub-threshold forms of the disorder are included in the estimate. BD often co-occurs with other psychiatric disorders. Approximately 65% of people with BD meet diagnostic criteria for at least one additional psychiatric disorder, most commonly anxiety disorders and substance use disorders (McElroy et al., 2001). The co-occurrence of BD with other psychiatric disorders is associated with poorer illness course, including higher rates of suicidality (Leverich et al., 2003). A recent cross-

persistently ill than European Americans (Breslau et al., 2005). Compared with European Americans with BD, African Americans tend to be underdiagnosed for BD (and over-diagnosed for schizophrenia) (Kilbourne, Haas, Mulsant, Bauer, & Pincus, 2004; Minsky, Vega, Miskimen, Gara, & Escobar, 2003), and Hispanic Americans with BD have been shown to receive fewer psychiatric medication prescriptions and specialty treatment visits (Gonzalez et al., 2007). Misdiagnosis of BD can result in the underutilization of treatment or the utilization of inappropriate treatment, and thus profoundly impact the course of illness.

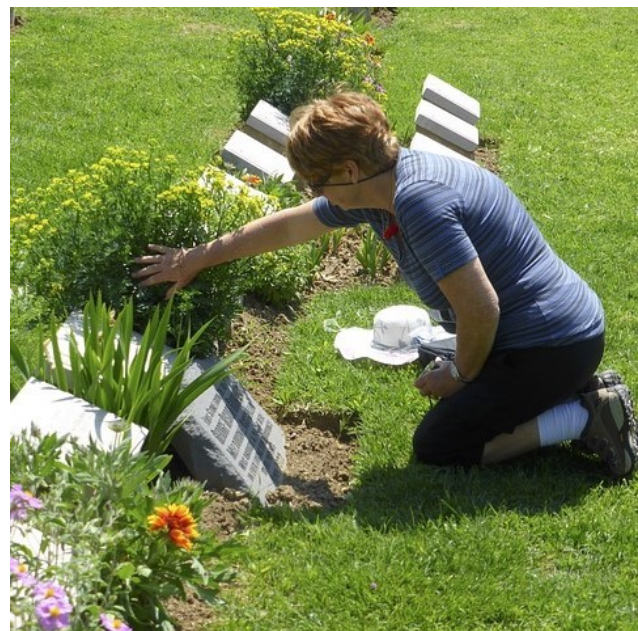
As with MDD, adolescence is known to be a significant risk period for BD; mood symptoms start by adolescence in roughly half of BD cases (Leverich et al., 2007; Perlis et al., 2004). Longitudinal studies show that those diagnosed with BD prior to adulthood experience a more pernicious course of illness relative to those with adult onset, including more episode recurrence, higher rates of suicidality, and profound social, occupational, and economic repercussions (e.g., Lewinsohn, Seeley, Buckley, & Klein, 2002). The prevalence of BD is substantially lower in older adults compared with younger adults (1% vs. 4%) (Merikangas et al., 2007).

What Are Some of the Factors Implicated in the Development and Course of Mood Disorders?

Mood disorders are complex disorders resulting from multiple factors. Causal explanations can be attempted at various levels, including biological and psychosocial levels. Below are several of the key factors that contribute to onset and course of mood disorders are highlighted.

Depressive Disorders

Research across family and twin studies has provided support that genetic factors are implicated in the development of MDD. Twin studies suggest that familial influence on MDD is mostly due to genetic effects and that individual-specific environmental effects (e.g., romantic relationships) play



Romantic relationships can affect mood as in the case of divorce or the death of a spouse. [Image: CC0 Public Domain]

an important role, too. By contrast, the contribution of shared environmental effect by siblings is negligible (Sullivan, Neale & Kendler, 2000). The mode of inheritance is not fully understood although no single genetic variation has been found to increase the risk of MDD significantly. Instead, several genetic variants and environmental factors most likely contribute to the risk for MDD (Lohoff, 2010).

One environmental stressor that has received much support in relation to MDD is stressful life events. In particular, severe stressful life events—those that have long-term consequences and involve loss of a significant relationship (e.g., divorce) or economic stability (e.g., unemployment) are strongly related to depression (Brown & Harris, 1989; Monroe et al., 2009). Stressful life events are more likely to predict the first MDE than subsequent episodes (Lewinsohn, Allen, Seeley, & Gotlib, 1999). In contrast, minor events may play a larger role in subsequent episodes than the initial episodes (Monroe & Harkness, 2005).

Depression research has not been limited to examining reactivity to stressful life events. Much research, particularly brain imaging research using functional magnetic resonance imaging (fMRI), has centered on examining neural circuitry—the interconnections that allow multiple brain regions to perceive, generate, and encode information in concert. A meta-analysis of neuroimaging studies showed that when viewing negative stimuli (e.g., picture of an angry face, picture of a car accident), compared with healthy control participants, participants with MDD have greater activation in brain regions involved in stress response and reduced activation of brain regions involved in positively motivated behaviors (Hamilton, Etkin, Furman, Lemus, Johnson, & Gotlib, 2012).

Other environmental factors related to increased risk for MDD include experiencing early adversity (e.g., childhood abuse or neglect; Widom, DuMont, & Czaja, 2007), chronic stress (e.g., poverty) and interpersonal factors. For example, marital dissatisfaction predicts increases in depressive symptoms in both men and women. On the other hand, depressive symptoms also predict increases in marital dissatisfaction (Whisman & Uebelacker, 2009). Research has found that people with MDD generate some of their interpersonal stress (Hammen, 2005). People with MDD whose relatives or spouses can be described as critical and emotionally overinvolved have higher relapse rates than do those living with people who are less critical and emotionally overinvolved (Butzlaff & Hooley, 1998).

People's attributional styles or their general ways of thinking, interpreting, and recalling information have also been examined in the etiology of MDD (Gotlib & Joormann, 2010). People with a pessimistic attributional style tend to make internal (versus external), global (versus specific), and stable (versus unstable) attributions to negative events, serving as a vulnerability to developing MDD. For example, someone who when he fails an exam thinks

that it was his fault (internal), that he is stupid (global), and that he will always do poorly (stable) has a pessimistic attribution style. Several influential theories of depression incorporate attributional styles (Abramson, Metalsky, & Alloy, 1989; Abramson Seligman, & Teasdale, 1978).

Bipolar Disorders

Although there have been important advances in research on the etiology, course, and treatment of BD, there remains a need to understand the mechanisms that contribute to episode onset and relapse. There is compelling evidence for biological causes of BD, which is known to be highly heritable (McGuffin, Rijdsdijk, Andrew, Sham, Katz, & Cardno, 2003). It may be argued that a high rate of heritability demonstrates that BD is fundamentally a biological phenomenon. However, there is much variability in the course of BD both within a person across time and across people (Johnson, 2005). The triggers that determine how and when this genetic vulnerability is expressed are not yet understood; however, there is evidence to suggest that psychosocial triggers may play an important role in BD risk (e.g., Johnson et al., 2008; Malkoff-Schwartz et al., 1998).

In addition to the genetic contribution, biological explanations of BD have also focused on brain function. Many of the studies using fMRI techniques to characterize BD have focused on the processing of emotional stimuli based on the idea that BD is fundamentally a disorder of emotion (APA, 2000). Findings show that regions of the brain thought to be involved in emotional processing and regulation are activated differently in people with BD relative to healthy controls (e.g., Altshuler et al., 2008; Hassel et al., 2008; Lennox, Jacob, Calder, Lupson, & Bullmore, 2004).

However, there is little consensus as to whether a particular brain region becomes more or less active in response to an emotional stimulus among people with BD compared with healthy controls. Mixed findings are in part due to samples consisting of participants who are at various phases of illness at the time of testing (manic, depressed, inter-episode). Sample sizes tend to be relatively small, making comparisons between subgroups difficult. Additionally, the use of a standardized stimulus (e.g., facial expression of anger) may not elicit a sufficiently strong response. Personally engaging stimuli, such as recalling a memory, may be more effective in inducing strong emotions (Isacowitz, Gershon, Allard, & Johnson, 2013).

Within the psychosocial level, research has focused on the environmental contributors to BD. A series of studies show that environmental stressors, particularly severe stressors (e.g., loss of a significant relationship), can adversely impact the course of BD. People with BD have substantially increased risk of relapse (Ellicott, Hammen, Gitlin, Brown, & Jamison, 1990) and

suffer more depressive symptoms (Johnson, Winett, Meyer, Greenhouse, & Miller, 1999) following a severe life stressor. Interestingly, positive life events can also adversely impact the course of BD. People with BD suffer more manic symptoms after life events involving attainment of a desired goal (Johnson et al., 2008). Such findings suggest that people with BD may have a hypersensitivity to rewards.

Evidence from the life stress literature has also suggested that people with mood disorders may have a circadian vulnerability that renders them sensitive to stressors that disrupt their sleep or rhythms. According to [social zeitgeber](#) theory (Ehlers, Frank, & Kupfer, 1988; Frank et al., 1994), stressors that disrupt sleep, or that disrupt the daily routines that entrain the biological clock (e.g., meal times) can trigger episode relapse. Consistent with this theory, studies have shown that life events that involve a disruption in sleep and daily routines, such as overnight travel, can increase bipolar symptoms in people with BD (Malkoff-Schwartz et al., 1998).

What Are Some of the Well-Supported Treatments for Mood Disorders?

Depressive Disorders



A number of medications are effective in treating mood disorders. Meditation, exercise, counseling and other therapies also show effectiveness for some disorders. [Image: CC0 Public Domain]

There are many treatment options available for people with MDD. First, a number of antidepressant medications are available, all of which target one or more of the neurotransmitters implicated in depression. The earliest antidepressant medications were monoamine oxidase inhibitors (MAOIs). MAOIs inhibit monoamine oxidase, an enzyme involved in deactivating dopamine, norepinephrine, and serotonin. Although effective in treating depression, MAOIs can have serious side effects. Patients taking MAOIs may develop dangerously high blood pressure if they take certain drugs (e.g., antihistamines) or eat foods containing tyramine, an amino acid commonly found in foods such as aged cheeses, wine, and soy sauce. Tricyclics, the second-oldest class of

antidepressant medications, block the reabsorption of norepinephrine, serotonin, or dopamine at synapses, resulting in their increased availability. Tricyclics are most effective for treating vegetative and somatic symptoms of depression. Like MAOIs, they have serious side effects, the most concerning of which is being cardiotoxic. Selective serotonin reuptake inhibitors (SSRIs; e.g., Fluoxetine) and serotonin and norepinephrine reuptake inhibitors (SNRIs; e.g., Duloxetine) are the most recently introduced antidepressant medications. SSRIs, the most commonly prescribed antidepressant medication, block the reabsorption of serotonin, whereas SNRIs block the reabsorption of serotonin and norepinephrine. SSRIs and SNRIs have fewer serious side effects than do MAOIs and tricyclics. In particular, they are less cardiotoxic, less lethal in overdose, and produce fewer cognitive impairments. They are not, however, without their own side effects, which include but are not limited to difficulty having orgasms, gastrointestinal issues, and insomnia. It should be noted that antidepressant medication may not work equally for all people. This approach to treatment often involves experimentation with several medications and dosages, and may be more effective when paired with physical exercise and psychotherapy.

Other biological treatments for people with depression include electroconvulsive therapy (ECT), transcranial magnetic stimulation (TMS), and deep brain stimulation. ECT involves inducing a seizure after a patient takes muscle relaxants and is under general anesthesia. ECT is a viable treatment for patients with severe depression or who show resistance to antidepressants although the mechanisms through which it works remain unknown. A common side effect is confusion and memory loss, usually short-term (Schulze-Rauschenbach, Harms, Schlaepfer, Maier, Falkai, & Wagner, 2005). Repetitive TMS is a noninvasive technique administered while a patient is awake. Brief pulsating magnetic fields are delivered to the cortex, inducing electrical activity. TMS has fewer side effects than ECT (Schulze-Rauschenbach et al., 2005), and while outcome studies are mixed, there is evidence that TMS is a promising treatment for patients with MDD who have shown resistance to other treatments (Rosa et al., 2006). Most recently, deep brain stimulation is being examined as a treatment option for patients who did not respond to more traditional treatments like those already described. Deep brain stimulation involves implanting an electrode in the brain. The electrode is connected to an implanted neurostimulator, which electrically stimulates that particular brain region. Although there is some evidence of its effectiveness (Mayberg et al., 2005), additional research is needed.

Several psychosocial treatments have received strong empirical support, meaning that independent investigations have achieved similarly positive results—a high threshold for examining treatment outcomes. These treatments include but are not limited to behavior therapy, cognitive therapy, and interpersonal therapy. Behavior therapies focus on increasing the frequency and quality of experiences that are pleasant or help the patient achieve mastery.

Cognitive therapies primarily focus on helping patients identify and change distorted automatic thoughts and assumptions (e.g., Beck, 1967). Cognitive-behavioral therapies are based on the rationale that thoughts, behaviors, and emotions affect and are affected by each other. Interpersonal Therapy for Depression focuses largely on improving interpersonal relationships by targeting problem areas, specifically unresolved grief, interpersonal role disputes, role transitions, and interpersonal deficits. Finally, there is also some support for the effectiveness of Short-Term Psychodynamic Therapy for Depression (Leichsenring, 2001). The short-term treatment focuses on a limited number of important issues, and the therapist tends to be more actively involved than in more traditional psychodynamic therapy.

Bipolar Disorders

Patients with BD are typically treated with pharmacotherapy. Antidepressants such as SSRIs and SNRIs are the primary choice of treatment for depression, whereas for BD, lithium is the first line treatment choice. This is because SSRIs and SNRIs have the potential to induce mania or hypomania in patients with BD. Lithium acts on several neurotransmitter systems in the brain through complex mechanisms, including reduction of excitatory (dopamine and glutamate) neurotransmission, and increasing of inhibitory (GABA) neurotransmission (Lenox & Hahn, 2000). Lithium has strong efficacy for the treatment of BD (Geddes, Burgess, Hawton, Jamison, & Goodwin, 2004). However, a number of side effects can make lithium treatment difficult for patients to tolerate. Side effects include impaired cognitive function (Wingo, Wingo, Harvey, & Baldessarini, 2009), as well as physical symptoms such as nausea, tremor, weight gain, and fatigue (Dunner, 2000). Some of these side effects can improve with continued use; however, medication noncompliance remains an ongoing concern in the treatment of patients with BD. Anticonvulsant medications (e.g., carbamazepine, valproate) are also commonly used to treat patients with BD, either alone or in conjunction with lithium.

There are several adjunctive treatment options for people with BD. Interpersonal and social rhythm therapy (IPSRT; Frank et al., 1994) is a psychosocial intervention focused on addressing the mechanism of action posited in social *zeitgeber* theory to predispose patients who have BD to relapse, namely sleep disruption. A growing body of literature provides support for the central role of sleep dysregulation in BD (Harvey, 2008). Consistent with this literature, IPSRT aims to increase rhythmicity of patients' lives and encourage vigilance in maintaining a stable rhythm. The therapist and patient work to develop and maintain a healthy balance of activity and stimulation such that the patient does not become overly active (e.g., by taking on too many projects) or inactive (e.g., by avoiding social contact). The efficacy of IPSRT has been demonstrated in that patients who received this treatment show reduced risk of episode recurrence and are more likely to remain well (Frank et al., 2005).

Conclusion

Everyone feels down or euphoric from time to time. For some people, these feelings can last for long periods of time and can also co-occur with other symptoms that, in combination, interfere with their everyday lives. When people experience an MDE or a manic episode, they see the world differently. During an MDE, people often feel hopeless about the future, and may even experience suicidal thoughts. During a manic episode, people often behave in ways that are risky or place them in danger. They may spend money excessively or have unprotected sex, often expressing deep shame over these decisions after the episode. MDD and BD cause significant problems for people at school, at work, and in their relationships and affect people regardless of gender, age, nationality, race, religion, or sexual orientation. If you or someone you know is suffering from a mood disorder, it is important to seek help. Effective treatments are available and continually improving. If you have an interest in mood disorders, there are many ways to contribute to their understanding, prevention, and treatment, whether by engaging in research or clinical work.

Outside Resources

Books: Recommended memoirs include *A Memoir of Madness* by William Styron (MDD); *Noonday Demon: An Atlas of Depression* by Andrew Solomon (MDD); and *An Unquiet Mind: A Memoir of Moods and Madness* by Kay Redfield (BD).

Web: Visit the **Association for Behavioral and Cognitive Therapies** to find a list of the recommended therapists and evidence-based treatments.

<http://www.abct.org>

Web: Visit the **Depression and Bipolar Support Alliance** for educational information and social support options.

<http://www.dbsalliance.org/>

Discussion Questions

1. What factors might explain the large gender difference in the prevalence rates of MDD?
2. Why might American ethnic minority groups experience more persistent BD than European Americans?
3. Why might the age of onset for MDD be decreasing over time?
4. Why might overnight travel constitute a potential risk for a person with BD?
5. What are some reasons positive life events may precede the occurrence of manic episode?

Vocabulary

Anhedonia

Loss of interest or pleasure in activities one previously found enjoyable or rewarding.

Attributional style

The tendency by which a person infers the cause or meaning of behaviors or events.

Chronic stress

Discrete or related problematic events and conditions which persist over time and result in prolonged activation of the biological and/or psychological stress response (e.g., unemployment, ongoing health difficulties, marital discord).

Early adversity

Single or multiple acute or chronic stressful events, which may be biological or psychological in nature (e.g., poverty, abuse, childhood illness or injury), occurring during childhood and resulting in a biological and/or psychological stress response.

Grandiosity

Inflated self-esteem or an exaggerated sense of self-importance and self-worth (e.g., believing one has special powers or superior abilities).

Hypersomnia

Excessive daytime sleepiness, including difficulty staying awake or napping, or prolonged sleep episodes.

Psychomotor agitation

Increased motor activity associated with restlessness, including physical actions (e.g., fidgeting, pacing, feet tapping, handwringing).

Psychomotor retardation

A slowing of physical activities in which routine activities (e.g., eating, brushing teeth) are performed in an unusually slow manner.

Social zeitgeber

Zeitgeber is German for “time giver.” Social zeitgebers are environmental cues, such as meal times and interactions with other people, that entrain biological rhythms and thus sleep-wake cycle regularity.

Socioeconomic status (SES)

A person's economic and social position based on income, education, and occupation.

Suicidal ideation

Recurring thoughts about suicide, including considering or planning for suicide, or preoccupation with suicide.

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Personality Disorders

Cristina Crego & Thomas Widiger

The purpose of this module is to define what is meant by a personality disorder, identify the five domains of general personality (i.e., neuroticism, extraversion, openness, agreeableness, and conscientiousness), identify the six personality disorders proposed for retention in the 5th edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) (i.e., borderline, antisocial, schizotypal, avoidant, obsessive-compulsive, and narcissistic), summarize the etiology for antisocial and borderline personality disorder, and identify the treatment for borderline personality disorder (i.e., dialectical behavior therapy and mentalization therapy).

Learning Objectives

- Define what is meant by a personality disorder.
- Identify the five domains of general personality.
- Identify the six personality disorders proposed for retention in DSM-5.
- Summarize the etiology for antisocial and borderline personality disorder.
- Identify the treatment for borderline personality disorder.

Introduction

Everybody has their own unique **personality**; that is, their characteristic manner of thinking, feeling, behaving, and relating to others (John, Robins, & Pervin, 2008). Some people are typically introverted, quiet, and withdrawn; whereas others are more extraverted, active, and

outgoing. Some individuals are invariably conscientiousness, dutiful, and efficient; whereas others might be characteristically undependable and negligent. Some individuals are consistently anxious, self-conscious, and apprehensive; whereas others are routinely relaxed, self-assured, and unconcerned. Personality traits refer to these characteristic, routine ways of thinking, feeling, and relating to others. There are signs or indicators of these traits in childhood, but they become particularly evident when the person is an adult. Personality traits are integral to each person's sense of self, as they involve what people value, how they think and feel about things, what they like to do, and, basically, what they are like most every day throughout much of their lives.

There are literally hundreds of different personality traits. All of these traits can be organized into the broad dimensions referred to as the **Five-Factor Model** (John, Naumann, & Soto, 2008). These five broad domains are inclusive; there does not appear to be any traits of

Neuroticism (Emotional Instability) fearful, apprehensive, angry, bitter, pessimistic, glum, timid, embarrassed, tempted, urgency, helpless, fragile	VS	Emotional Stability relaxed, unconcerned, cool, even-tempered, optimistic, self-assured, glib, shameless, controlled, restrained, clear-thinking, fearless, unflappable
Extraversion cordial, affectionate, attached, sociable, outgoing, dominant, forceful, vigorous, energetic, active, reckless, daring, high-spirited, excitement-seeking	VS	Introversion cold, aloof, indifferent, withdrawn, isolated, unassuming, quiet, resigned, passive, lethargic, cautious, monotonous, dull, placid, anhedonic
Openness (unconventionality) dreamer, unrealistic, imaginative, aberrant, aesthetic, self-aware, eccentric, strange, odd, peculiar, creative, permissive, broad-minded	VS	Closedness (conventionality) practical, concrete, uninvolved, no aesthetic interest, constricted, unaware, alexythymic, routine, predictable, habitual, stubborn, pragmatic, rigid, traditional, inflexible, dogmatic
Agreeableness gullible, naive, trusting, confiding, honest, sacrificial, giving, docile, cooperative, meek, self-effacing, humble, soft, empathetic	VS	Antagonism skeptical, cynical, suspicious, paranoid, cunning, manipulative, deceptive, stingy, selfish, greedy, exploitative, oppositional, combative, aggressive, confident, boastful, arrogant, tough, callous, ruthless
Conscientiousness perfectionistic, efficient, ordered, methodical, organized, rigid, reliable, dependable, workaholic, ambitious, dogged, devoted, cautious, ruminative, reflective	VS	Disinhibition lax, negligent, haphazard, disorganized, sloppy, casual, undependable, unethical, aimless, desultory, hedonistic, negligent, hasty, careless, rash

Table 1: Illustrative traits for both poles across Five-Factor Model personality dimensions.

personality that lie outside of the Five-Factor Model. This even applies to traits that you may use to describe yourself. Table I provides illustrative traits for both poles of the five domains of this model of personality. A number of the traits that you see in this table may describe you. If you can think of some other traits that describe yourself, you should be able to place them somewhere in this table.

DSM-5 Personality Disorders

When personality traits result in significant distress, social impairment, and/or occupational impairment, they are considered to be a personality disorder (American Psychiatric Association, 2013). The authoritative manual for what constitutes a personality disorder is provided by the American Psychiatric Association's (APA) *Diagnostic and Statistical Manual of Mental Disorders* (DSM), the current version of which is DSM-5 (APA, 2013). The DSM provides a common language and standard criteria for the classification and diagnosis of mental disorders. This manual is used by clinicians, researchers, health insurance companies, and policymakers. DSM-5 includes 10 **personality disorders**: antisocial, avoidant, borderline, dependent, histrionic, narcissistic, obsessive-compulsive, paranoid, schizoid, and schizotypal. All 10 of these personality disorders will be included in the next edition of the diagnostic manual, DSM-5.

This list of 10 though does not fully cover all of the different ways in which a personality can be maladaptive. DSM-5 also includes a "wastebasket" diagnosis of other specified personality disorder (OSPD) and unspecified personality disorder (UPD). This diagnosis is used when a clinician believes that a patient has a personality disorder but the traits that constitute this disorder are not well covered by one of the 10 existing diagnoses. OSPD and UPD or as they used to be referred to in previous editions - PDNOS (personality disorder not otherwise specified) are often one of the most frequently used diagnoses in clinical practice, suggesting that the current list of 10 is not adequately comprehensive (Widiger & Trull, 2007).

Description

Each of the 10 DSM-5 (and DSM-IV-TR) personality disorders is a constellation of maladaptive personality traits, rather than just one particular personality trait (Lynam & Widiger, 2001). In this regard, personality disorders are "syndromes." For example, **avoidant** personality disorder is a pervasive pattern of social inhibition, feelings of inadequacy, and hypersensitivity to negative evaluation (APA, 2013), which is a combination of traits from introversion (e.g., socially withdrawn, passive, and cautious) and neuroticism (e.g., self-consciousness, apprehensiveness, anxiousness, and worrisome). **Dependent** personality disorder includes

submissiveness, clinging behavior, and fears of separation (APA, 2013), for the most part a combination of traits of neuroticism (anxious, uncertain, pessimistic, and helpless) and maladaptive agreeableness (e.g., gullible, guileless, meek, subservient, and self-effacing). **Antisocial** personality disorder is, for the most part, a combination of traits from antagonism (e.g., dishonest, manipulative, exploitative, callous, and merciless) and low conscientiousness (e.g., irresponsible, immoral, lax, hedonistic, and rash).

Some of the DSM-5 personality disorders are confined largely to traits within one of the basic domains of personality. For example, **obsessive-compulsive** personality disorder is largely a disorder of maladaptive conscientiousness, including such traits as workaholism, perfectionism, punctilious, ruminative, and dogged; **schizoid** is confined largely to traits of introversion (e.g., withdrawn, cold, isolated, placid, and anhedonic); **borderline** personality disorder is largely a disorder of neuroticism, including such traits as emotionally unstable, vulnerable, overwhelmed, rageful, depressive, and self-destructive ; and **histrionic** personality disorder is largely a disorder of maladaptive extraversion, including such traits as attention-seeking, seductiveness, melodramatic emotionality, and strong attachment needs.



A person with an obsessive compulsive personality disorder may have a hard time relaxing, always feel under pressure, and believe that there isn't enough time to accomplish important tasks. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

It should be noted though that a complete description of each DSM-5 personality disorder would typically include at least some traits from other domains. For example, antisocial personality disorder (or psychopathy) also includes some traits from low neuroticism (e.g., fearlessness and glib charm) and extraversion (e.g., excitement-seeking and assertiveness); borderline includes some traits from antagonism (e.g., manipulative and oppositional) and low conscientiousness (e.g., rash); and histrionic includes some traits from antagonism (e.g., vanity) and low conscientiousness (e.g., impressionistic). **Narcissistic** personality disorder includes traits from neuroticism (e.g., reactive anger, reactive shame, and need for admiration), extraversion (e.g., exhibitionism and authoritativeness), antagonism (e.g., arrogance, entitlement, and lack of empathy), and conscientiousness (e.g., acclaim-seeking). **Schizotypal** personality disorder includes traits from neuroticism (e.g., social anxiousness and social

discomfort), introversion (e.g., social withdrawal), unconventionality (e.g., odd, eccentric, peculiar, and aberrant ideas), and antagonism (e.g., suspiciousness).

The APA currently conceptualizes personality disorders as qualitatively distinct conditions; distinct from each other and from normal personality functioning. However, included within an appendix to DSM-5 is an alternative view that personality disorders are simply extreme and/or maladaptive variants of normal personality traits, as suggested herein. Nevertheless, many leading personality disorder researchers do not hold this view (e.g., Gunderson, 2010; Hopwood, 2011; Shedler et al., 2010). They suggest that there is something qualitatively unique about persons suffering from a personality disorder, usually understood as a form of pathology in sense of self and interpersonal relatedness that is considered to be distinct from personality traits (APA, 2012; Skodol, 2012). For example, it has been suggested that antisocial personality disorder includes impairments in identity (e.g., egocentrism), self-direction, empathy, and capacity for intimacy, which are said to be different from such traits as arrogance, impulsivity, and callousness (APA, 2012).

Validity

It is quite possible that in future revisions of the DSM some of the personality disorders included in DSM-5 and DSM-IV-TR will no longer be included. In fact, for DSM-5 it was originally proposed that four be deleted. The personality disorders that were slated for deletion were histrionic, schizoid, **paranoid**, and dependent (APA, 2012). The rationale for the proposed deletions was in large part because they are said to have less empirical support than the diagnoses that were at the time being retained (Skodol, 2012). There is agreement within the field with regard to the empirical support for the borderline, antisocial, and schizotypal personality disorders (Mullins-Sweat, Bernstein, & Widiger, 2012; Skodol, 2012). However, there is a difference of opinion with respect to the empirical support for the dependent personality disorder (Bornstein, 2012; Livesley, 2011; Miller, Widiger, & Campbell, 2010; Mullins-Sweat et al., 2012).

Little is known about the specific etiology for most of the DSM-5 personality disorders. Because each personality disorder represents a constellation of personality traits, the etiology for the syndrome will involve a complex interaction of an array of different neurobiological vulnerabilities and dispositions with a variety of environmental, psychosocial events. Antisocial personality disorder, for instance, is generally considered to be the result of an interaction of genetic dispositions for low anxiousness, aggressiveness, impulsivity, and/or callousness, with a tough, urban environment, inconsistent parenting, poor parental role modeling, and/or peer support (Hare, Neumann, & Widiger, 2012). Borderline personality

disorder is generally considered to be the result of an interaction of a genetic disposition to negative affectivity interacting with a malevolent, abusive, and/or invalidating family environment (Hooley, Cole, & Gironde, 2012).

To the extent that one considers the DSM-5 personality disorders to be maladaptive variants of general personality structure, as described, for instance, within the Five-Factor Model, there would be a considerable body of research to support the validity for all of the personality disorders, including even the histrionic, schizoid, and paranoid. There is compelling multivariate behavior genetic support with respect to the precise structure of the Five-Factor Model (e.g., Yamagata et al., 2006), childhood antecedents (Caspi, Roberts, & Shiner, 2005), universality (Allik, 2005), temporal stability across the lifespan (Roberts & DelVecchio, 2000), ties with brain structure (DeYoung, Hirsh, Shane, Papademetris, Rajeevan, & Gray, 2010), and even molecular genetic support for neuroticism (Widiger, 2009).

Treatment

Personality disorders are relatively unique because they are often “ego-syntonic;” that is, most people are largely comfortable with their selves, with their characteristic manner of behaving,

feeling, and relating to others. As a result, people rarely seek treatment for their antisocial, narcissistic, histrionic, paranoid, and/or schizoid personality disorder. People typically lack insight into the maladaptivity of their personality.



Many people with personality disorders do not seek treatment. Those with borderline personality disorder and avoidant personality disorder are exceptions. High levels of neuroticism and emotional pain may motivate them to seek help. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

One clear exception though is borderline personality disorder (and perhaps as well avoidant personality disorder). Neuroticism is the domain of general personality structure that concerns inherent feelings of emotional pain and suffering, including feelings of distress, anxiety, depression, self-consciousness, helplessness, and vulnerability. Persons who have very high elevations on neuroticism (i.e., persons with borderline personality disorder) experience life as one of pain and suffering, and they will seek treatment to alleviate

this severe emotional distress. People with avoidant personality may also seek treatment for their high levels of neuroticism (anxiousness and self-consciousness) and introversion (social isolation). In contrast, narcissistic individuals will rarely seek treatment to reduce their arrogance; paranoid persons rarely seek treatment to reduce their feelings of suspiciousness; and antisocial people rarely (or at least willfully) seek treatment to reduce their disposition for criminality, aggression, and irresponsibility.

Nevertheless, maladaptive personality traits will be evident in many individuals seeking treatment for other mental disorders, such as anxiety, mood, or substance use. Many of the people with a substance use disorder will have antisocial personality traits; many of the people with mood disorder will have borderline personality traits. The prevalence of personality disorders within clinical settings is estimated to be well above 50% (Torgersen, 2012). As many as 60% of inpatients within some clinical settings are diagnosed with borderline personality disorder (APA, 2000). Antisocial personality disorder may be diagnosed in as many as 50% of inmates within a correctional setting (Hare et al., 2012). It is estimated that 10% to 15% of the general population meets criteria for at least one of the 10 DSM-IV-TR personality disorders (Torgersen, 2012), and quite a few more individuals are likely to have maladaptive personality traits not covered by one of the 10 DSM-5 diagnoses.

The presence of a personality disorder will often have an impact on the treatment of other mental disorders, typically inhibiting or impairing responsivity. Antisocial persons will tend to be irresponsible and negligent; borderline persons can form intensely manipulative attachments to their therapists; paranoid patients will be unduly suspicious and accusatory; narcissistic patients can be dismissive and denigrating; and dependent patients can become overly attached to and feel helpless without their therapists.

It is a misnomer, though, to suggest that personality disorders cannot themselves be treated. Personality disorders are among the most difficult of disorders to treat because they involve well-established behaviors that can be integral to a client's self-image (Millon, 2011). Nevertheless, much has been written on the treatment of personality disorder (e.g., Beck, Freeman, Davis, & Associates, 1990; Gunderson & Gabbard, 2000), and there is empirical support for clinically and socially meaningful changes in response to psychosocial and pharmacologic treatments (Perry & Bond, 2000). The development of an ideal or fully healthy personality structure is unlikely to occur through the course of treatment, but given the considerable social, public health, and personal costs associated with some of the personality disorders, such as the antisocial and borderline, even just moderate adjustments in personality functioning can represent quite significant and meaningful change.

Nevertheless, manualized and/or empirically validated treatment protocols have been

developed for only one personality disorder, borderline (APA, 2001).

Focus Topic: Treatment of Borderline Personality Disorder

Dialectical behavior therapy (Lynch & Cuyper, 2012) and mentalization therapy (Bateman & Fonagy, 2012): Dialectical behavior therapy is a form of cognitive-behavior therapy that draws on principles from Zen Buddhism, dialectical philosophy, and behavioral science. The treatment has four components: individual therapy, group skills training, telephone coaching, and a therapist consultation team, and will typically last a full year. As such, it is a relatively expensive form of treatment, but research has indicated that its benefits far outweighs its costs, both financially and socially.

It is unclear why specific and explicit treatment manuals have not been developed for the other personality disorders. This may reflect a regrettable assumption that personality disorders are unresponsive to treatment. It may also reflect the complexity of their treatment. As noted earlier, each DSM-5 disorder is a heterogeneous constellation of maladaptive personality traits. In fact, a person can meet diagnostic criteria for the antisocial, borderline, schizoid, schizotypal, narcissistic, and avoidant personality disorders and yet have only one diagnostic criterion in common. For example, only five of nine features are necessary for the diagnosis of borderline personality disorder; therefore, two persons can meet criteria for this disorder and yet have only one feature in common. In addition, patients meeting diagnostic criteria for one personality disorder will often meet diagnostic criteria for another. This degree of diagnostic overlap and heterogeneity of membership hinders tremendously any effort to identify a specific etiology, pathology, or treatment for a respective personality disorder as there is so much variation within any particular group of patients sharing the same diagnosis (Smith & Zapski, 2009).

Of course, this diagnostic overlap and complexity did not prevent researchers and clinicians from developing dialectical behavior therapy and mentalization therapy. A further reason for the weak progress in treatment development is that, as noted earlier, persons rarely seek treatment for their personality disorder. It would be difficult to obtain a sufficiently large group of people with, for instance, narcissistic or obsessive-compulsive disorder to participate in a treatment outcome study, one receiving the manualized treatment protocol, the other receiving treatment as usual.

Conclusions

It is evident that all individuals have a personality, as indicated by their characteristic way of thinking, feeling, behaving, and relating to others. For some people, these traits result in a considerable degree of distress and/or impairment, constituting a personality disorder. A considerable body of research has accumulated to help understand the etiology, pathology, and/or treatment for some personality disorders (i.e., antisocial, schizotypal, borderline, dependent, and narcissistic), but not so much for others (e.g., histrionic, schizoid, and paranoid). However, researchers and clinicians are now shifting toward a more dimensional understanding of personality disorders, wherein each is understood as a maladaptive variant of general personality structure, thereby bringing to bear all that is known about general personality functioning to an understanding of these maladaptive variants.

Outside Resources

Structured Clinical Interview for DSM-5 (SCID-5)

<https://www.appi.org/products/structured-clinical-interview-for-dsm-5-scid-5>

Web: DSM-5 website discussion of personality disorders

<http://www.dsm5.org/ProposedRevision/Pages/PersonalityDisorders.aspx>

Discussion Questions

1. Do you think that any of the personality disorders, or some of their specific traits, are ever good or useful to have?
2. If someone with a personality disorder commits a crime, what is the right way for society to respond? For example, does or should meeting diagnostic criteria for antisocial personality disorder mitigate (lower) a person's responsibility for committing a crime?
3. Given what you know about personality disorders and the traits that comprise each one, would you say there is any personality disorder that is likely to be diagnosed in one gender more than the other? Why or why not?
4. Do you believe that personality disorders can be best understood as a constellation of maladaptive personality traits, or do you think that there is something more involved for individuals suffering from a personality disorder?
5. The authors suggested Clyde Barrow as an example of antisocial personality disorder and Blanche Dubois for histrionic personality disorder. Can you think of a person from the media or literature who would have at least some of the traits of narcissistic personality disorder?

Vocabulary

Antisocial

A pervasive pattern of disregard and violation of the rights of others. These behaviors may be aggressive or destructive and may involve breaking laws or rules, deceit or theft.

Avoidant

A pervasive pattern of social inhibition, feelings of inadequacy, and hypersensitivity to negative evaluation.

Borderline

A pervasive pattern of instability of interpersonal relationships, self-image, and affects, and marked impulsivity.

Dependent

A pervasive and excessive need to be taken care of that leads to submissive and clinging behavior and fears of separation.

Five-Factor Model

Five broad domains or dimensions that are used to describe human personality.

Histrionic

A pervasive pattern of excessive emotionality and attention seeking.

Narcissistic

A pervasive pattern of grandiosity (in fantasy or behavior), need for admiration, and lack of empathy.

Obsessive-compulsive

A pervasive pattern of preoccupation with orderliness, perfectionism, and mental and interpersonal control, at the expense of flexibility, openness, and efficiency.

Paranoid

A pervasive distrust and suspiciousness of others such that their motives are interpreted as malevolent.

Personality

Characteristic, routine ways of thinking, feeling, and relating to others.

Personality disorders

When personality traits result in significant distress, social impairment, and/or occupational impairment.

Schizoid

A pervasive pattern of detachment from social relationships and a restricted range of expression of emotions in interpersonal settings.

Schizotypal

A pervasive pattern of social and interpersonal deficits marked by acute discomfort with, and reduced capacity for, close relationships as well as perceptual distortions and eccentricities of behavior.

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Dissociative Disorders

Dalena van Heugten - van der Kloet

In psychopathology, dissociation happens when thoughts, feelings, and experiences of our consciousness and memory do not collaborate well with each other. This module provides an overview of dissociative disorders, including the definitions of dissociation, its origins and competing theories, and their relation to traumatic experiences and sleep problems.

Learning Objectives

- Define the basic terminology and historical origins of dissociative symptoms and dissociative disorders.
- Describe the posttraumatic model of dissociation and the sleep-dissociation model, and the controversies and debate between these competing theories.
- What is the innovative angle of the sleep-dissociation model?
- How can the two models be combined into one conceptual scheme?

Introduction

Think about the last time you were daydreaming. Perhaps it was while you were driving or attending class. Some portion of your attention was on the activity at hand, but most of your conscious mind was wrapped up in fantasy. Now imagine that you could not control your daydreams. What if they intruded your waking consciousness unannounced, causing you to lose track of reality or experience the loss of time. Imagine how difficult it would be for you. This is similar to what people who suffer from dissociative disorders may experience. Of the

many disorders listed in the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) (American Psychiatric Association, 2013), dissociative disorders rank as among the most puzzling and controversial. Dissociative disorders encompass an array of symptoms ranging from memory loss (**amnesia**) for autobiographical events, to changes in identity and the experience of everyday reality (American Psychiatric Association, 2013).

Is it real?



Dissociative disorders are often exaggerated when portrayed in television, books and movies; however, there is concrete evidence that people do suffer from these disorders. [Image: Library of Congress, <https://goo.gl/BrTchp>, CC BY-NC 3.0, <https://goo.gl/osSgSv>]

Let's start with a little history. Multiple personality disorder, or dissociative identity disorder—as it is known now—used to be a mere curiosity. This is a disorder in which people present with more than one personality. For example, at times they might act and identify as an adult while at other times they might identify and behave like a child. The disorder was rarely diagnosed until the 1980s. That's when multiple personality disorder became an official diagnosis in the DSM-III. From then on, the numbers of “multiples” increased rapidly. In the 1990s, there were hundreds of people diagnosed with multiple personality in every major

city in the United States (Hacking, 1995). How could this “epidemic” be explained?

One possible explanation might be the media attention that was given to the disorder. It all started with the book *The Three Faces of Eve* (Thigpen & Cleckley, 1957). This book, and later the movie, was one of the first to speak of multiple personality disorder. However, it wasn't until years later, when the fictional “as told to” book of *Sybil* (Schreiber, 1973) became known worldwide, that the prototype of what it was like to be a “multiple personality” was born. *Sybil* tells the story of how a clinician—Cornelia Wilbur—unravels the different personalities of her patient Sybil during a long course of treatment (over 2,500 office hours!). She was one of the first to relate multiple personality to childhood sexual abuse. Probably, this relation between childhood abuse and dissociation has fueled the increase of numbers of multiples from that time on. It motivated therapists to actively seek for clues of childhood abuse in their dissociative patients. This suited well within the mindset of the 1980s, as childhood abuse was a sensitive issue then in psychology as well as in politics (Hacking, 1995).

From then on, many movies and books were made on the subject of multiple personality, and nowadays, we see patients with dissociative identity disorder as guests visiting the Oprah Winfrey show, as if they were our modern-day circus acts.

Defining dissociation

The DSM-5 defines dissociation as “a disruption and/or discontinuity in the normal integration of consciousness, memory, identity, emotion, perception, body representation, motor control and behavior” (American Psychiatric Association, 2013, p. 291). A distinction is often made between dissociative *states* and dissociative *traits* (e.g., Bremner, 2010; Bremner & Brett, 1997). State dissociation is viewed as a transient symptom, which lasts for a few minutes or hours (e.g., dissociation during a traumatic event). Trait dissociation is viewed as an integral aspect of personality. Dissociative symptoms occur in patients but also in the general population, like you and me. Therefore, dissociation has commonly been conceptualized as ranging on a continuum, from nonsevere manifestations of daydreaming to more severe disturbances typical of dissociative disorders (Bernstein & Putnam, 1986). The dissociative disorders include:

1. Dissociative Amnesia (extensive forgetting typically associated with highly aversive events);
2. Dissociative Fugue (short-lived reversible amnesia for personal identity, involving unplanned travel or “bewildered wandering.” Dissociative fugue is not viewed as a separate disorder but is a feature of some, but not all, cases of dissociative amnesia);
3. Depersonalization/Derealization Disorder (feeling as though one is an outside observer of one’s body); and
4. Dissociative Identity Disorder (DID; experiencing two or more distinct identities that recurrently take control over one’s behavior) (American Psychiatric Association, 2000).

Although the concept of dissociation lacks a generally accepted definition, the Structural Clinical Interview for DSM-IV Dissociative Disorders (SCID-D) (Steinberg, 2001) assesses five symptom clusters that encompass key features of the dissociative disorders. These clusters are also found in the DSM-5:

1. depersonalization,
2. derealization,
3. dissociative amnesia,
4. identity confusion, and

5. identity alteration.

Depersonalization refers to a “feeling of detachment or estrangement from one’s self.” Imagine that you are outside of your own body, looking at yourself from a distance as though you were looking at somebody else. Maybe you can also imagine what it would be like if you felt like a robot, deprived of all feelings. These are examples of depersonalization. Derealization is defined as “an alteration in the perception of one’s surroundings so that a sense of reality of the external world is lost” (Steinberg, 2001, p. 101). Imagine that the world around you seems as if you are living in a movie, or looking through a fog. These are examples of derealization. Dissociative amnesia does not refer to permanent memory loss, similar to the erasure of a computer disk, but rather to the hypothetical disconnection of memories from conscious inspection (Steinberg, 2001). Thus, the memory is still there somewhere, but you cannot reach it. Identity confusion is defined by Steinberg as “... thoughts and feelings of uncertainty and conflict a person has related to his or her identity” (Steinberg, 2001, p. 101), whereas identity alteration describes the behavioral acting out of this uncertainty and conflict (Bernstein & Putnam, 1986).



Those experiencing depersonalization report “dreamlike feelings” and that their bodies, feelings, emotions, and behaviors are not their own. [Image: Janine, <https://goo.gl/MDpyRG>, CC BY-NC-ND 2.0, <https://goo.gl/62XJAl>]

Dissociative disorders are not as uncommon as you would expect. Several studies in a variety of patient groups show that dissociative disorders are prevalent in a 4%–29% range (Ross, Anderson, Fleischer, & Norton, 1991; Sar, Tutkun, Alyanak, Bakim, & Baral, 2000; Tutkun et al., 1998. For reviews see: Foote, Smolin, Kaplan, Legatt, & Lipschitz, 2006; Spiegel et al., 2011). Studies generally find a much lower **prevalence** in the general population, with rates in the order of 1%–3% (Lee, Kwok, Hunter, Richards, & David, 2010; Rauschenberger & Lynn, 1995; Sandberg & Lynn, 1992). Importantly, dissociative symptoms are not limited to the dissociative disorders. Certain diagnostic groups,

notably patients with borderline personality disorder, posttraumatic stress disorder (PTSD), **obsessive-compulsive disorder** (Rufer, Fricke, Held, Cremer, & Hand, 2006), and schizophrenia (Allen & Coyne, 1995; Merckelbach, à Campo, Hardy, & Giesbrecht, 2005; Yu et al., 2010) also display heightened levels of dissociation.

Measuring dissociation

The Dissociative Experiences Scale (**DES**) (Bernstein & Putnam, 1986; Carlson & Putnam, 2000; Wright & Loftus, 1999) is the most widely used self-report measure of dissociation. A self-report measure is a type of **psychological test** in which a person completes a survey or **questionnaire** with or without the help of an investigator. This scale measures dissociation with items such as (a) "Some people sometimes have the experience of feeling as though they are standing next to themselves or watching themselves do something, and they actually see themselves as if they were looking at another person" and (b) "Some people find that sometimes they are listening to someone talk, and they suddenly realize that they did not hear part or all of what was said."

The DES is suitable only as a screening tool. When somebody scores a high level of dissociation on this scale, this does not necessarily mean that he or she is suffering from a dissociative disorder. It does, however, give an indication to investigate the symptoms more extensively. This is usually done with a structured clinical interview, called the Structured Clinical Interview for DSM-IV Dissociative Disorders (Steinberg, 1994), which is performed by an experienced clinician. With the publication of the new DSM-5 there has been an updated version of this instrument.

Dissociation and Trauma

The most widely held perspective on dissociative symptoms is that they reflect a defensive response to highly aversive events, mostly **trauma** experiences during the childhood years (Bremner, 2010; Spiegel et al., 2011; Spitzer, Vogel, Barnow, Freyberger, & Grabe, 2007).

One prominent interpretation of the origins of dissociative disorders is that they are the direct result of exposure to traumatic experiences. We will refer to this interpretation as the posttraumatic model (**PTM**). According to the PTM, dissociative symptoms can best be understood as mental strategies to cope with or avoid the impact of highly aversive experiences (e.g., Spiegel et al., 2011). In this view, individuals rely on dissociation to escape from painful memories (Gershuny & Thayer, 1999). Once they have learned to use this **defensive coping mechanism**, it can become automatized and habitual, even emerging in response to minor stressors (Van der Hart & Horst, 1989). The idea that dissociation can serve a defensive function can be traced back to Pierre Janet (1899/1973), one of the first scholars to link dissociation to psychological trauma (Hacking, 1995).

The PTM casts the clinical observation that dissociative disorders are linked to a trauma history

in straightforward causal terms, that is, one *causes* the other (Gershuny & Thayer, 1999). For example, Vermetten and colleagues (Vermetten, Schmahl, Lindner, Loewenstein, & Bremner, 2006) found that the DID patients in their study all suffered from posttraumatic stress disorder and concluded that DID should be conceptualized as an extreme form of early-abuse-related posttraumatic stress disorder (Vermetten et al., 2006).

Causality and evidence

The empirical evidence that trauma *leadsto* dissociative symptoms is the subject of intense debate (Kihlstrom, 2005; Bremner, 2010; Giesbrecht, Lynn, Lilienfeld & Merckelbach, 2010). Three limitations of the PTM will be described below.

First, the majority of studies reporting links between self-reported trauma and dissociation are based on cross-sectional designs. This means that the data are collected at one point in time. When analyzing this type of data, one can only state whether scoring high on a particular questionnaire (for example, a trauma questionnaire) is indicative of also scoring high on another questionnaire (for example, the DES). This makes it difficult to state if one thing led to another, and therefore if the relation between the two is *causal*. Thus, the data that these designs yield do not allow for strong causal claims (Merckelbach & Muris, 2002).

Second, whether somebody has experienced a trauma is often established using a questionnaire that the person completes himself or herself. This is called a self-report measure. Herein lies the problem. Individuals suffering from dissociative symptoms typically have high fantasy proneness. This is a character trait to engage in extensive and vivid fantasizing. The tendency to fantasize a lot may increase the risk of exaggerating or understating self-reports of traumatic experiences (Merckelbach et al., 2005; Giesbrecht, Lynn, Lilienfeld, & Merckelbach, 2008).

Third, high dissociative individuals report more cognitive failures than low dissociative individuals. Cognitive failures are everyday slips and lapses, such as failing to notice signposts on the road, forgetting appointments, or bumping into people. This can be seen, in part, in the DSM-5 criteria for DID, in which people may have difficulty recalling everyday events as well as those that are traumatic. People who frequently make such slips and lapses often mistrust their own cognitive capacities. They also tend to overvalue the hints and cues provided by others (Merckelbach, Horselenberg, & Schmidt, 2002; Merckelbach, Muris, Rassin, & Horselenberg, 2000). This makes them vulnerable to suggestive information, which may distort self-reports, and thus limits conclusions that can be drawn from studies that rely solely on self-reports to investigate the trauma-dissociation link (Merckelbach & Jelicic, 2004).

Most important, however, is that the PTM does not tell us *how* trauma produces dissociative symptoms. Therefore, workers in the field have searched for other explanations. They proposed that due to their dreamlike character, dissociative symptoms such as derealization, depersonalization, and absorption are associated with sleep-related experiences. They further noted that sleep-related experiences can explain the relation between highly aversive events and dissociative symptoms (Giesbrecht et al., 2008; Watson, 2001). In the following paragraph, the relation between dissociation and sleep will be discussed.

Dissociation and Sleep

A little history



Those who have fallen asleep in class or on the bus have likely experienced those “micro-dreams” - that moment or two where reality kind of blends in with your dreams. For a long time, scientists thought dissociative disorders were simply this confusion of waking and dreaming states. [Image: kooklanekookla, <https://goo.gl/Cn3xul>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Researchers (Watson, 2001) have proposed that dissociative symptoms, such as absorption, derealization, and depersonalization originate from sleep. This idea is not entirely new. In the 19th century, double consciousness (or *dédoublement*), the historical precursor of dissociative identity disorder (DID; formerly known as multiple personality disorder), was often described as “somnambulism,” which refers to a state of sleepwalking. Patients suffering from this disorder were referred to as “somnambules” (Hacking, 1995). Many 19th-century scholars believed that these patients were switching between a “normal state” and a “somnambulistic state.” Hughlings Jackson, a well-known English neurologist from this era, viewed dissociation as the uncoupling of normal consciousness, which would result in what he termed “the dreamy state” (Meares, 1999). Interestingly, a century later,

Levitan (1967) hypothesized that “depersonalization is a compromise state between dreaming and waking” (p.157). Arlow (1966) observed that the dissociation between the “experiencing self” and the “observing self” serves as the basis of depersonalized states, emphasizing its occurrence, especially in dreams. Likewise, Franklin (1990) considered dreamlike thoughts,

the amnesia one usually has for dreams, and the lack of orientation of time, place, and person during dreams to be strikingly similar to the amnesia DID patients often report for their traumas. Related, Barrett (1994, 1995) described the similarity between dream characters and “alter personalities” in DID, with respect to cognitive and sensory abilities, movement, amnesia, and continuity with normal waking. The many similarities between dreaming states and dissociative symptoms are also a recurrent theme in the more recent clinical literature (e.g., Bob, 2004).

Sleep problems in patients with dissociative disorders

Anecdotal evidence supports the idea that sleep disruptions are linked to dissociation. For example, in patients with depersonalization, symptoms are worst when they are tired (Simeon & Abugel, 2006). Interestingly, among participants who report memories of childhood sexual abuse, experiences of sleep paralysis typically are accompanied by raised levels of dissociative symptoms (McNally & Clancy, 2005; Abrams, Mulligan, Carleton, & Asmundson, 2008).

Patients with mood disorders, anxiety disorders, schizophrenia, and borderline personality disorder—conditions with relatively high levels of dissociative symptoms—as a rule exhibit sleep abnormalities. Recent research points to fairly specific relationships between certain sleep complaints (e.g., insomnia, nightmares) and certain forms of psychopathology (e.g., depression, posttraumatic stress disorder) (Koffel & Watson, 2009).

Studying the relationship between dissociation and sleep

In the general population, both dissociative symptoms and sleep problems are highly prevalent. For example, 29 percent of American adults report sleep problems (National Sleep Foundation, 2005). This allows researchers to study the relationship between dissociation and sleep not only in patients but also in the general population. In a pioneering study, Watson (2001) showed that dissociative symptoms—measured by the DES—are linked to self-reports of vivid dreams, nightmares, recurrent dreams, and other unusual sleep phenomena. This relationship has been studied extensively ever since, leading to three important statements.

First, Watson’s (2001) basic findings have been reproduced time and again. This means that the same results (namely that dissociation and sleep problems are related) have been found in lots of different studies, using different groups, and different materials. All lead to the conclusion that unusual sleep experiences and dissociative symptoms are linked.

Second, the connection between sleep and dissociation is specific. It seems that unusual sleep

phenomena that are difficult to control, including nightmares and waking dreams, are related to dissociative symptoms, but lucid dreaming—dreams that are controllable—are only weakly related to dissociative symptoms. For example, dream recall frequency was related to dissociation (Suszek & Kopera, 2005). Individuals who reported three or more nightmares over a three-week period showed higher levels of dissociation compared to individuals reporting two nightmares or less (Levin & Fireman, 2002), and a relation was found between dream intensity and dissociation (Yu et al., 2010).

Third, the sleep-dissociation link is apparent not only in general population groups—people such as you and me—but also in patient groups. Accordingly, one group of researchers reported nightmare disorder in 17 out of 30 DID patients (Agargun et al., 2003). They also found a 27.5% prevalence of nocturnal dissociative episodes in patients with dissociative disorders (Agargun et al., 2001). Another study investigated a group of borderline personality disorder patients and found that 49% of them suffered from nightmare disorder. Moreover, the patients with nightmare disorder displayed higher levels of dissociation than patients not suffering from nightmare disorder (Semiz, Basoglu, Ebrinc, & Cetin, 2008). Additionally, Ross (2011) found that patients suffering from DID reported higher rates of sleepwalking compared to a group of psychiatric outpatients and a sample from the general population.

To sum up, there seems to be a strong relationship between dissociative symptoms and unusual sleep experiences that is evident in a range of phenomena, including waking dreams, nightmares, and sleepwalking.

Inducing and reducing sleep problems

Sleep problems can be induced in healthy participants by keeping them awake for a long duration of time. This is called **sleep deprivation**. If dissociative symptoms are fueled by a labile sleep-wake cycle, then sleep loss would be expected to intensify dissociative symptoms. Some evidence that this might work was already found in 2001, when soldiers who underwent a U.S. Army survival training, which included sleep deprivation, showed increases in dissociative symptoms (Morgan et al., 2001). Other researchers conducted a study that tracked 25 healthy volunteers during one day and one night of sleep loss. They found that dissociative symptoms increased substantially after one night of sleep loss (Giesbrecht, Smeets, Leppink, Jellic, & Merckelbach, 2007).

To further examine the causal link between dissociative experiences and sleep, we (van der Kloet, Giesbrecht, Lynn, Merckelbach, & de Zutter, 2011) investigated the relationship between unusual sleep experiences and dissociation in a patient group at a private clinic. They

completed questionnaires upon arrival at the clinic and again when they departed eight weeks later. During their stay, they followed a strict program designed to improve sleep problems. And it worked! In most patients, sleep quality was improved after eight weeks. We found a robust link between sleep experiences and dissociative symptoms and determined that sleep *normalization* was accompanied by a *reduction* in dissociative symptoms.

An exciting interpretation of the link between dissociative symptoms and unusual sleep phenomena (see also, Watson, 2001) may be this: A disturbed sleep–wake cycle may lead to dissociative symptoms. However, we should be cautious. Although studies support a causal arrow leading from sleep disruption to dissociative symptoms, the associations between sleep and dissociation may be more complex. For example, causal links may be bi-directional, such that dissociative symptoms may lead to sleep problems and vice versa, and other psychopathology may interfere in the link between sleep and dissociative symptoms (van der Kloet et al., 2011).

Implications and Conclusions



No longer are dissociative disorders untreatable illnesses. With the new methods developed by researchers, there is hope for curing or at least improving the lives of those with this debilitating disorder. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

The sleep-dissociation model offers a fresh and exciting perspective on dissociative symptoms. This model may seem remote from the PTM. However, both models can be integrated in a single conceptual scheme in which traumatic childhood experiences may lead to disturbed sleep patterns, which may be the final common pathway to dissociative symptoms. Accordingly, the sleep-dissociation model may explain both: (a) how traumatic experiences disrupt the sleep–wake cycle and increase vulnerability to dissociative symptoms, and (b) why dissociation, trauma, fantasy proneness, and cognitive failures overlap.

Future studies can also discern what characteristic sleep disruptions in the sleep–wake cycle are most reliably related to dissociative disorders, and then establish training programs, including medication regimens, to address these problems. This would constitute an entirely novel and exciting approach to the treatment of dissociative symptoms.

In closing, the sleep-dissociation model can serve as a framework for studies that address a wide range of fascinating questions about dissociative symptoms and disorders. We now have good reason to be confident that research on sleep and dissociative symptoms will inform psychiatry, clinical science, and psychotherapeutic practice in meaningful ways in the years to come.

Outside Resources

Article: Extreme Dissociative Fugue: A life, Interrupted - A recent case of extreme dissociative fugue. The article is particularly powerful as it relates the story of a seemingly typical person, a young teacher, who suddenly experiences a dissociative fugue.

http://www.nytimes.com/2009/03/01/nyregion/thecity/01miss.html?_r=0

Book: Schreiber, F. R. (1973). *Sybil*. Chicago: Regnery.

Film: Debate Persists Over Diagnosing Mental Health Disorders, Long After 'Sybil'. This short film would be useful to provide students with perspectives on the debate over diagnoses. It could be used to introduce the debate and provide students with evidence to argue for or against the diagnosis.

<http://www.nytimes.com/2014/11/24/us/debate-persists-over-diagnosing-mental-health-disorders-long-after-sybil.html>

Structured Clinical Interview for DSM-5 (SCID-5)

<https://www.appi.org/products/structured-clinical-interview-for-dsm-5-scid-5>

Video: Depiction of the controversy regarding the existence of DID and show you some debate between clinicians and researchers on the topics of brain imaging, recovered memories, and false memories. False memory syndrome.

http://www.youtube.com/watch?v=K4NZ7_Hn-rl

Video: Patient Switching on Command and in Brain Scanner - This eight-minute video depicts the controversy regarding the existence of DID and relates some of the debate between clinicians and researchers on the topics of brain imaging, recovered memories, and false memories.

<http://www.youtube.com/watch?v=zhM0xp5vXqY>

Discussion Questions

1. Why are dissociation and trauma related to each other?
2. How is dissociation related to sleep problems?
3. Are dissociative symptoms induced or merely increased by sleep disturbances?
4. Do you have any ideas regarding treatment possibilities for dissociative disorders?

5. Does DID really exist?

Vocabulary

Amnesia

The loss of memory.

Anxiety disorder

A group of diagnoses in the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV-TR) classification system where anxiety is central to the person's dysfunctioning. Typical symptoms include excessive rumination, worrying, uneasiness, apprehension, and fear about future uncertainties either based on real or imagined events. These symptoms may affect both physical and psychological health. The anxiety disorders are subdivided into panic disorder, specific phobia, social phobia, posttraumatic stress disorder, obsessive-compulsive disorder, and generalized anxiety disorder.

Borderline Personality Disorder

This personality disorder is defined by a chronic pattern of instability. This instability manifests itself in interpersonal relationships, mood, self-image, and behavior that can interfere with social functioning or work. It may also cause grave emotional distress.

Cognitive failures

Every day slips and lapses, also called absentmindedness.

Consciousness

The quality or state of being aware of an external object or something within oneself. It has been defined as the ability to experience or to feel, wakefulness, having a sense of selfhood, and the executive control system of the mind.

Cross-sectional design

Research method that involves observation of all of a population, or a representative subset, at one specific point in time.

Defensive coping mechanism

An unconscious process, which protects an individual from unacceptable or painful ideas, impulses, or memories.

DES

Dissociative Experiences Scale.

DID

Dissociative identity disorder, formerly known as multiple personality disorder, is at the far end of the dissociative disorder spectrum. It is characterized by at least two distinct, and dissociated personality states. These personality states – or ‘alters’ - alternately control a person’s behavior. The sufferer therefore experiences significant memory impairment for important information not explained by ordinary forgetfulness.

Dissociation

A disruption in the usually integrated function of consciousness, memory, identity, or perception of the environment.

Fantasy proneness

The tendency to extensive fantasizing or daydreaming.

General population

A sample of people representative of the average individual in our society.

Insomnia

A sleep disorder in which there is an inability to fall asleep or to stay asleep as long as desired. Symptoms also include waking up too early, experience many awakenings during the night, and not feeling rested during the day.

Lucid dreams

Any dream in which one is aware that one is dreaming.

Mood disorder

A group of diagnoses in the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV-TR) classification system where a disturbance in the person’s mood is the primary dysfunction. Mood disorders include major depressive disorder, bipolar disorder, dysthymic and cyclothymic disorder.

Nightmares

An unpleasant dream that can cause a strong negative emotional response from the mind, typically fear or horror, but also despair, anxiety, and great sadness. The dream may contain situations of danger, discomfort, psychological or physical terror. Sufferers usually awaken in a state of distress and may be unable to return to sleep for a prolonged period of time.

Obsessive-Compulsive Disorder

This anxiety disorder is characterized by intrusive thoughts (obsessions), by repetitive

behaviors (compulsions), or both. Obsessions produce uneasiness, fear, or worry. Compulsions are then aimed at reducing the associated anxiety. Examples of compulsive behaviors include excessive washing or cleaning; repeated checking; extreme hoarding; and nervous rituals, such as switching the light on and off a certain number of times when entering a room. Intrusive thoughts are often sexual, violent, or religious in nature...

Prevalence

The number of cases of a specific disorder present in a given population at a certain time.

PTM

Post-traumatic model of dissociation.

Recurrent dreams

The same dream narrative or dreamscape is experienced over different occasions of sleep.

Schizophrenia

This mental disorder is characterized by a breakdown of thought processes and emotional responses. Symptoms include auditory hallucinations, paranoid or bizarre delusions, or disorganized speech and thinking. Sufferers from this disorder experience grave dysfunctions in their social functioning and in work.

SCID-D

Structural Clinical Interview for DSM-IV Dissociative Disorders.

Self-report measure

A type of psychological test in which a person fills out a survey or questionnaire with or without the help of an investigator.

Sleep deprivation

A sufficient lack of restorative sleep over a cumulative period so as to cause physical or psychiatric symptoms and affect routine performances of tasks.

Sleep paralysis

Sleep paralysis occurs when the normal paralysis during REM sleep manifests when falling asleep or awakening, often accompanied by hallucinations of danger or a malevolent presence in the room.

Sleep-wake cycle

A daily rhythmic activity cycle, based on 24-hour intervals, that is exhibited by many organisms.

State

When a symptom is acute, or transient, lasting from a few minutes to a few hours.

Trait

When a symptom forms part of the personality or character.

Trauma

An event or situation that causes great distress and disruption, and that creates substantial, lasting damage to the psychological development of a person.

Vivid dreams

A dream that is very clear, where the individual can remember the dream in great detail.

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Autism: Insights from the Study of the Social Brain

Kevin A. Pelphrey

People with autism spectrum disorder (ASD) suffer from a neurodevelopmental disorder. Social neuroscience is the study of the parts of the brain that support social interactions or the “social brain.” This module provides an overview of ASD and focuses on understanding how social brain dysfunction leads to ASD. Our increasing understanding of the social brain and its dysfunction in ASD will allow us to better identify the genes that cause ASD and will help us to create and pick out treatments to better match individuals. Because social brain systems emerge in infancy, social neuroscience can help us to figure out how to diagnose ASD even before the symptoms of ASD are clearly present. This is a hopeful time because social brain systems remain malleable well into adulthood and thus open to creative new interventions that are informed by state-of-the-art science.

Learning Objectives

- Know the basic symptoms of ASD.
- Distinguish components of the social brain and understand their dysfunction in ASD.
- Appreciate how social neuroscience may facilitate the diagnosis and treatment of ASD.

Defining Autism Spectrum Disorder

Autism Spectrum Disorder (ASD) is a developmental disorder that usually emerges in the first

three years and persists throughout the individual's life. Though the key symptoms of ASD fall into three general categories (see below), each person with ASD exhibits symptoms in these domains in different ways and to varying degrees. This *phenotypic heterogeneity* reflects the high degree of variability in the genes underlying ASD (Geschwind & Levitt, 2007). Though we have identified genetic differences associated with individual cases of ASD, each accounts for only a small number of the actual cases, suggesting that no single genetic cause will apply in the majority of people with ASD. There is currently no biological test for ASD.

Autism is in the category of *neurodevelopmental disorders*, which includes Intellectual Disabilities, Autism Spectrum Disorder, Attention-Deficit/Hyperactivity Disorder (ADHD), and Learning Disorders, among others. *Autism spectrum disorder* (ASD) is defined by the presence of profound difficulties in social interactions and communication combined with the presence of repetitive or restricted interests, cognitions and behaviors. The diagnostic process involves a combination of parental report and clinical observation. Children with significant

impairments across the social/communication domain who also exhibit repetitive behaviors can qualify for the ASD diagnosis. There is wide variability in the precise symptom profile an individual may exhibit.

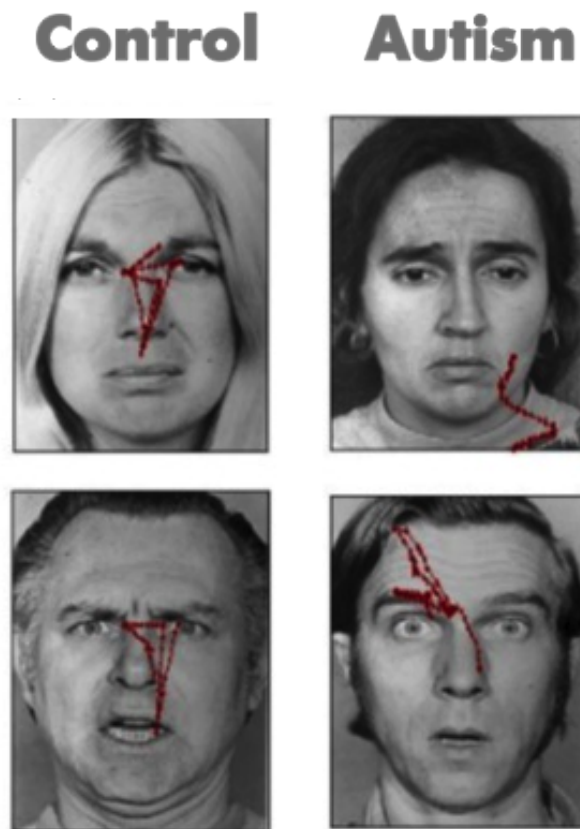


Figure 1. The red lines indicate the scan paths (collection of eye movements) used by people with (right column) and without (left column) autism to explore faces. Modified from Pelphrey et al., (2002).

Since Kanner first described ASD in 1943, important commonalities in symptom presentation have been used to compile criteria for the diagnosis of ASD. These diagnostic criteria have evolved during the past 70 years and continue to evolve (e.g., see the recent changes to the diagnostic criteria on the American Psychiatric Association's website, <http://www.dsm5.org/>), yet impaired social functioning remains a required symptom for an ASD diagnosis. Deficits in social functioning are present in varying degrees for simple behaviors such as eye contact, and complex behaviors like navigating the give and take of a group conversation for individuals of all functioning levels (i.e.

high or low IQ). Moreover, difficulties with social information processing occur in both visual (e.g., Pelphrey et al., 2002) and auditory (e.g., Dawson, Meltzoff, Osterling, Rinaldi, & Brown, 1998) sensory modalities.

Consider the results of an eye tracking study in which Pelphrey and colleagues (2002) observed that individuals with autism did not make use of the eyes when judging facial expressions of emotion (see right panels of Figure 1). While repetitive behaviors or language deficits are seen in other disorders (e.g., obsessive-compulsive disorder and specific language impairment, respectively), basic social deficits of this nature are unique to ASD. Onset of the social deficits appears to precede difficulties in other domains (Osterling, Dawson, & Munson, 2002) and may emerge as early as 6 months of age (Maestro et al., 2002).

Defining the Social Brain

Within the past few decades, research has elucidated specific brain circuits that support perception of humans and other species. This *social perception* refers to “the initial stages in the processing of information that culminates in the accurate analysis of the dispositions and intentions of other individuals” (Allison, Puce, & McCarthy, 2000). Basic social perception is a critical building block for more sophisticated social behaviors, such as thinking about the motives and emotions of others. Brothers (1990) first suggested the notion of a social brain, a set of interconnected neuroanatomical structures that process social information, enabling the recognition of other individuals and the evaluation their mental states (e.g., intentions, dispositions, desires, and beliefs).

The social brain is hypothesized to consist of the amygdala, the orbital frontal cortex (OFC), fusiform gyrus (FG), and the posterior superior temporal sulcus (STS) region, among other structures. Though all areas work in coordination to support social processing, each appears to serve a distinct role. The amygdala helps us recognize the emotional states of others (e.g., Morris et al., 1996) and also to experience and regulate our own emotions (e.g., LeDoux, 1992). The OFC supports the “reward” feelings we have when we are around other people (e.g., Rolls, 2000). The FG, located at the bottom of the surface of the temporal lobes detects faces and supports face recognition (e.g., Puce, Allison, Asgari, Gore, & McCarthy, 1996). The posterior STS region recognizes the biological motion, including eye, hand and other body movements, and helps to interpret and predict the actions and intentions of others (e.g., Pelphrey, Morris, Michelich, Allison, & McCarthy, 2005).

Current Understanding of Social Perception in ASD

The social brain is of great research interest because the social difficulties characteristic of ASD are thought to relate closely to the functioning of this brain network. Functional magnetic resonance imaging (fMRI) and event-related potentials (ERP) are complementary brain imaging methods used to study activity in the brain across the lifespan. Each method measures a distinct facet of brain activity and contributes unique information to our understanding of brain function.

fMRI uses powerful magnets to measure the levels of oxygen within the brain, which vary according to changes in neural activity. As the neurons in specific brain regions “work harder”, they require more oxygen. fMRI detects the brain regions that exhibit a relative increase in blood flow (and oxygen levels) while people listen to or view social stimuli in the MRI scanner. The areas of the brain most crucial for different social processes are thus identified, with spatial information being accurate to the millimeter.

In contrast, ERP provides direct measurements of the firing of groups of neurons in the cortex. Non-invasive sensors on the scalp record the small electrical currents created by this neuronal activity while the subject views stimuli or listens to specific kinds of information. While fMRI provides information about *where* brain activity occurs, ERP specifies *when* by detailing the timing of processing at the millisecond pace at which it unfolds.

ERP and fMRI are complementary, with fMRI providing excellent *spatial resolution* and ERP offering outstanding *temporal resolution*. Together, this information is critical to understanding the nature of social perception in ASD. To date, the most thoroughly investigated areas of the social brain in ASD are the superior temporal sulcus (STS), which underlies the perception and interpretation of biological motion, and the fusiform gyrus (FG), which supports face perception. Heightened sensitivity to biological motion (for humans, motion such as walking) serves an essential role in the development of humans and other highly social species. Emerging in the first days of life, the ability to detect biological motion helps to orient vulnerable young to critical sources of sustenance, support, and learning, and develops independent of visual experience with biological motion (e.g., Simion, Regolin, & Bulf, 2008).



The human brain has specialized functions to help guide our social interactions. [Image: Allan Ajifo, <https://goo.gl/jv4iXf>, CC BY 2.0, <https://goo.gl/BRvSA7>]

This inborn “life detector” serves as a foundation for the subsequent development of more complex social behaviors (Johnson, 2006).



From an evolutionary standpoint, it was incredibly important for our survival to maintain social relationships. Therefore, it makes sense that we would be able to recognize faces within the first few days of our infancy. [Image: donnierayjones, <https://goo.gl/obr12x>, CC BY 2.0, <https://goo.gl/v4Y0Zv>]

three months of age (e.g., de Haan, Johnson, & Halit, 2003) and continues throughout the lifespan (e.g., Bentin et al., 1996). Children with ASD, however, tend to show decreased attention to human faces by six to 12 months (Osterling & Dawson, 1994). Children with ASD also show reduced activity in the FG when viewing faces (e.g., Schultz et al., 2000). Slowed processing of faces (McPartland, Dawson, Webb, Panagiotides, & Carver, 2004) is a characteristic of people with ASD that is shared by parents of children with ASD (Dawson, Webb, & McPartland, 2005) and infants at increased risk for developing ASD because of having a sibling with ASD (McCleery, Akshoomoff, Dobkins, & Carver, 2009). Behavioral and attentional differences in face perception and recognition are evident in children and adults with ASD as well (e.g., Hobson, 1986).

Exploring Diversity in ASD

Because of the limited quality of the behavioral methods used to diagnose ASD and current clinical diagnostic practice, which permits similar diagnoses despite distinct symptom profiles

From very early in life, children with ASD display reduced sensitivity to biological motion (Klin, Lin, Gorrindo, Ramsay, & Jones, 2009). Individuals with ASD have reduced activity in the STS during biological motion perception. Similarly, people at increased genetic risk for ASD but who do not develop symptoms of the disorder (i.e. unaffected siblings of individuals with ASD) show increased activity in this region, which is hypothesized to be a compensatory mechanism to offset genetic vulnerability (Kaiser et al., 2010).

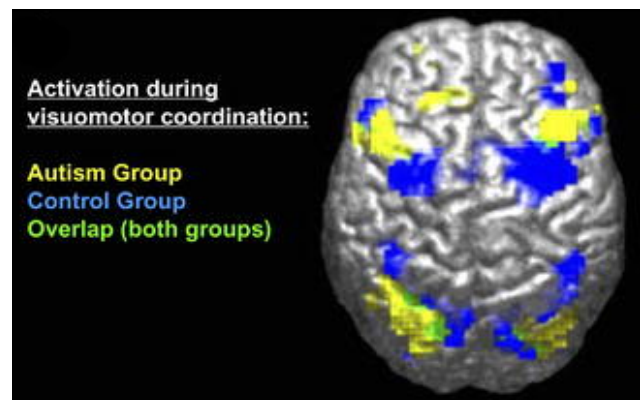
In typical development, preferential attention to faces and the ability to recognize individual faces emerge in the first days of life (e.g., Goren, Sarty, & Wu, 1975). The special way in which the brain responds to faces usually emerges by

(McPartland, Webb, Keehn, & Dawson, 2011), it is possible that the group of children currently referred to as having ASD may actually represent different syndromes with distinct causes. Examination of the social brain may well reveal diagnostically meaningful subgroups of children with ASD. Measurements of the “where” and “when” of brain activity during social processing tasks provide reliable sources of the detailed information needed to profile children with ASD with greater accuracy. These profiles, in turn, may help to inform treatment of ASD by helping us to match specific treatments to specific profiles.

The integration of imaging methods is critical for this endeavor. Using face perception as an example, the combination of fMRI and ERP could identify who, of those individuals with ASD, shows anomalies in the FG and then determine the stage of information processing at which these impairments occur. Because different processing stages often reflect discrete cognitive processes, this level of understanding could encourage treatments that address specific processing deficits at the neural level.

For example, differences observed in the early processing stages might reflect problems with low-level visual perception, while later differences would indicate problems with higher-order processes, such as emotion recognition. These same principles can be applied to the broader network of social brain regions and, combined with measures of behavioral functioning, could offer a comprehensive profile of brain-behavior performance for a given individual. A fundamental goal for this kind of subgroup approach is to improve the ability to tailor treatments to the individual.

Another objective is to improve the power of other scientific tools. Most studies of individuals with ASD compare groups of individuals, for example, individuals on with ASD compared to typically developing peers. However, studies have also attempted to compare children across the autism spectrum by group according to differential diagnosis (e.g., Asperger’s disorder versus autistic disorder), or by other behavioral or cognitive characteristics (e.g., cognitively able versus intellectually disabled or anxious versus non-anxious). Yet, the power of a scientific study to detect these kinds of significant, meaningful, individual differences is only as strong as



Trying to diagnose the precise autism disorder can be difficult; many cases share similar symptoms. However, burgeoning technology, like the fMRI, allows clinicians a glimpse into the patient’s brain and thus a better understanding of his or her disorder. [Image: Ralph-Axel Müller, <https://goo.gl/WwxCV1>, CC BY 2.5, <https://goo.gl/0QtWcf>]

the accuracy of the factor used to define the compared groups.

The identification of distinct subgroups within the autism spectrum according to information about the brain would allow for a more accurate and detailed exposition of the individual differences seen in those with ASD. This is especially critical for the success of investigations into the genetic basis of ASD. As mentioned before, the genes discovered thus far account for only a small portion of ASD cases. If meaningful, quantitative distinctions in individuals with ASD are identified; a more focused examination into the genetic causes specific to each subgroup could then be pursued. Moreover, distinct findings from neuroimaging, or *biomarkers*, can help guide genetic research. **Endophenotypes**, or characteristics that are not immediately available to observation but that reflect an underlying genetic liability for disease, expose the most basic components of a complex psychiatric disorder and are more stable across the lifespan than observable behavior (Gottesman & Shields, 1973). By describing the key characteristics of ASD in these objective ways, neuroimaging research will facilitate identification of genetic contributions to ASD.

Atypical Brain Development Before the Emergence of Atypical Behavior

Because autism is a developmental disorder, it is particularly important to diagnose and treat ASD early in life. Early deficits in attention to biological motion, for instance, derail subsequent experiences in attending to higher level social information, thereby driving development toward more severe dysfunction and stimulating deficits in additional domains of functioning, such as language development. The lack of reliable predictors of the condition during the first year of life has been a major impediment to the effective treatment of ASD. Without early predictors, and in the absence of a firm diagnosis until behavioral symptoms emerge, treatment is often delayed for two or more years, eclipsing a crucial period in which intervention may be particularly successful in ameliorating some of the social and communicative impairments seen in ASD.

In response to the great need for sensitive (able to identify subtle cases) and specific (able to distinguish autism from other disorders) early indicators of ASD, such as biomarkers, many research groups from around the world have been studying patterns of infant development using prospective longitudinal studies of infant siblings of children with ASD and a comparison group of infant siblings without familial risks. Such designs gather longitudinal information about developmental trajectories across the first three years of life for both groups followed by clinical diagnosis at approximately 36 months.

These studies are problematic in that many of the social features of autism do not emerge in



If autism is diagnosed early enough, treatments have developed to the point that children with ASD can learn and grow to have more intensive social interactions. [Image: hepingting, <https://goo.gl/TloAcY>, CC BY-SA 2.0, <https://goo.gl/rxiUsF>]

typical development until after 12 months of age, and it is not certain that these symptoms will manifest during the limited periods of observation involved in clinical evaluations or in pediatricians' offices. Moreover, across development, but especially during infancy, behavior is widely variable and often unreliable, and at present, behavioral observation is the only means to detect symptoms of ASD and to confirm a diagnosis. This is quite problematic because, even highly sophisticated behavioral methods, such as eye tracking (see Figure 1), do not necessarily reveal reliable differences in infants with ASD (Ozonoff et al., 2010). However, measuring the brain activity associated with social

perception can detect differences that do not appear in behavior until much later. The identification of biomarkers utilizing the imaging methods we have described offers promise for earlier detection of atypical social development.

ERP measures of brain response predict subsequent development of autism in infants as young as six months old who showed normal patterns of visual fixation (as measured by eye tracking) (Elsabbagh et al., 2012). This suggests the great promise of brain imaging for earlier recognition of ASD. With earlier detection, treatments could move from addressing existing symptoms to preventing their emergence by altering the course of abnormal brain development and steering it toward normality.

Hope for Improved Outcomes

The brain imaging research described above offers hope for the future of ASD treatment. Many of the functions of the social brain demonstrate significant *plasticity*, meaning that their functioning can be affected by experience over time. In contrast to theories that suggest difficulty processing complex information or communicating across large expanses of cortex (Minshew & Williams, 2007), this malleability of the social brain is a positive prognosticator for the development of treatment. The brains of people with ASD are not wired to process optimally social information. But this does not mean that these systems are irretrievably *broken*. Given the observed plasticity of the social brain, remediation of these difficulties may

be possible with appropriate and timely intervention.

What's more, being diagnosed on the autism spectrum does not necessarily mean that a person cannot have a fulfilling life. The social environment in which a person lives, especially social support from parents, friends, and instructors, can have a positive impact on the lives of people diagnosed with autism. Although research reveals that employment is lower among young adults with autism, the social environment of a job can positively impact their social development (Taylor et al, 2014). Similarly, research often reveals that it is not the symptoms of autism itself that interferes with a person's wellbeing, but the bullying, stigma, and concealment of symptoms that sometimes accompanies autism (Hong et al, 2016). Taken together, these lines of research suggest that supportive social environments are crucial for the wellbeing of those with autism.

Outside Resources

Web: American Psychiatric Association's website for the 5th edition of the Diagnostic and Statistical Manual of Mental Disorders

<http://www.dsm5.org>

Web: Autism Science Foundation - organization supporting autism research by providing funding and other assistance to scientists and organizations conducting, facilitating, publicizing and disseminating autism research. The organization also provides information about autism to the general public and serves to increase awareness of autism spectrum disorders and the needs of individuals and families affected by autism.

<http://www.autismsciencefoundation.org/>

Web: Autism Speaks - Autism science and advocacy organization

<http://www.autismspeaks.org/>

Discussion Questions

1. How can neuroimaging inform our understanding of the causes of autism?
2. What are the ways in which neuroimaging, including fMRI and ERP, may benefit efforts to diagnosis and treat autism?
3. How can an understanding of the social brain help us to understand ASD?
4. What are the core symptoms of ASD, and why is the social brain of particular interest?
5. What are some of the components of the social brain, and what functions do they serve?

Vocabulary

Endophenotypes

A characteristic that reflects a genetic liability for disease and a more basic component of a complex clinical presentation. Endophenotypes are less developmentally malleable than overt behavior.

Event-related potentials (ERP)

Measures the firing of groups of neurons in the cortex. As a person views or listens to specific types of information, neuronal activity creates small electrical currents that can be recorded from non-invasive sensors placed on the scalp. ERP provides excellent information about the timing of processing, clarifying brain activity at the millisecond pace at which it unfolds.

Functional magnetic resonance imaging (fMRI)

Entails the use of powerful magnets to measure the levels of oxygen within the brain that vary with changes in neural activity. That is, as the neurons in specific brain regions “work harder” when performing a specific task, they require more oxygen. By having people listen to or view social percepts in an MRI scanner, fMRI specifies the brain regions that evidence a relative increase in blood flow. In this way, fMRI provides excellent spatial information, pinpointing with millimeter accuracy, the brain regions most critical for different social processes.

Social brain

The set of neuroanatomical structures that allows us to understand the actions and intentions of other people.

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Psychopathy

Chris Patrick

Psychopathy (or “psychopathic personality”) is a topic that has long fascinated the public at large as well as scientists and clinical practitioners. However, it has also been subject to considerable confusion and scholarly debate over the years. This module reviews alternative conceptions of psychopathy that have been proposed historically, and reviews major instruments currently in use for the assessment of psychopathic tendencies in clinical and nonclinical samples. An integrative theoretic framework, the Triarchic model, is presented that provides a basis for reconciling differing historic conceptions and assessment approaches. Implications of the model for thinking about causal hypotheses of psychopathy, and for resolving longstanding points of contention in the field, are discussed.

Learning Objectives

- Learn about Cleckley’s classic account of psychopathy, presented in his book *The Mask of Sanity*, along with other historic conceptions.
- Compare and contrast differing inventories currently in use for assessing psychopathy in differing samples (e.g., adults and younger individuals, within clinical-forensic and community settings).
- Become familiar with the Triarchic model of psychopathy and its constituent constructs of boldness, meanness, and disinhibition.
- Learn about alternative theories regarding the causal origins of psychopathy.
- Consider how longstanding matters of debate regarding the nature, definition, and origins of psychopathy can be addressed from the perspective of the Triarchic model.

Introduction



Popular media have developed the character of the psychopath into a popular genre. But often these portrayals – psychopaths as criminal monsters preying on innocent people – are misleading. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

For many in the public at large, the term “psychopath” conjures up images of ruthless homicidal maniacs and criminal masterminds. This impression is reinforced on an ongoing basis by depictions of psychopathic individuals in popular books and films, such as *No Country for Old Men*, *Silence of the Lambs*, and *Catch Me if You Can*, and by media accounts of high-profile criminals ranging from Charles Manson to Jeffrey Dahmer to Bernie Madoff. However, the concept of **psychopathy** (“psychopathic personality”) held by experts in the mental health field differs sharply from this common public perception—emphasizing distinct dispositional tendencies as opposed to serious criminal acts of one sort or another. This module reviews historic and contemporary conceptions

of psychopathy as a clinical disorder, describes methods for assessing it, and discusses how a new conceptual model can help to address key questions regarding its nature and origins that have long been debated. It will be seen from this review that the topic remains no less fascinating or socially relevant when considered from a clinical–scientific perspective.

Historic Conceptions

Early writers characterized psychopathy as an atypical form of mental illness in which rational faculties appeared normal but everyday behavior and social relationships are markedly disrupted. French physician Philippe Pinel (1806/1962) documented cases of what he called *manie sans delire* (“insanity without delirium”), in which dramatic episodes of recklessness and aggression occurred in individuals not suffering from obvious clouding of the mind. German psychiatrist Julius Koch (1888) introduced the disease-oriented term *psychopathic* to convey the idea that conditions of this type had a strong constitutional-heritable basis. In his seminal book *The Mask of Sanity*, which focused on patients committed for hospital treatment, American psychiatrist Hervey Cleckley (1941/1976) described psychopathy as a deep-rooted emotional pathology concealed by an outward appearance of good mental health. In contrast

with other psychiatric patients, psychopathic individuals present as confident, sociable, and well adjusted. However, their underlying disorder reveals itself over time through their actions and attitudes. To facilitate identification of psychopathic individuals in clinical settings, Cleckley provided 16 diagnostic criteria distilled from his clinical case summaries, encompassing indicators of apparent psychological stability (e.g., charm and intelligence, absence of nervousness) along with symptoms of behavioral deviancy (e.g., irresponsibility, failure to plan) and impaired affect and social connectedness (e.g., absence of remorse, deceptiveness, inability to love).

Notably, Cleckley did not characterize psychopathic patients as inherently cruel, violent, or dangerous. Although some engaged in repetitive violent acts, more often the harm they caused was nonphysical and the product of impulsive self-centeredness as opposed to viciousness. Indeed, Cleckley's case histories included examples of "successful psychopaths" who ascended to careers as professors, medical doctors, or businessmen, along with examples of more aimless dysfunctional types. In contrast with this, other writers from Cleckley's time who were concerned with criminal expressions of psychopathy placed greater emphasis on symptoms of emotional coldness, aggression, and predatory victimization. For example, McCord and McCord (1964) described the condition in more generally pathologic terms, highlighting "guiltlessness" (lack of remorse) and "lovelessness" (lack of attachment capacity) as central defining features.



Hervey Cleckley, the man who devised the original tests for psychopathy, asserted that some psychopaths may appear as well-adjusted, successful people who maintain respectable careers in fields like business and medicine. [Image: Jonna Fransa, CC0 Public Domain, <https://goo.gl/m25gce>]

Cleckley's conception served as a referent for the diagnosis of psychopathy in the first two editions of the official American psychiatric nosology, the *Diagnostic and Statistical Manual of Mental Disorders* (DSM). However, a dramatic shift occurred in the third edition of the DSM, with the introduction of behaviorally oriented symptom definitions for most disorders to address longstanding problems of reliability. The Cleckley-oriented conception of psychopathy in prior editions was replaced by **antisocial personality disorder** (ASPD), defined by specific indicants of behavioral deviancy in childhood (e.g., fighting, lying, stealing, truancy)

continuing into adulthood (manifested as repeated rulebreaking, impulsiveness, irresponsibility, aggressiveness, etc.). Concerns with this new conception were expressed by psychopathy experts, who noted that ASPD provided limited coverage of interpersonal-affective symptoms considered essential to psychopathy (e.g., charm, deceitfulness, selfishness, shallow affect; Hare, 1983). Nonetheless, ASPD was retained in much the same form in the fourth edition of the DSM (DSM-IV; American Psychiatric Association [APA], 2000), and remained unchanged in the fifth edition of the DSM (American Psychiatric Association, 2013). That said, the DSM-5 does include a new, dimensional-trait approach to characterizing personality pathology (Strickland, Drislane, Lucy, Krueger, & Patrick, in press).

Contemporary assessment methods

Modern approaches to the assessment of psychopathy, consisting of rating instruments and self-report scales, reflect the foregoing historic conceptions to differing degrees.

Psychopathy in adult criminal offenders

The most widely used instrument for diagnosing psychopathy in correctional and forensic settings is the Psychopathy Checklist-Revised (PCL-R; Hare, 2003), which comprises 20 items rated on the basis of interview and file-record information. The items of the PCL-R effectively capture the interpersonal-affective deficits and behavioral deviance features identified by Cleckley, but include only limited, indirect coverage of positive adjustment features. The manual for the PCL-R recommends the use of a cutoff score of 30 out of 40 for assigning a diagnosis of psychopathy. High overall PCL-R scores are correlated with impulsive and aggressive tendencies, low empathy, Machiavellianism, lack of social connectedness, and persistent violent offending. Given these correlates, and the omission of positive adjustment indicators, psychopathy as assessed by the PCL-R appears more similar to the predatory-aggressive conception of McCord and McCord than to Cleckley's conception.

Although the PCL-R was developed to index psychopathy as a unitary condition, structural analyses of its items reveal distinct *interpersonal-affective* and *antisocial deviance* subdimensions (factors). Although moderately (about .5) correlated, these factors show contrasting relations with external criterion measures. The interpersonal-affective factor relates to indices of narcissism, low empathy, and proactive aggression (Hare, 2003), and to some extent (after controlling for its overlap with the antisocial factor) adaptive tendencies such as high social assertiveness and low fear, distress, and depression (Hicks & Patrick, 2006). High scores on the antisocial deviance factor, by contrast, are associated mainly with maladaptive tendencies and behaviors, including impulsiveness, sensation seeking, alienation

and mistrust, reactive aggression, early and persistent antisocial deviance, and substance-related problems.

Psychopathy in noncriminal adults



One of the key factors that the Psychopathic Personality Inventory (PPI) aims to assess is something called Fearless Dominance which includes social potency, immunity to stress, and lack of normal levels of fear. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Psychopathy has most typically been assessed in noncriminal adult samples using self-report-based measures. Older measures of this type emphasized the antisocial deviancy component of psychopathy with limited coverage of interpersonal-affective features (Hare, 2003). Some newer instruments provide more balanced coverage of both. One example is the now widely used Psychopathic Personality Inventory (PPI; Lilienfeld & Andrews, 1996), which was developed to index personality dispositions embodied within historic conceptions of psychopathy. Its current revised form (PPI-R; Lilienfeld & Widows, 2005) contains 154 items, organized into eight facet scales.

Like the items of the PCL-R, the subscales of the PPI cohere around two distinguishable

factors: a fearless dominance (FD) factor reflecting social potency, stress immunity, and fearlessness, and a self-centered impulsivity (SCI) factor reflecting egocentricity, exploitativeness, hostile rebelliousness, and lack of planning. However, unlike the factors of the PCL-R, the two PPI factors are uncorrelated, and thus even more distinct in their external correlates. Scores on PPI-FD are associated with indices of positive psychological adjustment (e.g., higher well-being; lower anxiety and depression) and measures of narcissism (low empathy, and thrill/adventure seeking (Benning, Patrick, Blonigen, Hicks, & Iacono, 2005). Given this, PPI-FD has been interpreted as capturing a more adaptive expression of dispositional fearlessness (i.e., *boldness*; see below) than the interpersonal-affective factor of the PCL-R—which can be viewed as tapping a more pathologic (antagonistic or “mean”) expression of fearlessness. Scores on PPI-SCI, like Factor 2 of the PCL-R, are associated with multiple indicators of deviancy—including impulsivity and aggressiveness, child and adult antisocial behavior, substance abuse problems, heightened distress and dysphoria, and

suicidal ideation.

Psychopathy in child and adolescent clinical samples

Different inventories exist for assessing psychopathic tendencies in children and adolescents. The best-known consist of rating-based measures developed, using the PCL-R as a referent, to identify psychopathic individuals among youth convicted of crimes or referred for treatment of conduct problems. The emphasis in work of this type has been on the importance of psychopathic features for predicting greater severity and persistence of conduct problems. Termed “callous-unemotional” traits, these features encompass low empathy, deficient remorse or guilt, shallow affect, and lack of concern about performance in school and other contexts (Frick & Moffitt, 2010).

One extensively researched measure for assessing psychopathic tendencies in youth is the Antisocial Process Screening Device (APSD; Frick & Marsee, 2006), used with clinic-referred children ages 6 through 13. The APSD includes 20 items completed by parents or teachers. As with the PCL-R and PPI, the items of the APSD tap two distinct factors: a Callous-Unemotional (CU) traits factor, reflecting emotional insensitivity and disregard for others; and an Impulsive/Conduct Problems (I/CP) factor, reflecting impulsivity, behavioral deviancy, and inflated self-importance. Children high on the I/CP factor alone show below-average intelligence, heightened emotional responsiveness to stressors, and angry (reactive) aggression (Frick & Marsee, 2006). By contrast, children high on both APSD factors show average or above-average intelligence, low reported levels of anxiety and nervousness, reduced reactivity to stressful events, and preference for activities entailing novelty and risk. They also learn less readily from punishment and engage in high levels of premeditated as well as reactive aggression and exhibit more persistent violent behavior across time. Given the documented importance of CU traits in moderating the expression of conduct disorder, the upcoming fifth edition of the DSM will include criteria for designating a distinct CU variant of child conduct disorder (Frick & Moffitt, 2010).

Core ingredients of psychopathy: disinhibition, boldness, and meanness

The foregoing material highlights the fact that historic conceptions of psychopathy and available instruments for assessing it place differing emphasis on different symptomatic features. This has contributed to longstanding disagreements among scholars about what psychopathy entails and what causes it. A theoretic conceptualization formulated recently to

reconcile alternative perspectives is the **Triarchic model** (Patrick, Fowles, & Krueger, 2009). This model conceives of psychopathy as encompassing three separable symptomatic components—disinhibition, boldness, and meanness—that can be viewed as thematic building blocks for differing conceptions of psychopathy.

Definitions

Disinhibition as described in the Triarchic model encompasses tendencies toward impulsiveness, weak behavioral restraint, hostility and mistrust, and difficulties in regulating emotion. *Meanness* entails deficient empathy, lack of affiliative capacity, contempt toward others, predatory exploitativeness, and empowerment through cruelty and destructiveness. Referents for disinhibition and meanness include the finding of distinct I/CP and CU factors in the child psychopathy literature and corresponding evidence for distinct disinhibitory and callous-aggression factors underlying impulse control (externalizing) problems in adults (Krueger, Markon, Patrick, Benning, & Kramer, 2007). The third construct in the model, *Boldness*, encompasses dominance, social assurance, emotional resiliency, and venturesomeness. Referents for this construct include the “mask” elements of Cleckley’s conception, Lykken’s (1995) low fear theory of psychopathy, the FD factor of the PPI, and developmental research on fearless temperament as a possible precursor to psychopathy (Patrick et al., 2009).



Cruelty and lack of empathy are tendencies found in those with psychopathic personality. [Image: Thomas Ricker, <https://goo.gl/igmuzh>, CC BY 2.0, <https://goo.gl/BRvSA7>]

From the perspective of the Triarchic model, Cleckley’s conception of psychopathy emphasized boldness and disinhibition, whereas criminally oriented conceptions (and affiliated measures, including the PCL-R and APSD) emphasize meanness and disinhibition more so. According to the model, individuals high in disinhibitory tendencies would warrant a diagnosis of psychopathy if also high in boldness or meanness (or both), but individuals high on only one of these tendencies would not. Individuals with differing relative elevations on these three symptomatic components would account for contrasting variants (subtypes) of psychopathy

as described in the literature (Hicks, Markon, Patrick, Krueger, & Newman, 2004; Karpman, 1941; Skeem, Johansson, Andershed, Kerr, & Loudon, 2007).

An inventory designed specifically to operationalize this model is the Triarchic Psychopathy Measure (TriPM; Patrick, 2010). The TriPM contains 58 items comprising three subscales that correspond to the constructs of the model (see Table 1). The items of the Disinhibition and Meanness scales (20 and 19 items, respectively) are taken from the Externalizing Spectrum Inventory (ESI; Krueger et al., 2007), a measure of problems and traits associated with externalizing psychopathology. The TriPM Boldness scale was developed to index fearless tendencies in social, affective-experiential, and activity preference domains, with reference to the FD factor of the PPI and the general factor shown to underlie differing scale measures of fear and fearlessness (Kramer, Patrick, Gasperi, & Krueger, 2012).

TriPM Subscale	Sample Item
Boldness	32. I can get over things that would traumatize others.
	38. I can convince people to do what I want.
	47. I stay away from physical danger as much as I can. (False)
Meanness	2. How other people feel is important to me (False)
	23. I enjoy pushing people around sometimes.
	33. I've injured people to see them in pain.
Disinhibition	3. I often act on immediate needs.
	21. I have good control over myself. (False)
	56. I have had problems at work because I was irresponsible.

Table 1. Sample items from the Triarchic Psychopathy Measure (TriPM; Patrick, 2010)

Although the TriPM is relatively new, promising evidence for its convergent and discriminant validity has begun to appear (e.g., Sellbom & Phillips, 2013; Strickland et al., in press; see also Venables & Patrick, 2012). Given that the inventory is freely available online, and that several foreign-language translations now exist (including Brazilian-Portuguese, Dutch, Finnish, German, Italian, Portuguese, Swedish, and Spanish), it can be expected that additional validity

data will accumulate rapidly over time. Work is also being done to evaluate whether effective scale measures of the Triarchic constructs can be derived from items of other existing psychopathy inventories such as the PPI. As discussed in the last part of this section, research examining the common and distinctive correlates of these three components of psychopathy is likely to be helpful for addressing and perhaps resolving ongoing points of uncertainty and debate in the field.

Causal factors

Considerable research has been devoted over many years to investigation of causal factors in psychopathy. Existing theories are of two types: (1) theories emphasizing core deficits in emotional sensitivity or responsiveness, and (2) theories positing basic impairments in cognitive-attentional processing (Patrick & Bernat, 2009). In support of these alternative theories, differing neurobiological correlates of psychopathy have been reported. One of the most consistent entails a lack of normal enhancement of the startle blink reflex to abrupt noises occurring during viewing of aversive foreground stimuli (e.g., scary or disturbing pictorial images) as compared with neutral or pleasant stimuli (see Figure 1). This result, akin to a failure to “jump” upon hearing a trash can tip while walking alone in a dark alley, has been interpreted as reflecting a lack of normal defensive (fear) reactivity. Another fairly consistent finding involves reduced amplitude of brain potential response to intermittent target stimuli, or following incorrect responses, within cognitive performance tasks—indicative of reduced cortical-attentional processing or impaired action monitoring (Patrick & Bernat, 2009). Yet other research using functional neuroimaging has demonstrated deficits in basic subcortical (amygdala) reactivity to interpersonal distress cues (e.g., fearful human faces) in high-psychopathic individuals (Jones, Laurens, Herba, Barker, & Viding, 2009; Marsh et al., 2008).

The Triarchic model may prove to be of use for reconciling alternative causal models of psychopathy that have been proposed based on contrasting neurobiological and behavioral findings. For example, lack of startle enhancement during aversive cuing has been tied specifically to the interpersonal-affective factor of the PCL-R and the counterpart FD factor of the PPI (Figure 1)—suggesting a link to the boldness component of psychopathy. By contrast, reduced brain potential responses in cognitive tasks appear more related to impulsive-externalizing tendencies associated with the disinhibition component of psychopathy (Carlson, Tháí, & McLaron, 2009; Patrick & Bernat, 2009). On the other hand, the finding of reduced subcortical response to affective facial cues has been tied to the CU traits factor of child/adolescent psychopathy, a referent for meanness in the Triarchic model. However, further research is needed to determine whether this finding reflects fear deficits common to meanness and boldness, or deficits in affiliative capacity or empathy specific to meanness.

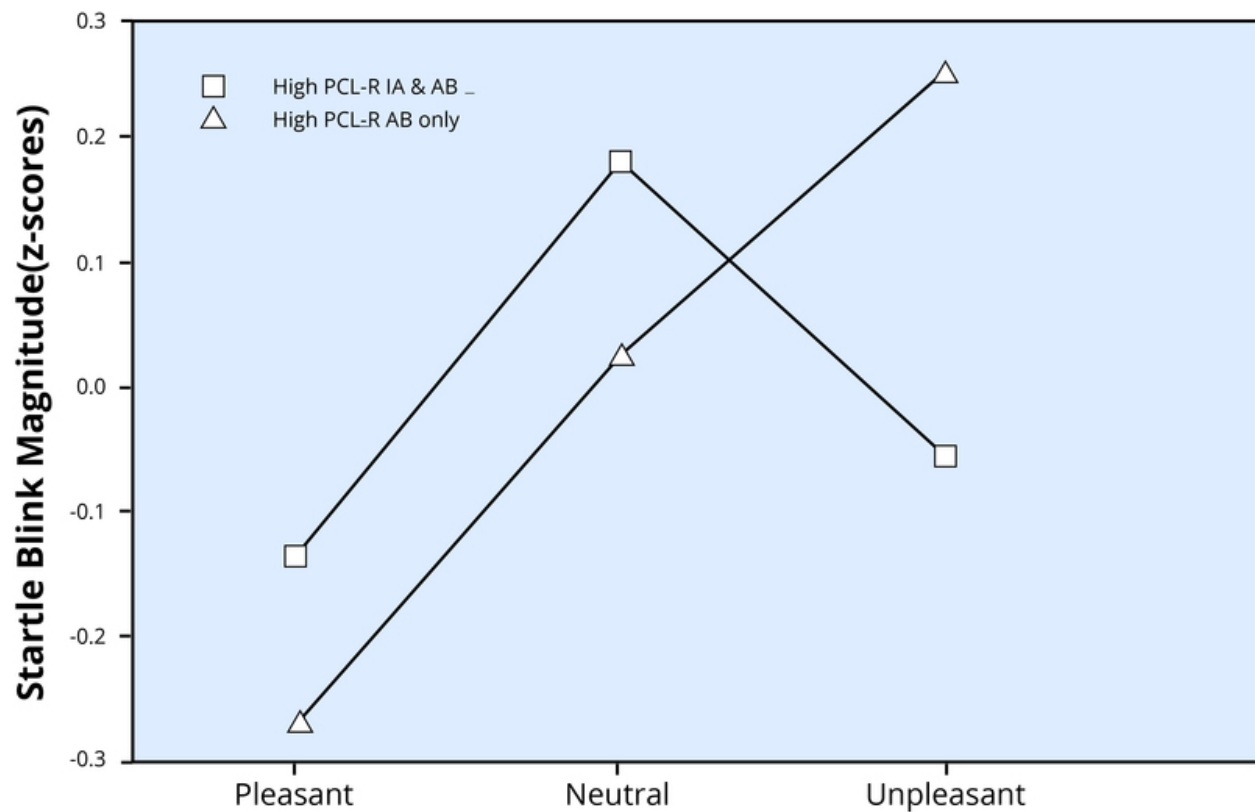
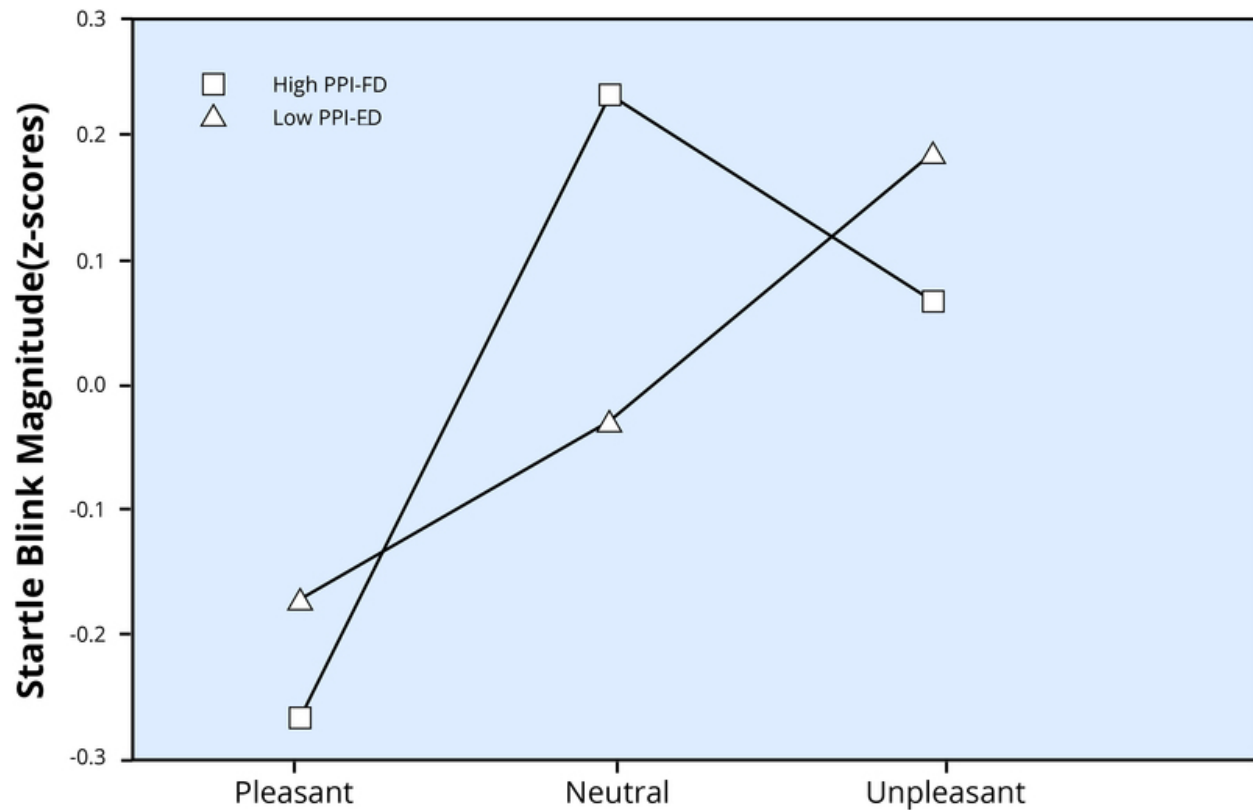


Figure 1. Evidence for lack of normal augmentation of the defensive startle reflex during viewing of aversive visual images in high-psychopathic individuals. Upper plot: Mean magnitude of startle blink responses to noise probes occurring during viewing of pleasant, neutral, and unpleasant picture stimuli in two male prisoner groups: (1) prisoners scoring high on the antisocial behavior (AB) factor of the Psychopathy Checklist-Revised (PCL-R; Hare, 2003) but not the interpersonal-affective (IA) factor (labeled "High PCL-R AB only" in the plot; $n = 18$), and (2) prisoners high on both factors of the PCL-R (labeled "High PCL-R IA & AB"; $n = 17$).

Triarchic model perspective on long-debated issues regarding psychopathy

As highlighted in the foregoing sections, scholars have grappled with issues of definition since psychopathy was first identified as a condition of clinical concern, and questions regarding its essential features and alternative expressions continue to be debated and studied. This final subsection discusses how some of the major issues of debate are addressed by the Triarchic model.

One key issue is whether psychological/emotional stability is characteristic or not of psychopathy. Cleckley's (1941/1976) view was that psychopathy entails a salient presentation of good mental health, and his diagnostic criteria included indicators of positive adjustment. By contrast, the dominant clinical assessment devices for psychopathy, the PCL-R and ASPD,



Lower plot: Mean magnitude of startle blink responses to noise probes occurring during viewing of pleasant, neutral, and unpleasant picture stimuli in two subgroups of young males from a large community sample (overall $N = 307$): (1) lowest 10% of scorers ($n = 31$) within the sample on the fearless dominance factor of the Psychopathic Personality Inventory (PPI; Lilienfeld & Andrews, 1996), labeled "Low PPI-FD" in the plot, and (2) highest 10% of scorers ($n = 31$) on the PPI fearless dominance factor of the PPI, labeled "High PPI-FD" in the plot. In both plots, blink means for each picture type are presented in z-score units ($M = 0$, $SD = 1$) derived by standardizing raw blink magnitude scores across trials for each individual subject. Data in the upper plot are from Patrick, Bradley, and Lang (1993); data in the lower plot are from Benning, Patrick, and Iacono (2005).

are heavily oriented toward deviancy and include no items that are purely indicative of adjustment. From a Triarchic model standpoint, the more adaptive elements of psychopathy are embodied in its boldness facet, which entails social poise, emotional stability, and enjoyment of novelty and adventure. At the same time, high boldness is also associated with narcissistic tendencies, reduced sensitivity to the feelings of others, and risk-taking (Benning et al., 2005). Thus, the concept of boldness provides a way to think about the intriguing "mask" element of psychopathy.

Related to this, another issue is whether lack of anxiety is central to psychopathy, as Cleckley and others (e.g., Fowles & Dindo, 2009; Lykken, 1995) have emphasized. This perspective is challenged by research showing either negligible or somewhat positive associations for overall scores on the PCL-R and other psychopathy measures with anxiety. The Triarchic model helps

to address this inconsistency by separating the disorder into subcomponents or facets, which relate differently to measures of trait anxiety: Boldness is correlated negatively with anxiousness (Benning et al., 2005), whereas Disinhibition and Meanness are correlated negatively and negligibly, respectively, with anxiety (Venables & Patrick, 2012). Related to this, cluster analytic studies of criminal offenders exhibiting high overall scores on the PCL-R have demonstrated one subtype characterized by low anxiety in particular, and another exhibiting high anxiety along with very high levels of impulsivity and aggression (Hicks et al., 2004; Skeem et al., 2007). The implication is that low anxiousness is central to one variant of criminal psychopathy (the bold-disinhibited, or “primary” type) but not to another variant (the “disinhibited-mean,” “aggressive-externalizing,” or “secondary” type).



Social psychologists classify a collection of three personality traits as the “dark triad”: Machiavellianism, psychopathy, and narcissism.

A further key question is whether violent/aggressive tendencies are typical of psychopathic individuals and should be included in the definition of the disorder. Cleckley’s (1941/1976) view was that “such tendencies should be regarded as the exception rather than as the rule” (p. 262). However, aggressiveness is central to criminally oriented conceptions of psychopathy, and the PCL-R includes an item reflecting hot-temperedness and aggression (“poor behavioral controls”) along with other items scored in part based on indications of cruelty and violence. In the Triarchic model, tendencies toward aggression are represented in both the disinhibition and meanness constructs, and a “mean-disinhibited” type of psychopath clearly exists, marked by the presence of salient aggressive behavior (Frick & Marsee, 2006; Hicks et al., 2004). Thus, Cleckley’s idea of aggression as ancillary to psychopathy may apply more to a variant of psychopathy that entails high boldness in conjunction with high disinhibition (Hicks et al., 2004).

Another question is whether criminal or antisocial behavior more broadly represents a defining feature of psychopathy, or a secondary manifestation (Cooke, Michie, Hart, & Clark, 2004). From the standpoint of the Triarchic model, antisocial behavior arises from the complex interplay of different “deviance-promoting” influences—including dispositional boldness, meanness, and disinhibition. However, whether approaches can be developed for classifying

antisocial behaviors in ways that relate more selectively to these and other distinct influences (e.g., through reference to underlying motives, spontaneity versus premeditation) is an important topic to be addressed in future research.

Another key question is whether differing subtypes of psychopathy exist. From the perspective of the Triarchic model, alternative variants of psychopathy reflect differing configurations of boldness, meanness, and disinhibition. Viewed this way, designations such as “bold-disinhibited” and “mean-disinhibited” may prove more useful for research and clinical purposes than labels like “primary” versus “secondary” or “low anxious” versus “high anxious.” An issue from this perspective is whether individuals who are high in boldness and/or meanness but low in disinhibition would qualify for a diagnosis of psychopathy. For example, should a high-bold/high-mean individual (e.g., a ruthless corporate executive, like the one portrayed by actor Michael Douglas in the film *Wall Street*; Pressman & Stone, 1987)—or an extremely mean/vicious but neither bold nor pervasively disinhibited individual, such as Russian serial murder Andrei Chikatilo (Cullen, 1993)—be considered psychopathic? Questions of this sort will need to be addressed through elaborations of existing theories in conjunction with further systematic research.

Yet another question is whether psychopathy differs in women as compared to men. Cleckley's descriptive accounts of psychopathic patients included two female case examples along with multiple male cases, and his view was that psychopathy clearly exists in women and reflects the same core deficit (i.e., absence of “major emotional accompaniments” of experience) as in men. However, men exhibit criminal deviance and ASPD at much higher rates than women (APA, 2000) and men in the population at large score higher in general on measures of psychopathy than women (Hare, 2003; Lilienfeld & Widows, 2005). From a Triarchic model perspective, these differences in prevalence may be attributable largely to differences between women and men in average levels of boldness, meanness, and disinhibition. Some supportive evidence exists for this hypothesis (e.g., findings of Hicks et al. [2007] demonstrating mediation of gender differences in ASPD symptoms by levels of externalizing proneness). Beyond this, it is important also to consider whether underlying psychopathic dispositions in men and women may be manifested differently in overt behavior (Verona & Vitale, 2006). Some intriguing evidence exists for this—including twin research findings demonstrating a genetic association between dispositional boldness (as indexed by estimated scores on PPI-FD) and a composite index of externalizing problems in male but not female participants (Blonigen, Hicks, Patrick, Krueger, Iacono, & McGue, 2005). However, more extensive research along these lines, examining all facets of the Triarchic model in relation to behavioral outcomes of differing kinds, will be required to effectively address the question of gender-moderated expression.

A final intriguing question is whether “successful” psychopaths exist. Hall and Benning (2006) hypothesized that successful psychopathy entails a preponderance of certain causal influences (resulting in particular symptomatic features) over others. Drawing on known correlates of PPI-FD (e.g., Benning et al., 2005; Ross, Benning, Patrick, Thompson, & Thurston, 2009) and theories positing separate etiologic mechanisms for differing features of psychopathy (Fowles & Dindo, 2009; Patrick & Bernat, 2009), these authors proposed that the presence of dispositional fearlessness (boldness) may be conducive to success when not accompanied by high externalizing proneness (disinhibition). For example, high-bold/low-disinhibited individuals could be expected to achieve higher success in occupations calling for leadership and/or courage because their psychopathic tendencies are manifested mainly in terms of social effectiveness, affective resilience, and venturesomeness.



Some aspects of the psychopathic personality could be beneficial in certain professions requiring leadership and courage. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Data relevant to this idea come from an intriguing study by Lilienfeld, Waldman, Landfield, Rubenzer, and Faschingbauer (2012), who used personality trait ratings of former U.S. presidents provided by expert historians to estimate scores on the FD and SCI factors of the PPI (Ross et al., 2009). They found that higher estimated levels of PPI-FD (boldness) predicted higher ratings of presidential performance, persuasiveness, leadership, and crisis management ability, whereas higher estimated levels of SCI predicted adverse outcomes such as documented abuses of power and impeachment proceedings. Further research on outcomes associated with high levels of boldness and/or meanness in the absence of high disinhibition should yield valuable new insights into dispositional factors underlying psychopathy and alternative ways psychopathic tendencies can be expressed.

Outside Resources

Book: (Fictional novels or biographies/biographical novels) Capote, T. (1966). *In cold blood*. New York, NY: Random House.

Book: (Fictional novels or biographies/biographical novels) Highsmith, P. (1955). *The talented Mr. Ripley*. New York, NY: Coward-McCann.

Book: (Fictional novels or biographies/biographical novels) Kerouac, J. (1957). *On the road*. New York, NY: Viking Press.

Book: (Fictional novels or biographies/biographical novels) Mailer, N. (1979). *The executioner's song*. New York, NY: Little, Brown & Co.

Book: (Fictional novels or biographies/biographical novels) McMurtry, L. (1985). *Lonesome dove*. New York, NY: Simon & Schuster.

Book: (Fictional novels or biographies/biographical novels) Rule, A. (1988). *Small sacrifices*. New York, NY: Signet.

Book: (Fictional novels or biographies/biographical novels) Wolff, G. (1979). *The duke of deception*. New York, NY: Vintage.

Book: (Reference) Babiak, P., & Hare, R. D. (2006). *Snakes in suits*. New York, NY: HarperCollins.

Book: (Reference) Blair, R. J. R., Mitchell, D., & Blair, K. (2005). *The psychopath: Emotion and the brain*. Malden, MA: Blackwell.

Book: (Reference) Hare, R. D. (1993). *Without conscience*. New York, NY: Guilford Press.

Book: (Reference) Häkkänen-Nyholm, H., & Nyholm, J. (2012). *Psychopathy and law: A practitioner's guide*. New York, NY: Wiley.

Book: (Reference) Patrick, C. J. (2006). *Handbook of psychopathy*. New York, NY: Guilford Press.

Book: (Reference) Patrick, C. J. (2018). *Handbook of psychopathy* (2nd ed.). New York, NY: Guilford Press.

Book: (Reference) Raine, A. (2013). The anatomy of violence. New York, NY: Random House.

Book: (Reference) Salekin, R., & Lynam, D. T. (2010). Handbook of child and adolescent psychopathy. New York, NY: Guilford Press.

Measure: Online home of the Triarchic Psychopathy Measure. It serves as a great resource for students who wish to look at how psychopathy is measured a little more deeply.

<https://www.phenxtoolkit.org/index.php?pageLink=browse.protocoldetails&id=121601>

Movie: (Facets of psychopathy - Boldness) Bigelow, K. (Producer & Director), Boal, M., Chartier, N., & Shapiro, G. (Producers). (2008). The hurt locker. United States: Universal Studios.

Movie: (Facets of psychopathy - Disinhibition) Felner, E. (Producer), Cox, A. (Director). (1986). Sid and Nancy. United States: Samuel Goldwyn.

Movie: (Facets of psychopathy - Meanness) Coen, J., Coen, E. (Producers & Directors), & Rudin, S. (Producer). (2007). No country for old men. United States: Miramax.

Movie: (Psychopathic criminals) Demme, J., Saraf, P., Saxon, E. (Producers), & Sena, D. (Director). (1993). Kalifornia. United States: Gramercy.

Movie: (Psychopathic criminals) Jaffe, S. R., Lansing, S. (Producers), & Lyn, A. (Director). (1987). Fatal attraction. United States: Paramount.

Movie: (Psychopathic criminals) Scorsese, M., Harris, R. A., Painten, J. (Producers), & Feears, S. (Director). (1990). The grifters. United States: Miramax.

Movie: (Psychopathic criminals) Ward, F., Bozman, R., Utt, K., Demme, J. (Producers), & Armitrage, G. (Director). (1990). Miami blues. United States: Orion.

Movie: (Psychopathic hospital patients) Wick, D., Conrad, C. (Producers), & Mangold, J. (Director). (1990). Girl, interrupted. United States: Columbia.

Movie: (Psychopathic hospital patients) Zaentz, S., Douglas, M. (Producers), & Forman, M. (Director). (1975). One flew over the cuckoo's nest. United States: United Artists.

Movie: (Psychopaths in business/politics) Bachrach, D. (Producer), & Pierson, F. (Director). (1992). Citizen Cohn. United States: Home Box Office.

Movie: (Psychopaths in business/politics) Pressman, E. R. (Producer), & Stone, O. (Director). (1987). Wall Street. United States: 20th Century Fox.

Web: A new non-profit organization dedicated to dispelling myths about psychopathy, sharing solutions, and advocating for the funding needed to find more effective, targeted treatments.

<https://psychopathyis.org>

Web: A valuable online resource containing information of various types including detailed reference lists is Dr. Robert Hare's website on the topic of psychopathy. Dr. Hare is the person who created the PCL-R. It has a number of interesting articles to read as well as helpful psychopathy links from about the web.

<http://www.hare.org/>

Web: Hervey Cleckley's classic book *The Mask of Sanity* is no longer in print at this time, but a version authorized for nonprofit educational use by his estate can be viewed online at

http://www.quantumfuture.net/store/sanity_1.PdF

Web: The Triarchic Psychopathy Measure can be accessed online at

<https://www.phenxtoolkit.org/index.php?pageLink=browse.protocoldetails&id=121601>

Web: The website for the Aftermath Foundation, a nonprofit organization that provides information and support for victims and family members of psychopathic individuals, is

<http://www.aftermath-surviving-psychopathy.org/>

Discussion Questions

1. What did Cleckley mean when he characterized psychopathy as involving a "Mask of Sanity"?
2. Compare and contrast the Psychopathy Checklist-Revised (PCL-R), the Antisocial Process Screening Device (APSD), and the Psychopathic Personality Inventory (PPI), in terms of the samples they are designed for, the way in which they are administered, and the content and factor structure of their items.
3. Identify and define the three facet constructs of the Triarchic model of psychopathy. Discuss how these facet constructs relate to the factors of the PCL-R, ASPD, and PPI. Discuss how U.S. President Teddy Roosevelt and fictional character Anton Chigurh from the film *No Country for Old Men* might compare in terms of scores on the three Triarchic constructs.

4. Identify alternative types of theories that have been proposed regarding the cause of psychopathy, and how these can be viewed from the perspective of the Triarchic model.
5. Identify two longstanding issues of debate regarding the nature/definition of psychopathy and how these issues are addressed by the Triarchic model.

Vocabulary

Antisocial personality disorder

Counterpart diagnosis to psychopathy included in the third through fifth editions of the Diagnostic and Statistical Manual of Mental Disorders (DSM; APA, 2000). Defined by specific symptoms of behavioral deviancy in childhood (e.g., fighting, lying, stealing, truancy) continuing into adulthood (manifested as repeated rule-breaking, impulsiveness, irresponsibility, aggressiveness, etc.).

Psychopathy

Synonymous with psychopathic personality, the term used by Cleckley (1941/1976), and adapted from the term psychopathic introduced by German psychiatrist Julius Koch (1888) to designate mental disorders presumed to be heritable.

Triarchic model

Model formulated to reconcile alternative historic conceptions of psychopathy and differing methods for assessing it. Conceives of psychopathy as encompassing three symptomatic components: boldness, involving social efficacy, emotional resiliency, and venturesomeness; meanness, entailing lack of empathy/emotional-sensitivity and exploitative behavior toward others; and disinhibition, entailing deficient behavioral restraint and lack of control over urges/emotional reactions.

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ADHD and Behavior Disorders in Children

Richard Milich & Walter Roberts

Attention-Deficit/Hyperactivity Disorder (ADHD) is a psychiatric disorder that is most often diagnosed in school-aged children. Many children with ADHD find it difficult to focus on tasks and follow instructions, and these characteristics can lead to problems in school and at home. How children with ADHD are diagnosed and treated is a topic of controversy, and many people, including scientists and nonscientists alike, hold strong beliefs about what ADHD is and how people with the disorder should be treated. This module will familiarize the reader with the scientific literature on ADHD. First, we will review how ADHD is diagnosed in children, with a focus on how mental health professionals distinguish between ADHD and normal behavior problems in childhood. Second, we will describe what is known about the causes of ADHD. Third, we will describe the treatments that are used to help children with ADHD and their families. The module will conclude with a brief discussion of how we expect that the diagnosis and treatment of ADHD will change over the coming decades.

Learning Objectives

- Distinguish childhood behavior disorders from phases of typical child development.
- Describe the factors contributing to Attention-Deficit/Hyperactivity Disorder (ADHD)
- Understand the controversies surrounding the legitimacy and treatment of childhood behavior disorders
- Describe the empirically supported treatments for Attention-Deficit/Hyperactivity Disorder (ADHD)

Introduction

Childhood is a stage of life characterized by rapid and profound development. Starting at birth, children develop the skills necessary to function in the world around them at a rate that is faster than any other time in life. This is no small accomplishment! By the end of their first decade of life, most children have mastered the complex cognitive operations required to comply with rules, such as stopping themselves from acting impulsively, paying attention to parents and teachers in the face of distraction, and sitting still despite boredom. Indeed, acquiring self-control is an important developmental task for children (Mischel, Shoda, & Rodriguez, 1989), because they are expected to comply with directions from adults, stay on task at school, and play appropriately with peers. For children with Attention-Deficit/Hyperactivity Disorder (ADHD), however, exercising self-control is a unique challenge. These children, oftentimes despite their best intentions, struggle to comply with adults' instructions, and they are often labeled as "problem children" and "rule breakers." Historically, people viewed these children as willfully noncompliant due to moral or motivational defect (Still, 1902). However, scientists now know that the noncompliance observed in children with ADHD can be explained by a number of factors, including neurological dysfunction.



Although we commonly think of children as "little bundles of energy", one of the defining characteristics of children diagnosed with ADHD is that they are perpetually in motion even during times when they are expected to be still. [Image: Chicago's North Shore Conventions & Visitors Bureau, <https://goo.gl/U2Zl18>, CC BY 2.0, <https://goo.gl/9uSnqN>]

The goal of this module is to review the classification, causes, consequences, and treatment of ADHD. ADHD is somewhat unique among the psychiatric disorders in that most people hold strong opinions about the disorder, perhaps due to its more controversial qualities. When applicable, we will discuss some of the controversial beliefs held by social critics and laypeople, as well as scientists who study the disorder. Our hope is that a discussion of these controversies will allow you to reach your own conclusions about the legitimacy of the disorder.

Why Diagnose Children's Behavior Problems?

When a family is referred to a mental health professional for help dealing with their child's problematic behaviors, the clinician's first goal is to identify the nature and cause of the child's problems. Accurately diagnosing children's behavior problems is an important step in the intervention process, because a child's diagnosis can guide clinical decision making. Childhood behavior problems often arise from different causes, require different methods for treating, and have different developmental courses. Arriving at a diagnosis will allow the clinician to make inferences about how each child will respond to different treatments and provide predictive information to the family about how the disorder will affect the child as he or she develops.



On the one hand, diagnosing a child with ADHD can help him or her get beneficial treatment; however, the diagnosis can also have potentially negative effects on peer relationships and how children perceive themselves. [Image: gibsonsgolfer, <https://goo.gl/WWZfbQ>, CC BY-NC 2.0, <https://goo.gl/Fllc2e>]

Despite the utility of the current diagnostic system, the practice of diagnosing children's behavior problems is controversial. Many adults feel strongly that labeling children as "disordered" is stigmatizing and harmful to children's self-concept. There is some truth in this concern. One study found that children have more negative attitudes toward a play partner if they are led to believe that their partner has ADHD, regardless of whether or not their partner actually has the disorder (Harris, Milich, Corbitt, Hoover, & Brady, 1992). Others have criticized the use of the diagnostic system because they believe it **pathologizes** normal behavior in children. Despite these criticisms, the diagnostic system has played a central role in research and treatment of child behavior disorders, and it is unlikely to change substantially in the near future. This

section will describe ADHD as a diagnostic category and discuss controversies surrounding the legitimacy of this disorder.

ADHD is the most commonly diagnosed childhood behavior disorder. It affects 3% to 7% of children in the United States (American Psychiatric Association, 2000), and approximately 65% of children diagnosed with ADHD will continue to experience symptoms as adults (Faraone,

Biederman, & Mick, 2006). The core symptoms of ADHD are organized into two clusters, including clusters of hyperactivity/impulsivity and inattention. The hyperactive symptom cluster describes children who are perpetually in motion even during times when they are expected to be still, such as during class or in the car. The impulsive symptom cluster describes difficulty in delaying response and acting without considering the repercussions of behavior. Hyperactive and impulsive symptoms are closely related, and boys are more likely than girls to experience symptoms from this cluster (Hartung & Widiger, 1998). Inattentive symptoms describe difficulty with organization and task follow-through, as well as a tendency to be distracted by external stimuli. Two children diagnosed with ADHD can have very different symptom presentations. In fact, children can be diagnosed with different subtypes of the disorder (i.e., Combined Type, Predominantly Inattentive Type, or Predominantly Hyperactive-Impulsive Type) according to the number of symptoms they have in each cluster.

Are These Diagnoses Valid?

Many laypeople and social critics argue that ADHD is not a “real” disorder. These individuals claim that children with ADHD are only “disordered” because parents and school officials have trouble managing their behavior. These criticisms raise an interesting question about what constitutes a psychiatric disorder in children: How do scientists distinguish between clinically significant ADHD symptoms and normal instances of childhood impulsivity, hyperactivity, and inattention? After all, many 4-year-old boys are hyperactive and cannot focus on a task for very long. To address this issue, several criteria are used to distinguish between normal and disordered behavior:

1. The symptoms must significantly impair the child’s functioning in important life domains (e.g., school, home).
2. The symptoms must be inappropriate for the child’s developmental level.

One goal of this module will be to examine whether ADHD meets the criteria of a “true” disorder. The first criterion states that children with ADHD should show impairment in major functional domains. This is certainly true for children with ADHD. These children have lower academic achievement compared with their peers. They are more likely to repeat a grade or be suspended and less likely to graduate from high school (Loe & Feldman, 2007). Children with ADHD are often unpopular among their peers, and many of these children are actively disliked and socially rejected (Landau, Milich, & Diener, 1998). Children with ADHD are likely to experience comorbid psychological problems such as learning disorders, depression, anxiety, and oppositional defiant disorder. As they grow up, adolescents and adults with ADHD are at risk to abuse alcohol and other drugs (Molina & Pelham, 2003) and experience other

adverse outcomes (see Focus Topic 1). In sum, there is sufficient evidence to conclude that children diagnosed with ADHD are significantly impaired by their symptoms.

Focus Topic 1: Adult outcomes of children with ADHD

Children with ADHD often continue to experience symptoms of the disorder as adults. Historically, this fact was not recognized by the medical community; instead, they believed that children “matured out” of their symptoms as they entered adulthood. Fortunately, opinions have changed over time, and it is now generally accepted that ADHD can be present among adults. A recent prevalence estimate suggests that 4.4% of adults in the United States meet criteria for ADHD (Kessler et al., 2006). This study also found that the majority of adults with ADHD are not receiving treatment for their disorder. Adult ADHD, if left untreated, can cause numerous negative outcomes, including:

- Depression and poor self-concept, personality disorder, and other psychiatric comorbidity (Kessler et al., 2006)
- Substance abuse (Molina & Pelham, 2003)
- Poor work performance, termination from jobs, chronic unemployment, and poor academic achievement (Barkley, Fischer, Smallish, & Fletcher, 2006)
- Divorce and problems with interpersonal relationships (Biederman et al., 2006)
- High-risk sexual behaviors and early parenthood (Barkley et al., 2006; Flory, Molina, Pelham, Gnagy, & Smith, 2006)
- Impairments in driving ability (Weafer, Fillmore, & Milich, 2009)
- Obesity (Cortese et al., 2008)

Despite the list of negative outcomes associated with adult ADHD, adults with the disorder are not doomed to live unfulfilling lives of limited accomplishment. Many adults with ADHD have benefited from treatment and are able to overcome their symptoms. For example, pharmacological treatment of adult ADHD has been shown to reduce risk of criminal behavior (Lichtenstein et al., 2012). Others have succeeded by avoiding careers in which their symptoms would be particularly problematic (e.g., those with heavy organizational demands). In any case, it is important that people with ADHD are identified and treated early, because early treatment predicts more positive outcomes in adulthood (Kessler et al., 2006).

It is also important to determine that a child's symptoms are not caused by normal patterns of development. Many of the behaviors that are diagnostic of ADHD in some children would be considered developmentally appropriate for a younger child. This is true for many psychological and psychiatric disorders in childhood. For example, bedwetting is quite common in 3-year-old children; at this age, most children have not gained control over nighttime urination. For this reason, a 3-year-old child who wets the bed would not be diagnosed with enuresis (i.e., the clinical term for chronic bedwetting), because his or her behavior is developmentally appropriate. Bedwetting in an 8-year-old child, however, is developmentally *inappropriate*. At this age, children are expected to remain dry overnight, and failure to master this skill would prevent children from sleeping over at friends' houses or attending overnight camps. A similar example of developmentally appropriate versus inappropriate hyperactivity and noncompliance is provided in Focus Topic 2.

Focus Topic 2: Two children referred for problems with noncompliance and hyperactivity

Case 1 - Michael

Michael, a 4-year-old boy, was referred to a child psychologist to be evaluated for ADHD. His parents reported that Michael would not comply with their instructions. They also complained that Michael would not remain seated during "quality time" with his father. The evaluating psychologist interviewed the family, and by all accounts Michael was noncompliant and often left his seat. Specifically, when Michael's mother asked him to prepare his preschool lunch, Michael would leave the kitchen and play with his toys soon after opening his lunch box. Further, the psychologist found that quality time involved Michael and his father sitting down for several hours to watch movies. In other settings, such as preschool, Michael was compliant with his teacher's request and no more active than his peers.

In this case, Michael's parents held unrealistic expectations for a child at Michael's developmental level. The psychologist would likely educate Michael's parents about normative child development rather than diagnosing Michael with ADHD.

Case 2 - Jake

Jake, a 10-year-old boy, was referred to the same psychologist as Michael. Jake's mother was concerned because Jake was not getting ready for school on time. Jake also had trouble remaining seated during dinner, which interrupted mealtime for the rest of the family. The psychologist found that in the morning, Jake would complete one or two steps of his routine before he became distracted and switched activities, despite his mother's constant reminders. During dinnertime, Jake would leave his seat between 10 and 15 times over the course of the meal. Jake's teachers were worried because Jake was only able to complete 50% of his homework. Further, his classmates would not pick Jake for team sports during recess because he often became distracted and wandered off during the game.

In this case, Jake's symptoms would not be considered developmentally appropriate for a 10-year-old child. Further, his symptoms caused him to experience impairment at home and school. Unlike Michael, Jake probably would be diagnosed with ADHD.

Why Do Some Children Develop Behavior Disorders?



Studies of twins have shown that genetics are primarily responsible for ADHD. [Image: donnierayjones, <https://goo.gl/dgPvFx>, CC BY 2.0, <https://goo.gl/9uSnqN>]

The reasons that some children develop ADHD are complex, and it is generally recognized that a single cause is insufficient to explain why an individual child does or does not have the disorder. Researchers have attempted to identify risk factors that predispose a child to develop ADHD. These risk factors range in scope from genetic (e.g., specific gene polymorphisms) to familial (e.g., poor parenting) to cultural (e.g., low socioeconomic status). This section will identify some of the risk factors that are thought to contribute to ADHD. It will conclude by reviewing some of the more controversial ideas about the causes of ADHD, such as poor parenting and children's diets, and

review some of the evidence pertaining to these causes.

Most experts believe that genetic and neurophysiological factors cause the majority of ADHD cases. Indeed, ADHD is primarily a genetic disorder—twin studies find that whether or not a child develops ADHD is due in large part (75%) to genetic variations (Faraone et al., 2005). Further, children with a family history of ADHD are more likely to develop ADHD themselves (Faraone & Biederman, 1994). Specific genes that have been associated with ADHD are linked to neurotransmitters such as dopamine and serotonin. In addition, neuroimaging studies have found that children with ADHD show reduced brain volume in some regions of the brain, such as the prefrontal cortex, the corpus callosum, the anterior cingulate cortex, the basal ganglia, and the cerebellum (Seidman, Valera, & Makris, 2005). Among their other functions, these regions of the brain are implicated in organization, impulse control, and motor activity, so the reduced volume of these structures in children with ADHD may cause some of their symptoms.

Although genetics appear to be a main cause of ADHD, recent studies have shown that environmental risk factors may cause a minority of ADHD cases. Many of these environmental risk factors increase the risk for ADHD by disrupting early development and compromising the integrity of the central nervous system. Environmental influences such as low birth weight, malnutrition, and maternal alcohol and nicotine use during pregnancy can increase the likelihood that a child will develop ADHD (Mick, Biederman, Faraone, Sayer, & Kleinman, 2002). Additionally, recent studies have shown that exposure to environmental toxins, such as lead and pesticides, early in a child's life may also increase risk of developing ADHD (Nigg, 2006).

Controversies on Causes of ADHD

Controversial explanations for the development of ADHD have risen and fallen in popularity since the 1960s. Some of these ideas arise from cultural folklore, others can be traced to "specialists" trying to market an easy fix for ADHD based on their proposed cause. Some other ideas contain a kernel of truth but have been falsely cast as causing the majority of ADHD cases.

Some critics have proposed that poor parenting is a major cause of ADHD. This explanation is popular because it is intuitively appealing—one can imagine how a child who is not being disciplined at home may be noncompliant in other settings. Although it is true that parents of children with ADHD use discipline less consistently, and a lack of structure and discipline in the home can exacerbate symptoms in children with ADHD (Campbell, 2002), it is unlikely that poor parenting alone causes ADHD in the first place. To the contrary, research suggests that the noncompliance and impulsivity on the child's part can cause caregivers to use

discipline less effectively.

In a classic series of studies, Cunningham and Barkley (1979) showed that mothers of children with ADHD were less attentive to their children and imposed more structure to their playtime relative to mothers of typically developing children. However, these researchers also showed that when the children were given stimulant medication, their compliance increased and their mothers' parenting behavior improved to the point where it was comparable to that of the mothers of children without ADHD (Barkley & Cunningham, 1979). This research suggests that instead of poor parenting causing children to develop ADHD, it is the stressful effects of managing an impulsive child that causes parenting problems in their caregivers. One can imagine how raising a child with ADHD could be stressful for parents. In fact, one study showed that a brief interaction with an impulsive and noncompliant child caused parents to increase their alcohol consumption—presumably these parents were drinking to cope with the stress of dealing with the impulsive child (Pelham et al., 1997). It is, therefore, important to consider the reciprocal effects of noncompliant children on parenting behavior, rather than assuming that parenting ability has a unidirectional effect on child behavior.



It is still a common belief that giving sugar to kids makes them hyperactive; however, a critical review of the research showed that such a belief is nothing more than a myth. [Image: courosa, <https://goo.gl/0NerUI>, CC BY-NC-SA 2.0, <https://goo.gl/HEXbAA>]

Other purported causes of ADHD are dietary. For example, it was long believed that excessive sugar intake can cause children to become hyperactive. This myth is largely disproven (Milich, Wolraich, & Lindgren, 1986). However, other diet-oriented explanations for ADHD, such as sensitivity to certain food additives, have been proposed (Feingold, 1976). These theories have received a bit more support than the sugar hypothesis (Pelsser et al., 2011). In fact, the possibility that certain food additives may cause hyperactivity in children led to a ban on several artificial food colorings in the United Kingdom, although the Food and Drug Administration rejected similar measures in the United States. Even if artificial food dyes do cause hyperactivity in a subgroup of children, research does not support these food additives as a primary cause of ADHD.

Further, research support for elimination diets as a treatment for ADHD has been inconsistent at best.

In sum, scientists are still working to determine what causes children to develop ADHD, and despite substantial progress over the past four decades, there are still many unanswered questions. In most cases, ADHD is probably caused by a combination of genetic and environmental factors. For example, a child with a genetic predisposition to ADHD may develop the disorder after his or her mother uses tobacco during her pregnancy, whereas a child without the genetic predisposition may not develop the disorder in the same environment. Fortunately, the causes of ADHD are relatively unimportant for the families of children with ADHD who wish to receive treatment, because what caused the disorder for an individual child generally does not influence how it is treated.

Methods of Treating ADHD in Children

There are several types of evidence-based treatment available to families of children with ADHD. The type of treatment that might be used depends on many factors, including the child's diagnosis and treatment history, as well as parent preference. To treat children with less severe noncompliance problems, parents can be trained to systematically use **contingency management** (i.e., rewards and punishments) to manage their children's behavior more effectively (Kazdin, 2005). For the children with ADHD, however, more intensive treatments often are necessary.

Medication

The most common method of treating ADHD is to prescribe stimulant medications such as Adderall™. These medications treat many of the core symptoms of ADHD—treated children will show improved impulse control, time-on-task, and compliance with adults, and decreased hyperactivity and disruptive behavior. However, there are also negative side effects to stimulant medication, such as growth and appetite suppression, increased blood pressure, insomnia, and changes in mood (Barkley, 2006). Although these side effects can be unpleasant for children, they can often be avoided with careful monitoring and dosage adjustments.



Some critics of medicating as a possible attempt to mitigate the effects of ADHD, are concerned that the medications to treat ADHD might be over-prescribed. [Image: Tony Webster, <https://goo.gl/qo2xNB>, CC BY 2.0, <https://goo.gl/9uSnqN>]

Opinions differ on whether stimulants should be used to treat children with ADHD. Proponents argue that stimulants are relatively safe and effective, and that untreated ADHD poses a much greater risk to children (Barkley, 2006). Critics argue that because many stimulant medications are similar to illicit drugs, such as cocaine and methamphetamine, long-term use may cause cardiovascular problems or predispose children to abuse illicit drugs. However, longitudinal studies have shown that people taking these medications are not more likely to experience cardiovascular problems or to abuse drugs (Biederman, Wilens, Mick, Spencer, & Faraone, 1999; Cooper et al., 2011). On the other hand, it is not entirely clear how long-term stimulant treatment can affect the brain, particularly in adults who have been medicated for ADHD since childhood.

Finally, critics of psychostimulant medication have proposed that stimulants are increasingly being used to manage energetic but otherwise healthy children. It is true that the percentage of children prescribed stimulant medication has increased since the 1980s. This increase in use is not unique to stimulant medication, however. Prescription rates have similarly increased for most types of psychiatric medication (Olfson, Marcus, Weissman, & Jensen, 2002). As parents and teachers become more aware of ADHD, one would expect that more children with ADHD will be identified and treated with stimulant medication. Further, the percentage of children in the United States being treated with stimulant medication is lower than the estimated prevalence of children with ADHD in the general population (Nigg, 2006).

Parent Management Training

Parenting children with ADHD can be challenging. Parents of these children are understandably frustrated by their children's misbehavior. Standard discipline tactics, such as warnings and privilege removal, can feel ineffective for children with ADHD. This often leads to ineffective parenting, such as yelling at or ridiculing the child with ADHD. This cycle can leave parents feeling hopeless and children with ADHD feeling alienated from their family. Fortunately, **parent management training** can provide parents with a number of tools to cope with and effectively manage their child's impulsive and oppositional behavior. Parent management training teaches parents to use immediate, consistent, and powerful consequences (i.e., rewards and punishment), because children with ADHD respond well to these types of behavioral contingencies (Luman, Oosterlaan, & Sergeant, 2005). Other, more intensive, psychosocial treatments use similar behavioral principles in summer camp-based settings (Pelham, Fabiano, Gnagy, Greiner, & Hoza, 2004), and school-based intervention programs are becoming more popular. A description of a school-based intervention program for ADHD is described in Focus Topic 3.

Focus Topic 3: Treating ADHD in Schools

Succeeding at school is one of the most difficult challenges faced by children with ADHD and their parents. Teachers expect students to attend to lessons, complete lengthy assignments, and comply with rules for approximately seven hours every day. One can imagine how a child with hyperactive and inattentive behaviors would struggle under these demands, and this mismatch can lead to frustration for the student and his or her teacher. Disruptions caused by the child with ADHD can also distract and frustrate peers. Succeeding at school is an important goal for children, so researchers have developed and validated intervention strategies based on behavioral principles of contingency management that can help children with ADHD adhere to rules in the classroom (described in DuPaul & Stoner, 2003). Illustrative characteristics of an effective school-based contingency management system are described below:

Token reinforcement program

This program allows a student to earn tokens (points, stars, etc.) by meeting behavioral goals and not breaking rules. These tokens act as secondary reinforcers because they can be redeemed for privileges or goods. Parents and teachers work with the students to identify problem behaviors and create concrete behavioral goals. For example, if a student is disruptive during silent reading time, then a goal might be for him or her to remain seated for at least 80% of reading time. Token reinforcement programs are most effective when tokens are provided for appropriate behavior and removed for inappropriate behavior.

Time out

Time out can be an effective punishment when used correctly. Teachers should place a student in time out only when they fail to respond to token removal or if they engage in a severely disruptive behavior (e.g., physical aggression). When placed in time out, the student should not have access to any type of reinforcement (e.g., toys, social interaction), and the teacher should monitor their behavior throughout time out.

Daily report card

The teacher keeps track of whether or not the student meets his or her goals and records this

information on a report card. This information is sent home with the student each day so parents can integrate the student's performance at school into a home-based contingency management program.

Educational services and accommodations

Students with ADHD often show deficits in specific academic skills (e.g., reading skills, math skills), and these deficits can be improved through direct intervention. Students with ADHD may spend several hours each week working one-on-one with an educator to improve their academic skills. Environmental accommodations can also help a student with ADHD be successful. For example, a student who has difficulty focusing during a test can be allowed extra time in a low-distraction setting.

What Works Best? The Multimodal Treatment Study

Recently, a large-scale study, the Multimodal Treatment Study (MTA) of Children with ADHD, compared pharmacological and behavioral treatment of ADHD (MTA Cooperative Group, 1999). This study compared the outcomes of children with ADHD in four different treatment conditions, including standard community care, intensive behavioral treatment, stimulant medication management, and the combination of intensive behavioral treatment and stimulant medication. In terms of core symptom relief, stimulant medication was the most effective treatment, and combined treatment was no more effective than stimulant medication alone (MTA Cooperative Group, 1999). Behavioral treatment was advantageous in other ways, however. For example, children who received combined treatment were less disruptive at school than children receiving stimulant medication alone (Hinshaw et al., 2000). Other studies have found that children who receive behavioral treatment require lower doses of stimulant medication to achieve the desired outcomes (Pelham et al., 2005). This is important because children are better able to tolerate lower doses of stimulant medication. Further, parents report being more satisfied with treatment when behavioral management is included as a component in the program (Jensen et al., 2001). In sum, stimulant medication and behavioral treatment each have advantages and disadvantages that complement the other, and the best outcomes likely occur when both forms of treatment are used to improve children's behavior.

The Future of ADHD

It is difficult to predict the future; however, based on trends in research and public discourse, we can predict how the field may change as time progresses. This section will discuss two areas of research and public policy that will shape how we understand and treat ADHD in the coming decades.

Controlling Access to Stimulant Medication

It is no secret that many of the drugs used to treat ADHD are popular drugs of abuse among high school and college students, and this problem seems to be getting worse. The rate of illicit stimulant use has steadily risen over the past several decades (Teter, McCabe, Cranford, Boyd, & Guthrie, 2005), and it is probably not a coincidence that prescription rates for stimulant medication have increased during the same time period (Setlik, Bond, & Ho, 2009). Students who abuse stimulants often report doing so because they act as an academic performance enhancer by boosting alertness and concentration. Although they may enhance performance in the short term, nonmedical use of these drugs can lead to dependence and other adverse health consequences, especially when taken in ways other than prescribed (e.g., crushed and snorted) (Volkow & Swanson, 2003). Stimulants can be particularly dangerous when they are taken without supervision from a physician, because this may lead to adverse drug interactions or side effects. Because this increase in prescription stimulant abuse represents a threat to public health, an important goal for policy makers will be to reduce the availability of prescription stimulants to those who would use them for nonmedical reasons.

One of the first steps for addressing prescription stimulant abuse will be understanding how illicit users gain access to medication. Probably the most common method of obtaining stimulants is through drug diversion. The majority of college students who abuse stimulants report obtaining them from peers with valid prescriptions (McCabe & Boyd, 2005). Another way that would-be abusers may gain access to medication is by malingering (i.e., faking) symptoms of ADHD (Quinn, 2003). These individuals will knowingly exaggerate their symptoms to a physician in order to obtain a prescription. Other sources of illicit prescription



Prescription stimulant abuse among young people is a growing concern. [Image: Jesse! S?, <https://goo.gl/GIFyCg>, CC BY 2.0, <https://goo.gl/v4Y0Zv>]

drugs have been identified (e.g., pharmacy websites) (Califano, 2004), but more research is needed to understand how much these sources contribute to the problem. As we gain an understanding of how people gain access to illicit medication, policy makers and researchers can make efforts to curtail the rate of stimulant misuse. For example, because drug diversion is a major source of illicit stimulants, policymakers have enacted prescription monitoring programs to keep track of patient's prescription-seeking behavior (Office of Drug Control Policy, 2011), and, in some cases, patients are required to pass drug screens before receiving their prescriptions. To address malingering, researchers are working to develop psychological tests that can identify individuals who are faking symptoms (Jasinski et al., 2011). Finally, pharmacologists are working to develop stimulant medications that do not carry the same risk of abuse as the currently available drugs (e.g., lisdexamfetamine) (Biederman et al., 2007).

Although all of these measures will reduce illicit users' access to stimulant medication, it is important to consider how the policies will affect access among people who need these medications to treat their ADHD symptoms. Prescription tracking programs may reduce physicians' willingness to prescribe stimulants out of fear of being investigated by law enforcement. Patients with ADHD with comorbid substance abuse problems may be denied access to stimulant medication because they are considered high risk for drug diversion. Similarly, lengthy psychological evaluations to assess for malingering and mandated drug screenings may be prohibitively expensive for less affluent individuals with ADHD. These measures to reduce illicit drug use are necessary from a public health perspective, but as we move forward and enact policies to reduce stimulant abuse, it will be equally important to consider impact of such legislation on patients' access to treatment.

The Role of Neuroscience and Behavioral Genetics in Understanding ADHD

Much of the research on ADHD has been conducted to answer several deceptively complex questions: What causes ADHD? How are people with ADHD different from their typically developing peers? How can ADHD be prevented or treated? Historically, our tools for answering these questions was limited to observing outward human behavior, and our ability to ask questions about the physiology of ADHD was severely limited by the technology of the time. In the past two decades, however, rapid advances in technology (e.g., functional magnetic resonance imaging, genetic analysis) have allowed us to probe the physiological bases of human behavior. An exciting application of this technology is that we are able to extend our understanding of ADHD beyond basic behavior; we are learning about the underlying neurophysiology and genetics of the disorder. As we gain a fuller understanding of ADHD, we may be able to apply this knowledge to improve prevention and treatment of the disorder.

Knowledge of the underlying physiology of ADHD may guide efforts to develop new nonstimulant medications, which may not carry the side effects or abuse potential of traditional stimulants. Similarly, these advances may improve our ability to diagnose ADHD. Although it is extremely unlikely that a perfectly accurate genetic or neuroimaging test for ADHD will ever be developed (Thome et al., 2012), such procedures could be used in conjunction with behavioral evaluation and questionnaires to improve diagnostic accuracy. Finally, identifying genetic traits that predispose children to develop ADHD may allow physicians to use targeted prevention programs that could reduce the chances that children at risk for developing the disorder will experience symptoms.

Discussion Questions

1. Does ADHD meet the definition of a psychiatric disorder?
2. Explain the difference between developmentally appropriate and developmentally inappropriate behavior problems.
3. Do you believe that it is ethical to prescribe stimulant medication to children? Why or why not? What are the risks associated with withholding stimulant medication from children with ADHD?
4. How should society balance the need to treat individuals with ADHD using stimulants with public health concerns about the abuse of these same medications?

Vocabulary

Contingency management

A reward or punishment that systematically follows a behavior. Parents can use contingencies to modify their children's behavior.

Drug diversion

When a drug that is prescribed to treat a medical condition is given to another individual who seeks to use the drug illicitly.

Malingering

Fabrication or exaggeration of medical symptoms to achieve secondary gain (e.g., receive medication, avoid school).

Oppositional defiant disorder

A childhood behavior disorder that is characterized by stubbornness, hostility, and behavioral defiance. This disorder is highly comorbid with ADHD.

Parent management training

A treatment for childhood behavior problems that teaches parents how to use contingencies to more effectively manage their children's behavior.

Pathologizes

To define a trait or collection of traits as medically or psychologically unhealthy or abnormal.

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Social Anxiety

Todd Kashdan

Social anxiety occurs when we are overly concerned about being humiliated, embarrassed, evaluated, or rejected by others in social situations. Everyone experiences social anxiety some of the time, but for a minority of people, the frequency and intensity of social anxiety is intense enough to interfere with meaningful activities (e.g., relationships, academics, career aspirations). When a person's level of social anxiety is excessive, social interactions are either dreaded or avoided, social cues and emotions are difficult to understand, and positive thoughts and emotions are rare, then that person may be diagnosed with social anxiety disorder (or social phobia). There are effective treatments—with both medications and psychotherapy—for this problem. Unfortunately, only a small proportion of people with social anxiety disorder actually seek treatment.

Learning Objectives

- Distinguish social anxiety from social anxiety disorder.
- Identify commonly feared social situations.
- Know the prevalence and treatment rates of social anxiety disorder.
- Understand how social anxiety influences thoughts, feelings, and behaviors.
- Identify effective treatments for social anxiety disorder.

Introduction

A public speaker waits backstage before her name is called. She visualizes what will happen

in a few moments: the audience will cheer as she walks out and then turn silent, with all eyes on her. She imagines this will cause her to feel uncomfortable and, instead of standing balanced, she will lean to one side, not quite sure what to do with her hands. And when her mouth opens, instead of words, guttural sounds will emerge from a parched throat before her mind goes blank. In front of friends, family, and strangers, she is paralyzed with fear and embarrassment. Physically, in the moments leading up to the performance, she sweats, trembles, has difficulty breathing, notices a racing heartbeat, and feels nauseated. When someone asks her a question, she loses her voice or its pitch rises a few octaves. She attempts to hide her anxiety by tensing her muscles or telling herself to breathe and stay calm. Behaviorally, she seeks ways to escape the audience's gaze (e.g., by playing a video and asking the audience questions), and she tries to get through the performance as quickly as possible (e.g., rushing off the stage). Later, she works hard to avoid similar situations, passing up future speaking opportunities.

Welcome to the often terrifying world of social anxiety.



Social anxiety can be accompanied by physical manifestations such as shaking hands or a racing heartbeat. [Image: Kai Schreiber, <https://goo.gl/XLYZHn>, CC BY-SA 2.0, <https://goo.gl/rxiUsF>]

People have a fundamental need to feel like they belong and are liked, so it is painful when we feel rejected or left out by those who matter to us. In response, we often become acutely aware of the impression we make on others, and we avoid doing things that may cause others to be upset with us. **Social anxiety** is the *excessive* concern about being in social situations where scrutiny is likely. When people are socially anxious, they become overly concerned about embarrassing themselves, and they tend to reveal these signs of discomfort through

sweating or blushing; they worry that their character flaws will be exposed and result in rejection. See Figure 1 for examples of situations that commonly evoke social anxiety.

The term **anxiety** describes a general apprehension about possible future danger, rather than a reaction to an immediate threat (i.e., fear). Nevertheless, like fear, the experience of social anxiety may involve *physical*, *emotional*, and *behavioral* symptoms like those described in the example above.

Nearly everyone experiences some social anxiety at one point or another. It is particularly common before performing in front of an audience or meeting new people on one's own, and this is normal. Social anxiety provides information about the demands required of us to handle an ongoing challenge (Frijda, 1996). It lets us know that the situation is meaningful, and the impression we make on other people may be important to our social standing. Most people are able to "power through" the situation, eventually feeling more comfortable and learning that it was not as bad as expected. This is a fundamentally important point: people think that their anxiety leading up to a situation (anticipatory feelings) will only increase further in the actual situation, when, in fact, our anxiety tends to peak in the moments before a situation.

Sometimes, people experience more than the "normal" amount of anxiety. For people with excessive social anxiety, their anxiety often arises in a broader array of situations, is more intense, and does not subside as quickly. For those people, negative social outcomes are viewed as highly probable and costly, and their attention during social interactions tends to be inwardly directed (e.g., "Did my comment sound stupid? Can she tell that I'm sweating?"). This running internal commentary prevents people from focusing on the situation at hand, and even simple social interactions may become overwhelming (Bögels & Mansell, 2004).

Social Anxiety Disorder

When social anxiety and avoidance interfere with a person's ability to function in important roles (e.g., as a student, worker, friend), the condition is called **social anxiety disorder** (SAD), also known as social phobia (American Psychiatric Association, 2013). In the United States, SAD affects approximately 12.1% of people in their lifetimes and 7.1% of adults in a given year (Ruscio et al., 2008). About 1 of every 4 people report at least one significant social fear in their lifetimes—most commonly, public speaking (see Figure 1). To be diagnosed with SAD, a person must report an impairing fear of multiple social situations that has persisted for at least six months. Most people with SAD fear eight or more distinct social situations such as initiating a conversation with a stranger, maintaining conversations, going on a first date, going to a work party/function, talking with an authority figure, talking in front of a group of

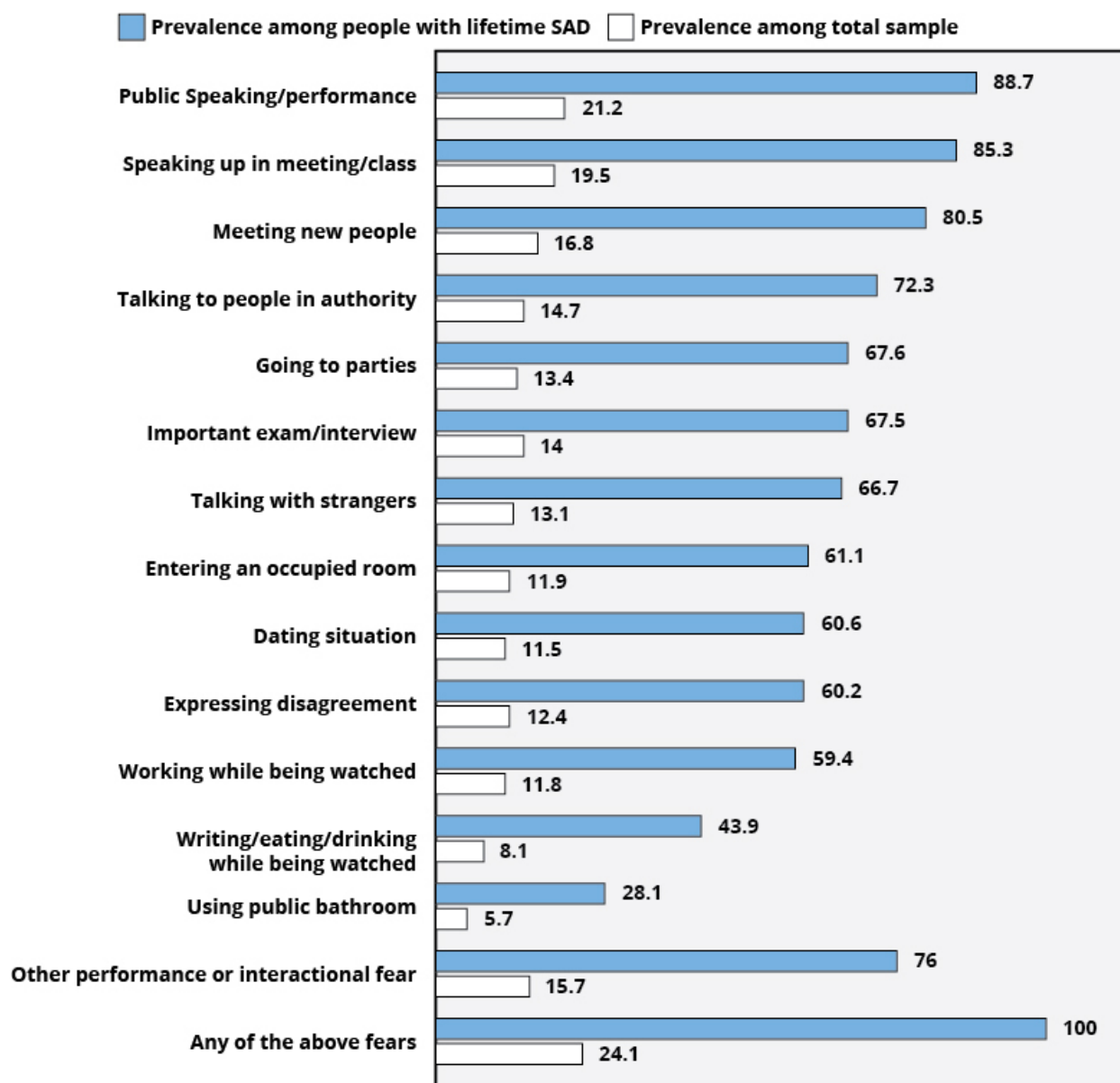


Figure 1: Lifetime Prevalence of Social Fears. Source: Ruscio, et al., 2008

people, and eating in front of other people (Ruscio et al., 2008).

SAD is one of the most common anxiety disorders recognized by the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5; APA, 2013). SAD affects men and women about equally, and the majority of people with SAD report that their fears began in early adolescence, typically around age 13 (Kessler et al., 2005). Unfortunately, this condition tends to be chronic and few people recover on their own without an intervention.

Despite the availability of effective treatments, few people seek help for their social fears (see Figure 2). In an epidemiological study, only 5.4% of people with SAD (and no other psychiatric



Fear of public performance is the most common of all social fears. [Image: Brisbane City Council, <https://goo.gl/QakNCF>, CC BY 2.0, <https://goo.gl/BRvSA7>]

disorders) ever received mental health treatment (Schneier, Johnson, Hornig, Liebowitz, & Weissman, 1992). There are several explanations for why people with SAD avoid treatment—for starters, the fear of being evaluated by a therapist and the stigma of seeking psychological services. Thus, the very features of the disorder may prevent a person from seeking treatment for it. Another explanation is that many physicians, teachers, parents, and peers do not believe that social anxiety disorder is a real condition and, instead, view it as nothing more than extreme shyness or inhibition. Finally, health care providers are often ill-equipped to assess

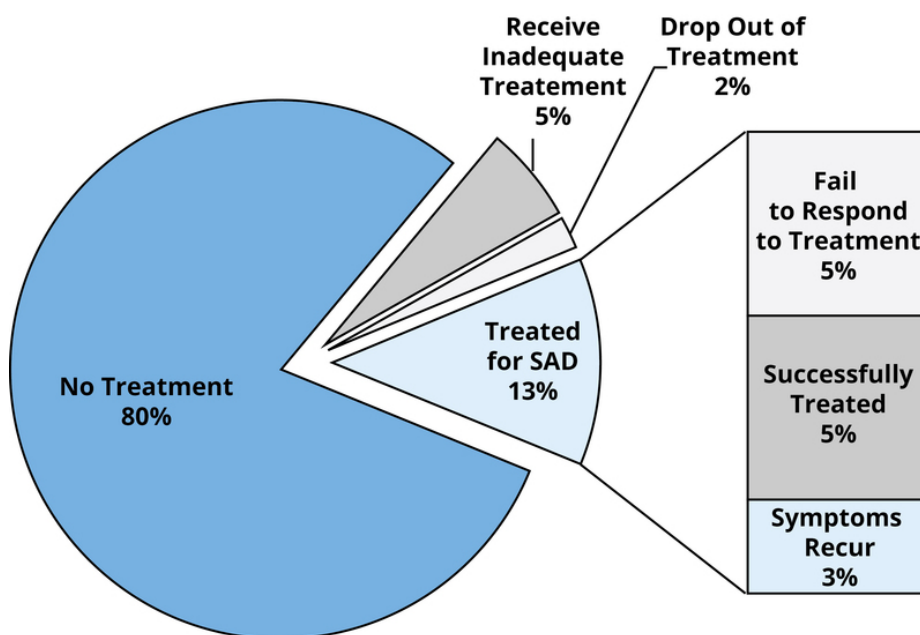


Figure 2: Treatment rate of adults with SAD based on data from Coles, et al., 2004; Feske and Chambless, 2005; Kessler, et al., 2005; and Ruscio, et al., 2008

SAD and may not be aware of evidence-based treatments (Kashdan, Christopher Frueh, Knapp, Hebert, & Magruder, 2006), and clients often do not know enough about social fears to discuss them with their doctors. Sadly, 60% to 80% of people with SAD suffer from symptoms for at least two decades (Ruscio et al., 2008). Thus, it is important to understand not only what social anxiety is but also what perpetuates social fears.

Fear of Evaluation

A central component of the social anxiety experience is how a person thinks about him- or herself, about others, and about social situations. According to the self-presentation theory of social anxiety (Leary & Kowalski, 1995), people feel socially anxious when they wish to make a good impression on others but doubt their ability to do so. People with excessive social anxiety are likely to view themselves as having more flaws or deficits, compared to those who rarely feel social anxiety (Clark & Wells, 1995); thus, for SAD sufferers, social interactions may seem like dangerous places where flaws can be observed and scrutinized (Moscovitch, 2009).

At first, researchers believed that the core feature of SAD was a fear of negative evaluation—being preoccupied with the possibility of being unfavorably judged or rejected by others (Watson & Friend, 1969). Recent evidence has suggested that people with SAD are actually concerned with both positive and negative evaluation. Fear of positive evaluation is the dread associated with success and public favorable evaluation, raising the expectations for subsequent social interactions. The fear of being positively evaluated is particularly relevant when a social comparison occurs, such as when a person gets a promotion at work (Weeks, Heimberg, Rodebaugh, & Norton, 2008; Weeks, Heimberg, & Rodebaugh, 2008). Both of these fears of evaluation contribute to social anxiety (Weeks, Heimberg, Rodebaugh, Goldin, & Gross, 2012).

Why might socially anxious people dread being praised? Gilbert's (2001) evolutionary theory suggests that social anxiety is a mechanism that evolved to facilitate group cohesion. When in society there are people of different social ranks, a person lower on the social hierarchy (e.g., an entry-level employee) would experience anxiety when interacting with higher-ranking group members (e.g., bosses). Such anxiety would lead a person to display submissive behavior (e.g., avoiding eye contact) and prompt them to avoid doing anything that could cause conflict. Anything that increases social status—such as receiving a promotion or dating an attractive romantic partner—can cause tension and conflict with others of higher status. Whereas fear of negative evaluation is relevant to other psychological conditions, such as depression and eating disorders, fear of positive evaluation is unique to SAD (Fergus et al., 2009; Weeks, Heimberg, Rodebaugh, et al., 2008). Furthermore, when people are successfully

treated for SAD, this fear of positive evaluation declines (Weeks et al., 2012).

Biased Attention and Interpretation

If you were to observe what people with SAD pay attention to in a social interaction, you would find that they are quick to recognize any signs of social threats. For instance, they are faster at detecting angry faces in a crowd (see Gilboa-Schechtman, Foa, & Amir, 1999). Imagine looking at the audience as you give a speech and the first faces you notice are scowling back!



People with SAD may apply interpretation biases to social situations that cause them to accentuate perceived negative interactions while failing to take notice of positive interactions. [Image: hobvias sudoneighm, <https://goo.gl/q46fPn>, CC BY 2.0, <https://goo.gl/BRvSA7>]

At the same time, SAD sufferers' attention is biased *away* from positive, rewarding information (see Taylor, Bomyea, & Amir, 2010). This means that people with SAD are unlikely to notice the smiling, nodding faces in the crowd, and they fail to pick up the subtle hints that somebody wants to spend more time with them or to be asked out on a romantic date. These interpretation and attention biases are obstacles to starting and maintaining social relationships. When you attend to only negativity, you start to believe that you are unlovable and that the world is a hostile, unfriendly place.

Complete the following sentence: "As I passed a group of people in the hall, they burst out in laughter, because . . ."

People with SAD are more likely to complete the sentence with a statement suggesting that there is something wrong with their behavior or appearance (e.g., “they thought I looked ridiculous”) as opposed to a neutral explanation (e.g., “one of them made a joke”). The problem is that when you assume people are attacking you, you feel more self-conscious and are less likely to stay in a situation and to interact with that group of people or others in the future. Our thoughts influence our behavior, and the negative interpretations and predictions of people with SAD only serve to feed their social avoidance patterns (Amir, Beard, & Bower, 2005).

Deficient Positive Experiences

The strongest predictor of a happy, meaningful, long-lasting life is the presence of satisfying, healthy relationships (Berscheid & Reis, 1998). Thus, the fact that people with SAD frequently avoid social interactions—even those with the potential for fun or intimacy—means that they miss out on an important source of positive experiences.

By studying people’s day-to-day experiences, researchers have discovered several positivity deficits in the lives of socially anxious people. For example, Kashdan and Collins (2010) gave participants portable electronic devices that randomly prompted them to describe what they were feeling and doing multiple times per day for several weeks. During such random assessments, socially anxious people reported less intense positive emotions (e.g., joy,



Individuals with SAD may suffer from "positivity deficits" that limit the frequency or intensity of positive emotions such as joy or calm. [Image: Jan Jespersen, <https://goo.gl/ohXeJY>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

happiness, calm), regardless of whether they were around other people (whereas, less anxious people report more intense positive emotions when socializing). Socially anxious people experience less frequent positive emotions even when spending time with close friends and family members (Brown, Silvia, Myin-Germeys, & Kwapil, 2007; Vittengl & Holt, 1998). In fact, even in the most intimate of situations—during sexual encounters with romantic partners—socially anxious people report less intense pleasure and less intimacy (Kashdan, Adams, et al., 2011). All of these findings highlight the vast reach of excessive social anxiety in people's lives and how it detracts from the relationships and activities that hold the greatest promise for happiness and meaning in life (Kashdan, Weeks, & Savostyanova, 2011).

Problematic Emotion Regulation

A possible explanation for the distress and diminished positive experiences seen in SAD is that the sufferers' ability to respond to and manage their emotions is impaired. **Emotion regulation** refers to how people recognize, interpret, experience, and attempt to alter emotional states (Gross, 1998). One symptom of SAD is the concern that the anxiety will be visible to others (APA, 2013). Given this concern, socially anxious people spend considerable time and effort preparing for and avoiding anxiety-related thoughts, sensations, and behaviors. They engage in **safety behaviors**, such as rehearsing exactly what to say in a conversation, asking questions of others to deflect attention from themselves, and holding a drink or food to have an excuse to pause before responding to a question (Clark & Wells, 1995). Because there is only so much we can pay attention to in a given moment, excessive self-focused attention detracts from a person's ability to be mindful in a social encounter. In effect, by devoting effort to controlling emotions and minimizing the potential for rejection, a person paradoxically increases the likelihood of misunderstanding others or appearing aloof. Such encounters are also less enjoyable and possess less potential for deepening relationships.

Socially anxious people believe that openly expressing emotions is likely to have negative consequences (Juretic & Zivcic-Becirevic, 2013). In turn, they are more apt to suppress or hide their negative emotions (Spokas, Luterek, & Heimberg, 2009) and avoid anything that is distressing (Kashdan, Morina, & Priebe, 2009). Emotion suppression is often ineffective: the more we try not to think about something, the more we end up thinking about it (Richards & Gross, 1999). Unfortunately, people with SAD report being less skilled at using more effective emotion regulation strategies, such as finding alternative, constructive ways of thinking about a situation (Werner, Goldin, Ball, Heimberg, & Gross, 2011).

Socially anxious people also respond to positive emotions in an unexpected way. Whereas

most people not only enjoy positive emotions but also seek them out and attempt to savor them, socially anxious people often fear intense positive emotions (Turk, Heimberg, Luterek, Mennin, & Fresco, 2005). When positive emotions arise, just like negative emotions, SAD sufferers make efforts to suppress them (Eisner, Johnson, & Carver, 2009; Farmer & Kashdan, 2012). Why downplay positive emotions? It is possible that avoiding public displays of positive emotions is another way that people with SAD can avoid scrutiny (e.g., not laughing because others might not find a joke funny) and prevent the wrath of powerful others (e.g., not expressing excitement about a personal triumph because others might be envious) (Weeks et al., 2008). A recent study sampled the day-to-day social interactions of people with and without SAD to uncover what distinguishes these two groups. What the researchers found was that the amount of social anxiety felt during social interactions was less important in distinguishing people with SAD from healthy adults than the intense effort put into avoid feeling anxious and infrequent positive emotions when spending time with other people (Kashdan et al., 2013).

The ego depletion model (Muraven & Baumeister, 2000) proposes that people have a limited capacity for physical and mental self-control (e.g., physical endurance, attention). When we perform tasks that require significant effort and energy (e.g., suppressing emotions), we deplete these self-control resources, leaving us with less capacity to focus on subsequent tasks or to make good decisions (Vohs, Baumeister, & Ciarocco, 2005). When depleted or mentally exhausted, we tend to opt for whatever is immediately rewarding as opposed to pursuing meaningful goals (Hayes, Luoma, Bond, Masuda, & Lillis, 2006). For socially anxious people, what is immediately rewarding tends to be escaping or avoiding social situations in order to minimize the potential for unpleasant feelings. In other words, the way people with high social anxiety control their emotions not only makes their social situations less pleasant in the moment but also limits their capacity for pursuing rewarding opportunities afterward. Consistent with this, Farmer and Kashdan (2012) demonstrated that over the course of two weeks, when socially anxious people used more emotion suppression, they experienced fewer pleasant social events and less intense positive emotions on the following day. Taken together, this research suggests that socially anxious people respond to their emotions in ways that have far-reaching effects on their well-being, likely maintaining fears associated with social anxiety.

Treatments

Although SAD tends to be a chronic condition if left untreated (Wittchen, Fuetsch, Sonntag, Müller, & Liebowitz, 2000), the good news is that there are effective treatments that reduce social fears. Currently, there are two gold-standard treatments for SAD: cognitive behavioral

therapy (CBT) and pharmacotherapy (Gould, Buckminster, Pollack, & Otto, 1997). The frontrunner among psychotherapy options, CBT, is an approach mental health professionals (e.g., licensed clinical psychologists) use to help people with SAD learn to think, behave, and feel differently so that they can feel more comfortable in social situations and improve their quality of life (Heimberg & Becker, 2002).

The most effective strategy to treat SAD is exposure (Feske & Chambless, 1995)—where clients repeatedly confront their feared situations without the use of safety behaviors, starting with situations that are only slightly anxiety provoking (e.g., imagining a conversation with an attractive stranger) and gradually working their way up to more frightening situations (e.g., starting conversations with trained actors during therapy sections). Additional exposures are then assigned between sessions so that people can experiment with feared situations in their daily lives (e.g., saying hello to a passing pedestrian). As another example, someone who avoids riding elevators due to fear of interacting with other riders might start out by taking an elevator during off-peak hours, then during popular times, and then practicing talking with a stranger while riding an elevator, and so on. After taking part in these exposures, people learn that feared social situations are not as probable or dangerous as previously believed.



The exposure strategy encourages people with SAD to gradually build up their exposure to feared situations - like having conversations in public - without safety behaviors. [Image: torbakhopper, <https://goo.gl/BqeFh2>, CC BY-NC-ND 2.0, <https://goo.gl/bhtmlY>]

Cognitive techniques form the other component of CBT. These are strategies therapists use to help people develop more realistic and helpful thoughts and expectations about social situations. For example, people with SAD often have unrealistic beliefs that contribute to anxiety (e.g., “Everyone can see that I’m sweating”). The therapist helps them challenge such thoughts and develop more helpful expectations about situations (e.g., from “If I pause while speaking, then everyone will think I’m stupid” to “It is OK to pause; the silence may seem longer to me than to others”). These techniques are most effective in combination with behavioral techniques that help clients test out some of their assumptions in real situations (Taylor, 1996). For instance, a behavioral experiment might involve giving a cashier the wrong change (making a mistake), to test whether the feared consequence (“the cashier will laugh at me”) actually arises and, if so, whether it is as painful as expected.

Pharmacotherapy for SAD involves using medications to reduce people’s anxiety level so that they are able to stop avoiding situations and enjoy a better quality of life. The current first-line prescribed medications are selective serotonin re-uptake inhibitors (SSRIs), such as escitalopram, paroxetine, and sertraline, as well as serotonin norepinephrine reuptake inhibitors (SNRIs) like venlafaxine (Bandelow et al., 2012). Both of these medications are typically used as antidepressants that act on the neurotransmitter serotonin, which plays a big role in how the amygdala responds to possibly threatening information. SNRIs also act on norepinephrine at higher doses. These medications have few side effects and also are likely to improve symptoms of other anxiety or depression symptoms that often co-occur with SAD. Other classes of medications with some evidence of helpfulness for SAD symptoms include benzodiazepines and monoamine oxidase inhibitors, though these medications often produce a number of negative side effects (Blanco, Bragdon, Schneier, & Liebowitz, 2013).

Both approaches—CBT and medications—are moderately helpful at reducing social anxiety symptoms in approximately 60% of clients; however, the majority of people with SAD do not fully remit, and many experience a return of symptoms after treatment ends (Fedoroff & Taylor, 2001). Notably, CBT tends to have more lasting effects, and there is some modest evidence for possible added benefit when combining medications with psychotherapy (Blanco et al., 2010). It is obvious, though, that existing treatments are insufficient.

Current recommended treatments may not address some of the deficits discussed earlier. Specifically, people may avoid fewer social situations, but they might still have fewer, less intense positive life experiences. Researchers are constantly improving available treatments via new techniques and medications. Some new developments in the treatment of SAD include the encouragement of mindful awareness (vs. self-focus) and acceptance (vs. avoidance) of experiences (Dalrymple & Herbert, 2007; Goldin & Gross, 2010). As new treatments develop, it will be important to see whether these treatments not only improve SAD symptoms but

also help sufferers achieve greater happiness, meaning in life, and success.

Conclusions

In this module, we discussed the normal experience of social anxiety as well as the clinically impairing distress suffered by people with SAD. It is important to remember that nearly every psychological experience and characteristic you will read about exists on a continuum. What appears to distinguish people with SAD from healthy adults is not the presence of intense social anxiety but the unwillingness to experience anxious thoughts, feelings, and sensations, and the immense effort put into avoiding this discomfort. Other problems linked to excessive social anxiety include infrequent positive events, diminished positive experiences, and a tendency to view benign and even positive social situations as threatening. Together, these symptoms prevent people from initiating and maintaining healthy social relationships, and lead to deficient well-being. When social fears become overwhelming, it is important to remember that effective treatments are available to improve one's quality of life.

Outside Resources

Institution: Andrew Kukes Foundation for Social Anxiety

<http://akfsa.org/>

Institution: Anxiety and Depression Association of America

<http://www.adaa.org/>

Video: Social Anxiety Documentary - Afraid of People

<http://www.youtube.com/watch?v=gmEJEfy5f50>

Web: CalmClinic

<http://www.calmclinic.com/>

Web: Which Celebrities Suffer with Social Anxiety?

<https://www.verywell.com/which-celebrities-suffer-with-social-anxiety-3024283>

Discussion Questions

1. What differentiates people who are shy from those with social anxiety disorder?
2. Because the most effective treatment for social anxiety disorder is exposure to feared situations, what kinds of exposures would you devise for someone who fears talking in front of an audience? Engaging in small talk? Writing or eating in front of others? Speaking up in a small group? Talking to strangers?
3. Why might social anxiety disorder typically begin in late childhood/early adolescence?
4. How does culture influence fears of negative and positive evaluation? After all, social groups differ in their adherence to a vertical social hierarchy.
5. What may be some reasons people with severe social anxiety might not seek or receive treatment? How would you remove these obstacles?

Vocabulary

Amygdala

A brain structure in the limbic system involved in fear reactivity and implicated in the biological basis for social anxiety disorder.

Anxiety

A state of worry or apprehension about future events or possible danger that usually involves negative thoughts, unpleasant physical sensations, and/or a desire to avoid harm.

Cognitive behavioral therapy (CBT)

Psychotherapy approach that incorporates cognitive techniques (targeting unhelpful thoughts) and behavioral techniques (changing behaviors) to improve psychological symptoms.

Ego depletion

The idea that people have a limited pool of mental resources for self-control (e.g., regulating emotions, willpower), and this pool can be used up (depleted).

Emotion regulation

The ability to recognize emotional experiences and respond to situations by engaging in strategies to manage emotions as necessary.

Exposure treatment

A technique used in behavior therapy that involves a patient repeatedly confronting a feared situation, without danger, to reduce anxiety.

Fear of negative evaluation

The preoccupation with and dread of the possibility of being judged negatively by others.

Fear of positive evaluation

The dread associated with favorable public evaluation or acknowledgment of success, particularly when it involves social comparison.

Pharmacotherapy

A treatment approach that involves using medications to alter a person's neural functioning to reduce psychological symptoms.

Safety behaviors

Actions people take to reduce likelihood of embarrassment or minimizing anxiety in a situation (e.g., not making eye contact, planning what to say).

Selective serotonin re-uptake inhibitors (SSRIs)

A class of antidepressant medications often used to treat SAD that increase the concentration of the neurotransmitter serotonin in the brain.

Serotonin norepinephrine reuptake inhibitors (SNRIs)

A class of antidepressant medications often used to treat SAD that increase the concentration of serotonin and norepinephrine in the brain.

Social anxiety

Excessive anticipation and distress about social situations in which one may be evaluated negatively, rejected, or scrutinized.

Social anxiety disorder (SAD)

An anxiety disorder marked by severe and persistent social anxiety and avoidance that interferes with a person's ability to fulfill their roles in important life domains.

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Psychopharmacology

Susan Barron

Psychopharmacology is the study of how drugs affect behavior. If a drug changes your perception, or the way you feel or think, the drug exerts effects on your brain and nervous system. We call drugs that change the way you think or feel psychoactive or psychotropic drugs, and almost everyone has used a psychoactive drug at some point (yes, caffeine counts). Understanding some of the basics about psychopharmacology can help us better understand a wide range of things that interest psychologists and others. For example, the pharmacological treatment of certain neurodegenerative diseases such as Parkinson's disease tells us something about the disease itself. The pharmacological treatments used to treat psychiatric conditions such as schizophrenia or depression have undergone amazing development since the 1950s, and the drugs used to treat these disorders tell us something about what is happening in the brain of individuals with these conditions. Finally, understanding something about the actions of drugs of abuse and their routes of administration can help us understand why some psychoactive drugs are so addictive. In this module, we will provide an overview of some of these topics as well as discuss some current controversial areas in the field of psychopharmacology.

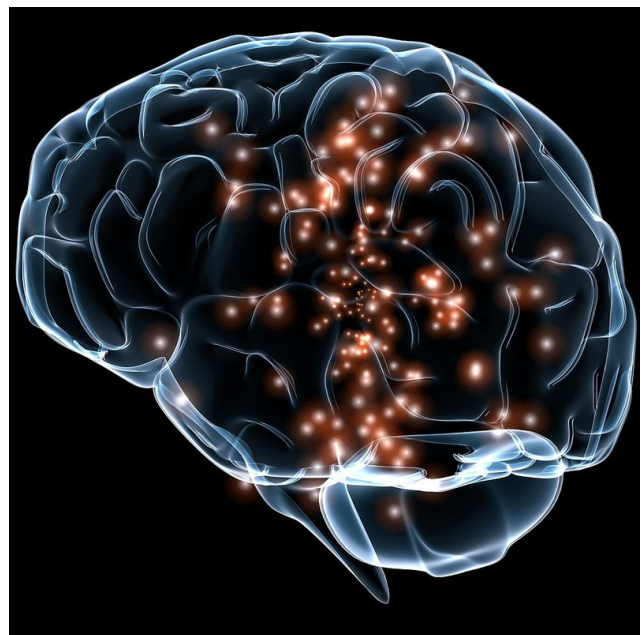
Learning Objectives

- How do the majority of psychoactive drugs work in the brain?
- How does the route of administration affect how rewarding a drug might be?
- Why is grapefruit dangerous to consume with many psychotropic medications?
- Why might individualized drug doses based on genetic screening be helpful for treating conditions like depression?
- Why is there controversy regarding pharmacotherapy for children, adolescents, and the elderly?

Introduction

Psychopharmacology, the study of how drugs affect the brain and behavior, is a relatively new science, although people have probably been taking drugs to change how they feel from early in human history (consider the of eating fermented fruit, ancient beer recipes, chewing on the leaves of the cocaine plant for stimulant properties as just some examples). The word *psychopharmacology* itself tells us that this is a field that bridges our understanding of behavior (and brain) and pharmacology, and the range of topics included within this field is extremely broad.

Virtually any drug that changes the way you feel does this by altering how neurons communicate with each other. Neurons (86 billion in your nervous system) communicate with each other by releasing a chemical (**neurotransmitter**) across a tiny space between two neurons (the **synapse**). When the neurotransmitter crosses the synapse, it binds to a postsynaptic receptor (protein) on the receiving neuron and the message may then be transmitted onward. Obviously, neurotransmission is far more complicated than this – links at the end of this module can provide some useful background if you want more detail – but the first step is understanding that virtually all **psychoactive drugs** interfere with or alter how neurons communicate with each other.



Drugs that alter our feelings and behavior do so by affecting the communication between neurons in the brain. [Image: <https://goo.gl/oQCafL>, CC0 Public Domain, <https://goo.gl/m25gce>]

There are many neurotransmitters. Some of the most important in terms of psychopharmacological treatment and drugs of abuse are outlined in Table 1. The neurons that release these neurotransmitters, for the most part, are localized within specific circuits of the brain that mediate these behaviors. Psychoactive drugs can either increase activity at the synapse (these are called **agonists**) or reduce activity at the synapse (**antagonists**). Different drugs do this by different mechanisms, and some examples of agonists and antagonists are presented in Table 2. For each example, the drug's trade name, which is the name of the drug provided by the drug company, and generic name (in parentheses) are

provided.

Neurotransmitter	Abbreviation	Behaviors or Diseases Related to These Neurotransmitter
Acetylcholine	ACh	Learning and memory; Alzheimer's disease; muscle movement in the peripheral nervous system
Dopamine	DA	Reward circuits; Motor circuits involved in Parkinson's disease; Schizophrenia
Norepinephrine	NE	Arousal; Depression
Serotonin	5HT	Depression; Aggression; Schizophrenia
Glutamate	GLU	Learning; Major excitatory neurotransmitter in the brain
GABA	GABA	Anxiety disorders; Epilepsy; Major inhibitory neurotransmitter in the brain
Endogenous Opioids	Endorphins, Enkephalins	Pain; Analgesia; Reward

Table 1

A very useful link at the end of this module shows the various steps involved in neurotransmission and some ways drugs can alter this.

Table 2 provides examples of drugs and their primary mechanism of action, but it is very important to realize that drugs also have effects on other neurotransmitters. This contributes to the kinds of side effects that are observed when someone takes a particular drug. The reality is that no drugs currently available work only exactly where we would like in the brain or only on a specific neurotransmitter. In many cases, individuals are sometimes prescribed one **psychotropic drug** but then may also have to take additional drugs to reduce the side effects caused by the initial drug. Sometimes individuals stop taking medication because the side effects can be so profound.

Pharmacokinetics: What Is It – Why Is It Important?

While this section may sound more like pharmacology, it is important to realize how important pharmacokinetics can be when considering psychoactive drugs. **Pharmacokinetics** refers to how the body handles a drug that we take. As mentioned earlier, psychoactive drugs exert

Drug	Mechanism	Use	Agonist/Antagonist
L-dopa	Increase synthesis of DA	Parkinson's disease	Agonist for DA
Adderall (mixed salts amphetamine)	Increase release of DA, NE	ADHD	Agonist for DA, NE
Ritalin (methylphenidate)	Blocks removal of DA, NE, and lesser (5HT) from synapse	ADHD	Agonist for DA, NE mostly
Aricept (donepezil)	Blocks removal of ACh from synapse	Alzheimer's disease	Agonist for ACh
Prozac (fluoxetine)	Blocks removal of 5HT from synapse	Depression, obsessive compulsive disorder	Agonist 5HT
Seroquel (quetiapine)	Blocks DA and 5HT receptors	Schizophrenia, bipolar disorder	Antagonist for DA, 5HT
Revia (naltrexone)	Blocks opiod post-synaptic receptors	Alcoholism, opiod addiction	Antagonist (for opioids)

Table 2

their effects on behavior by altering neuronal communication in the brain, and the majority of drugs reach the brain by traveling in the blood. The acronym ADME is often used with A

standing for absorption (how the drug gets into the blood), Distribution (how the drug gets to the organ of interest – in this module, that is the brain), Metabolism (how the drug is broken down so it no longer exerts its psychoactive effects), and Excretion (how the drug leaves the body). We will talk about a couple of these to show their importance for considering psychoactive drugs.



A drug delivered by IV reaches the brain more quickly than if the drug is taken orally. While rapid delivery has advantages, there are also risks involved with IV administration. [Image: Calleamanecer, <https://goo.gl/OX6Yj5>, CC BY-SA 3.0, <https://goo.gl/eLCn2O>]

Drug Administration

There are many ways to take drugs, and these routes of drug administration can have a significant impact on how quickly that drug reaches brain. The most common route of administration is oral administration, which is relatively slow and – perhaps

surprisingly – often the most variable and complex route of administration. Drugs enter the stomach and then get absorbed by the blood supply and capillaries that line the small intestine. The rate of absorption can be affected by a variety of factors including the quantity and the type of food in the stomach (e.g., fats vs. proteins). This is why the medicine label for some drugs (like antibiotics) may specifically state foods that you should or should NOT consume within an hour of taking the drug because they can affect the rate of absorption. Two of the most rapid routes of administration include inhalation (i.e., smoking or gaseous anesthesia) and intravenous (IV) in which the drug is injected directly into the vein and hence the blood supply. Both of these routes of administration can get the drug to brain in less than 10 seconds. IV administration also has the distinction of being the most dangerous because if there is an adverse drug reaction, there is very little time to administer any antidote, as in the case of an IV heroin overdose.

Why might how quickly a drug gets to the brain be important? If a drug activates the reward circuits in the brain AND it reaches the brain very quickly, the drug has a high risk for abuse and addiction. Psychostimulants like amphetamine or cocaine are examples of drugs that have high risk for abuse because they are agonists at DA neurons involved in reward AND because these drugs exist in forms that can be either smoked or injected intravenously. Some argue that cigarette smoking is one of the hardest addictions to quit, and although part of the reason for this may be that smoking gets the nicotine into the brain very quickly (and indirectly acts on DA neurons), it is a more complicated story. For drugs that reach the brain very quickly, not only is the drug very addictive, but so are the cues associated with the drug (see Rohsenow, Niaura, Childress, Abrams, & Monti, 1990). For a crack user, this could be the pipe that they use to smoke the drug. For a cigarette smoker, however, it could be something as normal as finishing dinner or waking up in the morning (if that is when the smoker usually has a cigarette). For both the crack user and the cigarette smoker, the cues associated with the drug may actually cause craving that is alleviated by (you guessed it) – lighting a cigarette or using crack (i.e., relapse). This is one of the reasons individuals that enroll in drug treatment programs, especially out-of-town programs, are at significant risk of relapse if they later find themselves in proximity to old haunts, friends, etc. But this is much *more* difficult for a cigarette smoker. How can someone avoid eating? Or avoid waking up in the morning, etc. These examples help you begin to understand how important the route of administration can be for psychoactive drugs.

Drug Metabolism

Metabolism involves the breakdown of psychoactive drugs, and this occurs primarily in the liver. The liver produces enzymes (proteins that speed up a chemical reaction), and these

enzymes help catalyze a chemical reaction that breaks down psychoactive drugs. Enzymes exist in “families,” and many psychoactive drugs are broken down by the same family of enzymes, the cytochrome P450 superfamily. There is not a unique enzyme for each drug; rather, certain enzymes can break down a wide variety of drugs. Tolerance to the effects of many drugs can occur with repeated exposure; that is, the drug produces less of an effect over time, so more of the drug is needed to get the same effect. This is particularly true for sedative drugs like alcohol or opiate-based painkillers. *Metabolic tolerance* is one kind of tolerance and it takes place in the liver. Some drugs (like alcohol) cause **enzyme induction** – an increase in the enzymes produced by the liver. For example, chronic drinking results in alcohol being broken down more quickly, so the alcoholic needs to drink more to get the same effect – of course, until so much alcohol is consumed that it damages the liver (alcohol can cause fatty liver or cirrhosis).

Recent Issues Related to Psychotropic Drugs and Metabolism

Grapefruit Juice and Metabolism



Grapefruit can interfere with enzymes in the liver that help the body to process certain drugs. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Certain types of food in the stomach can alter the rate of drug absorption, and other foods can also alter the rate of drug metabolism. The most well known is grapefruit juice. Grapefruit juice suppresses cytochrome P450 enzymes in the liver, and these liver enzymes normally break down a large variety of drugs (including some of the psychotropic drugs). If the enzymes are suppressed, drug levels can build up to potentially toxic levels. In this case, the effects can persist for extended periods of time after the consumption of grapefruit juice. As of 2013, there are at least 85 drugs shown to adversely interact with grapefruit juice (Bailey, Dresser, & Arnold, 2013). Some psychotropic drugs that are likely to interact with grapefruit juice include carbamazepine (Tegretol), prescribed for

bipolar disorder; diazepam (Valium), used to treat anxiety, alcohol withdrawal, and muscle spasms; and fluvoxamine (Luvox), used to treat obsessive compulsive disorder and

depression. A link at the end of this module gives the latest list of drugs reported to have this unusual interaction.

Individualized Therapy, Metabolic Differences, and Potential Prescribing Approaches for the Future

Mental illnesses contribute to more disability in western countries than all other illnesses including cancer and heart disease. Depression alone is predicted to be the second largest contributor to disease burden by 2020 (World Health Organization, 2004). The numbers of people affected by mental health issues are pretty astonishing, with estimates that 25% of adults experience a mental health issue in any given year, and this affects not only the individual but their friends and family. One in 17 adults experiences a serious mental illness (Kessler, Chiu, Demler, & Walters, 2005). Newer antidepressants are probably the most frequently prescribed drugs for treating mental health issues, although there is no “magic bullet” for treating depression or other conditions. Pharmacotherapy with psychological therapy may be the most beneficial treatment approach for many psychiatric conditions, but there are still many unanswered questions. For example, why does one antidepressant help one individual yet have no effect for another? Antidepressants can take 4 to 6 weeks to start improving depressive symptoms, and we don't really understand why. Many people do not respond to the first antidepressant prescribed and may have to try different drugs before finding something that works for them. Other people just do not improve with antidepressants (Ioannidis, 2008). As we better understand why individuals differ, the easier and more rapidly we will be able to help people in distress.

One area that has received interest recently has to do with an individualized treatment approach. We now know that there are genetic differences in some of the cytochrome P450 enzymes and their ability to break down drugs. The general population falls into the following 4 categories: 1) *ultra-extensive metabolizers* break down certain drugs (like some of the current antidepressants) very, very quickly, 2) *extensive metabolizers* are also able to break down drugs fairly quickly, 3) *intermediate metabolizers* break down drugs more slowly than either of the two above groups, and finally 4) *poor metabolizers* break down drugs much more slowly than all of the other groups. Now consider someone receiving a prescription for an antidepressant – what would the consequences be if they were either an ultra-extensive metabolizer or a poor metabolizer? The ultra-extensive metabolizer would be given antidepressants and told it will probably take 4 to 6 weeks to begin working (this is true), but they metabolize the medication so quickly that it will never be effective for them. In contrast, the poor metabolizer given the same daily dose of the same antidepressant may build up such high levels in their blood (because they are not breaking the drug down), that they will have a wide range of side effects

and feel really badly – also not a positive outcome. What if – instead – prior to prescribing an antidepressant, the doctor could take a blood sample and determine which type of metabolizer a patient actually was? They could then make a much more informed decision about the best dose to prescribe. There are new genetic tests now available to better individualize treatment in just this way. A blood sample can determine (at least for some drugs) which category an individual fits into, but we need data to determine if this actually is effective for treating depression or other mental illnesses (Zhou, 2009). Currently, this genetic test is expensive and not many health insurance plans cover this screen, but this may be an important component in the future of psychopharmacology.

Other Controversial Issues

Juveniles and Psychopharmacology

A recent Centers for Disease Control (CDC) report has suggested that as many as 1 in 5 children between the ages of 5 and 17 may have some type of mental disorder (e.g., ADHD, autism, anxiety, depression) (CDC, 2013). The incidence of bipolar disorder in children and adolescents has also increased 40 times in the past decade (Moreno, Laje, Blanco, Jiang, Schmidt, & Olfson, 2007), and it is now estimated that 1 in 88 children have been diagnosed with an autism spectrum disorder (CDC, 2011). Why has there been such an increase in these numbers? There is no single answer to this important question. Some believe that greater public awareness has contributed to increased teacher and parent referrals. Others argue that the increase stems from changes in criterion currently used for diagnosing. Still others suggest environmental factors, either prenatally or postnatally, have contributed to this upsurge.

We do not have an answer, but the question does bring up an additional controversy related to how we should treat this population of children and adolescents. Many psychotropic drugs used for treating psychiatric disorders have been tested in adults, but few have been tested for safety



There are concerns about both the safety and efficacy of drugs like Prozac for children and teens. [Image: zaza_bj, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

or efficacy with children or adolescents. The most well-established psychotropics prescribed for children and adolescents are the psychostimulant drugs used for treating attention deficit hyperactivity disorder (ADHD), and there are clinical data on how effective these drugs are. However, we know far less about the safety and efficacy in young populations of the drugs typically prescribed for treating anxiety, depression, or other psychiatric disorders. The young brain continues to mature until probably well after age 20, so some scientists are concerned that drugs that alter neuronal activity in the developing brain could have significant consequences. There is an obvious need for clinical trials in children and adolescents to test the safety and effectiveness of many of these drugs, which also brings up a variety of ethical questions about who decides what children and adolescents will participate in these clinical trials, who can give consent, who receives reimbursements, etc.

The Elderly and Psychopharmacology

Another population that has not typically been included in clinical trials to determine the safety or effectiveness of psychotropic drugs is the elderly. Currently, there is very little high-quality evidence to guide prescribing for older people – clinical trials often exclude people with multiple comorbidities (other diseases, conditions, etc.), which are typical for elderly populations (see Hilmer and Gnjidict, 2008; Pollock, Forsyth, & Bies, 2008). This is a serious issue because the elderly consume a disproportionate number of the prescription meds prescribed. The term **polypharmacy** refers to the use of multiple drugs, which is very common

Clinical Trial Subjects	Aged Patients Who Receive Drug Therapies
One drug	Drug of interest and medications
Single doses	Chronic administration
No disease	Multiple diseases
No alcohol, tobacco, OTC* drugs, nutraceuticals	OTC* drugs, nutraceuticals, alcohol, tobacco, and other
20-40 years (vs 60-75 years)	65-100+ years
Caucasians	Caucasians and minorities
Selection bias	All comers/socioeconomic basis

*OTC = Over the counter

Table 3. Characteristics of clinical trial subjects vs. actual patients. (Reprinted by permission from Schwartz & Abernethy, 2008.)

in elderly populations in the United States. As our population ages, some estimate that the proportion of people 65 or older will reach 20% of the U.S. population by 2030, with this group consuming 40% of the prescribed medications. As shown in Table 3 (from Schwartz and Abernethy, 2008), it is quite clear why the typical clinical trial that looks at the safety and effectiveness of psychotropic drugs can be problematic if we try to interpret these results for an elderly population.

Metabolism of drugs is often slowed considerably for elderly populations, so less drug can produce the same effect (or all too often, too much drug can result in a variety of side effects). One of the greatest risk factors for elderly populations is falling (and breaking bones), which can happen if the elderly person gets dizzy from too much of a drug. There is also evidence that psychotropic medications can reduce bone density (thus worsening the consequences if someone falls) (Brown & Mezuk, 2012). Although we are gaining an awareness about some of the issues facing pharmacotherapy in older populations, this is a very complex area with many medical and ethical questions.

This module provided an introduction of some of the important areas in the field of psychopharmacology. It should be apparent that this module just touched on a number of topics included in this field. It should also be apparent that understanding more about psychopharmacology is important to anyone interested in understanding behavior and that our understanding of issues in this field has important implications for society.

Outside Resources

Video: Neurotransmission

<http://www.youtube.com/watch?v=FR4S1BqdFG4>

Web: Description of how some drugs work and the brain areas involved - 1

<http://www.drugabuse.gov/news-events/nida-notes/2007/10/impacts-drugs-neurotransmission>

Web: Description of how some drugs work and the brain areas involved - 2

<http://learn.genetics.utah.edu/content/addiction/mouse/>

Web: Information about how neurons communicate and the reward pathways

<http://learn.genetics.utah.edu/content/addiction/rewardbehavior/>

Web: National Institute of Alcohol Abuse and Alcoholism

<http://www.niaaa.nih.gov/>

Web: National Institute of Drug Abuse

<http://www.drugabuse.gov/>

Web: National Institute of Mental Health

<http://www.nimh.nih.gov/index.shtml>

Web: Neurotransmission

https://science.education.nih.gov/supplements/nih2/Addiction/activities/lesson2_neurotransmission.html

Web: Report of the Working Group on Psychotropic Medications for Children and Adolescents: Psychopharmacological, Psychosocial, and Combined Interventions for Childhood Disorders: Evidence Base, Contextual Factors, and Future Directions (2008):

<http://www.apa.org/pi/families/resources/child-medications.pdf>

Web: Ways drugs can alter neurotransmission

http://thebrain.mcgill.ca/flash/d/d_03/d_03_m/d_03_m_par/d_03_m_par.html

Discussion Questions

1. What are some of the issues surrounding prescribing medications for children and

adolescents? How might this be improved?

2. What are some of the factors that can affect relapse to an addictive drug?
3. How might prescribing medications for depression be improved in the future to increase the likelihood that a drug would work and minimize side effects?

Vocabulary

Agonists

A drug that increases or enhances a neurotransmitter's effect.

Antagonist

A drug that blocks a neurotransmitter's effect.

Enzyme

A protein produced by a living organism that allows or helps a chemical reaction to occur.

Enzyme induction

Process through which a drug can enhance the production of an enzyme.

Metabolism

Breakdown of substances.

Neurotransmitter

A chemical substance produced by a neuron that is used for communication between neurons.

Pharmacokinetics

The action of a drug through the body, including absorption, distribution, metabolism, and excretion.

Polypharmacy

The use of many medications.

Psychoactive drugs

A drug that changes mood or the way someone feels.

Psychotropic drug

A drug that changes mood or emotion, usually used when talking about drugs prescribed for various mental conditions (depression, anxiety, schizophrenia, etc.).

Synapse

The tiny space separating neurons.

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